What Makes a Latent Fire? Discovering Causal Dependencies Between Sparse Autoencoder Latents

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Abstract

Circuit discovery typically explains an output: a logit, a probability, a behavioural metric. We ask instead what makes an internal latent fire. For a target sparse autoencoder (SAE) latent, *weighted circuit masking* (WCM) learns a circuit of upstream latents with a scaling coefficient for each, trained so that the circuit alone reproduces the target with everything else ablated: set to zero, to one of two means, or to its value on a real context where the target is silent. We fit WCM at one sparsity penalty on three models: 8,000 sampled latents of TuringLLM, a 12-layer language model trained on known data, and 512 each of Pythia-70m and GPT-2 small with their public dictionaries. With circuits of a median 609, 294, and 371 upstream latents, 88%, 82%, and 82% of the scored targets (82%, 81%, and 82% counting every excluded target as a fail) get a circuit that reproduces the target under zero and both mean ablations and is necessary for it, on held-out text, while random circuits of the same size, given the same coefficient fitting, pass for none of 366 targets tested. On TuringLLM, weaker penalties give larger circuits that pass somewhat more often, up to a plateau, and a size-only rule fixed before a pilot kept the same penalty on both public models. The coefficients are what keep these circuits small: without them the same nodes pass for 2–14% of targets, and on TuringLLM the smallest passing unweighted circuit is a median 6× larger than the weighted one, 14× for targets in the last four layers. The procedure needs no behavioural task and applies to any latent in an SAE bank.

1 Introduction

Consider a latent in layer 10 of a 12-layer language model that fires on the possessive in "Ohm's", "Snell's", or "Kirchhoff's", before the word "law" has appeared, but stays silent on "Darwin's", "Rutherford's", and "Picasso's". Given an invented name, it fires in a passage on electromagnetism and not in one on the history of painting. Something upstream supplies what the model knows about which names carry laws, and what the passage is about. What makes this latent fire?

Sparse autoencoders (SAEs) decompose a model's activations into latents like this one, which often track coherent concepts (Cunningham et al., 2024; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024), a working answer to superposition (Elhage et al., 2022). A latent says what the model represents; it does not say how the model computes it. That question belongs to circuit discovery, whose methods, over components or features and by search, attribution, or learned masks, predominantly target an output (Conmy et al., 2023; Syed et al., 2023; Marks et al., 2025;

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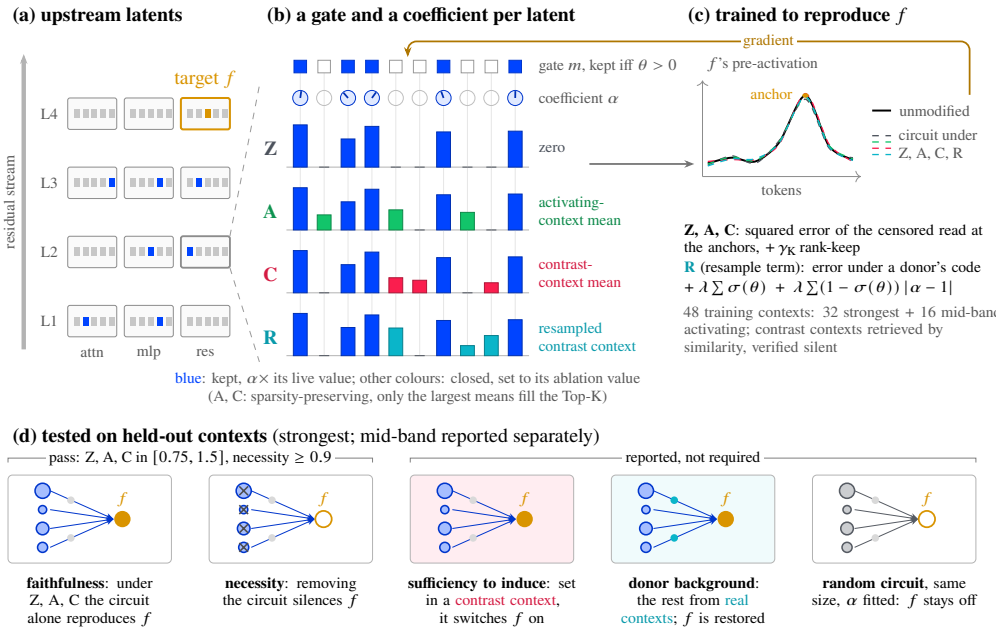


Figure 1: How weighted circuit masking works. **(a)** For a target latent f, every latent at every upstream SAE site is a candidate node. **(b)** Each candidate gets a gate m (kept when $\theta > 0$) and a scaling coefficient α: a kept latent contributes α times its live value, and a closed one is replaced by its ablation value: zero (Z), its activating- or contrast-context mean, only the largest means filling the Top-K (A, C), or a donor contrast context's code, redrawn every step (R). **(c)** They are trained so that, under all four, the circuit reproduces f's pre-activation at its anchors, with a sparsity penalty on every open gate. **(d)** The weighted circuit, typically 10^2–10^3 latents, passes on held-out contexts if it is faithful under Z, A, and C and necessary; sufficiency to induce, a donor-background test (its donors never R's), and same-size random circuits with fitted coefficients are reported.

Ameisen et al., 2025; Bhaskar et al., 2024; Caples et al., 2025). An output is the model's computation expressed in our vocabulary; targeting latents instead can capture variables that never show up as a token. Behavioural targets also start from tasks we thought to pose; latents, learned without labels, let the model's activations propose what to explain.

Internal targets have been approached by attribution, tracing upstream of a chosen SAE or transcoder latent, or a probe direction, and thresholding the scores (Ge et al., 2024; Golimblevskaia et al., 2025; Palit et al., 2026). To our knowledge, no method *optimises* a circuit of upstream latents for a single target latent and tests it by ablating everything outside the circuit and measuring that latent. We call this *latent-targeted circuit discovery* (Figure 1).

Our method, *weighted circuit masking* (WCM), learns two things for every upstream latent: a gate for whether it belongs to the circuit, and a scaling coefficient for how strongly it acts. Both are trained so that the circuit alone reproduces the target with everything else ablated in four ways at once: set to zero, to its mean over contexts where the target fires or where it is silent, or to its value on a real context where the target is silent. We test each circuit on held-out text for *faithfulness* (the circuit alone reproduces the target), *necessity* (removing the circuit removes the target), and *sufficiency to induce* (setting the circuit's values turns on the silent target).

We apply WCM to 8,000 sampled latents of TuringLLM, a 12-layer model trained on a documented synthetic corpus, and to 512 latents each of Pythia-70m and GPT-2 small with their public Top-K dictionaries. At one sparsity penalty, which a size-only rule fixed before a pilot kept on both public models, 82–88% of the scored targets get a circuit of a few hundred latents that passes a strict joint test: faithful under zero and both mean ablations and necessary, on held-out text (Section 4.1).

Random circuits of the same size, given the same coefficient fitting, pass for none of 366 targets. The coefficients are what keep circuits small: with every coefficient set to 1 the same nodes pass for 2–14% of targets, and on TuringLLM the smallest passing unweighted circuit is a median 6× larger than the weighted one (14× in the last four layers), while unweighted circuits never reach the 83% that weighted circuits reach on the same 173 sweep targets at the production penalty (Section 4.2). Node sets are not unique (a refit shares a median Jaccard of 0.39–0.51 with the original), so we report sizes and scores rather than single memberships (Section 5.2 and Appendix P). Section 4.4 opens one passing circuit, for a latent that reads a date or place as classical Greek antiquity; Appendix M returns to the law latent above, and Appendix N shows circuits for randomly drawn targets.

We contribute:

- **Circuits for arbitrary latents at scale, on three models** (Sections 3.2 and 4.1): weighted circuit masking, which learns a gate and a separate scaling coefficient for each upstream latent under four ablations at once, finds circuits of a few hundred latents that pass a held-out test for 82–88% of scored sampled latents, at one sparsity penalty that transfers across the three models, while size-matched random circuits with the same fitting never pass.
- **Coefficients keep circuits small** (Section 4.2): without fitted coefficients the same nodes pass for 2–14% of targets, and the smallest passing unweighted circuit for a deep TuringLLM target is a median 14× larger; unweighted circuits pass for at most 64% of deep targets at any size we fitted, against 76% for weighted circuits at the production penalty and up to 90% at weaker ones.

The appendices give the protocol, the production run in detail, robustness tests, and an earlier run and development experiments; Section 5 collects recommendations. We release the pipeline, discovery methods, and analysis code at `github.com/DanielJamesDavies/turing-explorer-circuit-discovery`.

2 Related Work

Circuit discovery. Manual analysis of components (Olah et al., 2020; Wang et al., 2023) gave way to automated search: ACDC (Conmy et al., 2023) and Edge Pruning (Bhaskar et al., 2024) prune components or edges against an output metric, and attribution patching (Nanda, 2023; Syed et al., 2023; Kramár et al., 2024) and its refinements (Hanna et al., 2024; Rezaei Jafari et al., 2025) score them to first order (the attribution baselines of Appendix C, whose comparison with WCM is still pending, adapt the activation-space IG variant of Marks et al. (2025), not EAP-IG). Sparse feature circuits (Marks et al., 2025) moved the nodes to SAE features with error nodes, and attribution graphs (Ameisen et al., 2025; Lindsey et al., 2025) build per-prompt graphs over cross-layer transcoder features (Dunefsky et al., 2024), a substrate we do not run. Their objectives are anchored at an output, and circuits for one task found on inputs of different token frequency differ in structure but share a core and one mechanism (Makou et al., 2026).

Circuits for internal features. Internal targets are not new. Hierarchical attribution extracts the subgraph behind a chosen SAE feature (Ge et al., 2024), dictionary features have been traced patch-free in Othello-GPT (He et al., 2024), transcoder features are reverse-engineered from their input circuits (Dunefsky et al., 2024), latent attribution debugs a chosen latent (Dupre la Tour and Mossing, 2025), CircuitLens explains a feature's activation from upstream interactions (Golimblevskaia et al., 2025), and attribution graphs have been extended to concept directions read by linear probes (Palit et al., 2026). Upstream features have also been ranked by single ablations (Cunningham et al., 2024) or linked by sparse regression (Yin et al., 2026). These methods attribute, ablate one node at a time, or regress; circuit probing is closer in spirit, optimising a mask over one block's neurons for a labelled intermediate variable (Lepori et al., 2023), and sparsely connected transcoders learn latent-to-latent masks while the dictionary is trained (jacob_drori, 2025).

Masks and scalars. Learned masks descend from subnetwork probing (Cao et al., 2021; Conmy et al., 2023) and edge pruning (Bhaskar et al., 2024; Li and Janson, 2024); over SAE latents they have

localised attributes (Chaudhary and Geiger, 2024) and found behavioural circuits, with per-position masks under one per-token mean ablation (Caples et al., 2025); our unweighted circuit masking moves this objective to a latent, with one shared mask, four training ablations, the sparsity-preserving fill, and rank-keep. Such masks keep a node at its natural scale or ablate it; learned per-unit scales serve other jobs, in pruning, steering, and regression (Liu et al., 2017; Kwon et al., 2022; Stoehr et al., 2024; Yin et al., 2026), and gate-times-magnitude parametrisations separate membership from strength outside circuit discovery (Louizos et al., 2018; Lai et al., 2025).

Evaluating circuits. What removing a latent means can decide the result (Miller et al., 2024; Mueller, 2024); optimal ablation learns it per component (Li and Janson, 2024), the dual of our per-node coefficients, while our ablations stay fixed by data; denoising and noising answer different questions (Heimersheim and Nanda, 2024), and our sufficiency to induce is a denoising-style test. Prior work evaluates under several ablations but discovers under one; the nearest objectives sum two directions of one counterfactual (Chen et al., 2025) or certify bounded patching perturbations (Hadad et al., 2026). Size-matched random circuits (Shi et al., 2024; Seth et al., 2026) go unfitted, like the circuits they test, while ours share our circuits' coefficient fitting; causal scrubbing (Chan et al., 2022) is stricter (Appendix L), and MIB standardises evaluation for output targets (Mueller et al., 2025).

What this paper adds is the conjunction: circuits of SAE latents *optimised* for a single target latent, with a gate and a separate *fitted* coefficient per node, trained under several ablations at once, and tested on held-out contexts against random circuits given the same fitting, for 8,000 sampled latents of one model and 512 of each of two public models (further related work in Appendix O).

3 Method

WCM applies to any model with a bank of Top-K sparse autoencoders, one per site (attention output, MLP output, and residual stream at each layer; GPT-2 small adds the residual stream between attention and MLP), each keeping K latents active per position. For a latent f and a sequence–position pair (x, t), write $a_f(x, t) \geq 0$ for its activation (after Top-K) and $\hat{a}_f(x, t)$ for its pre-activation (before it); pseudocode is in Appendix Q. As a running example we follow a layer-9 residual-stream latent of TuringLLM that fires where a passage places its subject in classical Greek antiquity: after "In the 5th century BC, the philosopher taught in the city of" it reaches 79% of its typical activation, and after "In the 19th century, . . . " it stays silent (Section 4.4).

3.1 Problem

Targets and contexts. Fix a target latent f and let U_f be the latents at every SAE site that causally precedes f's site. Each target has *activating contexts*, sequences on which it fires, each with an anchor at the target's peak and split into training and held-out sets, and *contrast contexts*, sequences close to the activating ones on which it is verified silent, outside its site's Top-K (Section 4 and Appendix R). For the running example they are history passages about other periods, such as the Enlightenment, Renaissance music, or the pre-Columbian era.

Circuit-only execution. A *weighted circuit* is a node set $C \subseteq U_f$ with a scaling coefficient $\alpha_{s,i} > 0$ for each node. Under an ablation method π, the forward pass is modified at every upstream site s: with c^s the site's Top-K code, computed from the (already modified) stream, gates $m_s \in [0, 1]^{d_{\text{sae}}}$, and ablation value c^s_π,

$$x \;\leftarrow\; x + W^s_{\text{dec}}\big[\, m_s \odot \alpha_s \odot c^s + (1 - m_s) \odot c^s_\pi - c^s \,\big]\,. \tag{1}$$

A circuit runs with binary gates, $m_{s,i} = 1$ exactly for its nodes: members are kept at α times their live value and every other latent is sent to its ablation value; WCM trains continuous gates and thresholds them to obtain C (Section 3.2). Editing only the decoded difference passes SAE errors through unchanged, and every downstream site, including the target's, reads the modified stream. Nodes are

recomputed, not clamped: a node's value at each position is whatever the modified stream gives it. The ablation value is zero (Z), the latent's mean over activating contexts (A), or its mean over contrast contexts (C), optionally sparsity-preserving, with the per-position Top-K re-imposed; training adds a fourth, resample ablation (R), which takes it from a real contrast context (Section 3.2). An unweighted circuit fixes $\alpha = 1$. Writing a_C for the target's mean activation at the anchors under this intervention, $a_\varnothing$ for the empty circuit, and a_{pos} for the unmodified model, *faithfulness* is

$$\phi_\pi^{\text{free}}(C) = \frac{a_C - a_\varnothing}{a_{\text{pos}} - a_\varnothing}.$$

The problem. Faithfulness depends on the ablation method (Miller et al., 2024; Mueller, 2024), and a circuit trained under one method can fail under another (Appendix V.3), so one weighted circuit is asked to serve all three:

$$n_\varepsilon(f) \;=\; \min\big\{\, |C| \;:\; \exists\, \alpha \;\text{ with }\; \phi_\pi^{\text{free}}(C, \alpha) \geq 1 - \varepsilon \;\text{ for every } \pi \in \{\text{Z}, \text{C}, \text{A}\} \,\big\}. \tag{P}$$

WCM solves a differentiable relaxation of (P); with $\alpha = 1$ the smallest faithful node sets are several times larger (Section 4.2). Problem (P) asks for *a* minimiser, and its near-optimal solutions form a large equivalence class (Appendix P), so only class-invariant quantities describe the target.

3.2 Weighted Circuit Masking

Rather than thresholding attribution scores (the rank-and-cut baselines it is compared against, Appendix C), WCM learns the circuit directly, adopting the optimisation framing of learned-mask circuit discovery (Bhaskar et al., 2024; Caples et al., 2025) with a continuous-sparsification gate (Savarese et al., 2020), and moving its objective from an output to a single latent. Each upstream latent gets two learned quantities: a gate, which decides whether it belongs to the circuit, and a coefficient, which decides how strongly it acts. The sparsity penalty charges for membership only, so strength is fitted separately from it. At every upstream site s, WCM sets $m_s = \sigma(\theta_s/T)$ and $\alpha_s = \text{softplus}(\psi_s)$ in (1), with parameters $\theta_s, \psi_s \in \mathbb{R}^{d_{\text{sae}}}$, and minimises

$$\begin{aligned}
\mathcal{L} = \frac{1}{\nu}\Big[\sum_{\pi \in \{\text{Z},\text{C},\text{A}\}} \gamma_\pi \Big(\text{mean}\big[(\bar{z}_\pi - \bar{z})^2\big] + \gamma_{\text{K}}\, \mathcal{K}_\pi \Big) + \gamma_{\text{R}}\, \text{mean}\big[(\tilde{z}_{\text{R}} - z)^2\big] \Big] \\
+ \lambda \sum_{s,i} \Big[\sigma(\theta_{s,i}) + \big(1 - \sigma(\theta_{s,i})\big)\, |\alpha_{s,i} - 1| \Big], \qquad (2) \\
\nu = \text{mean}\big[z^2\big], \qquad \mathcal{K}_\pi = \text{mean}\Big[\text{relu}\big(\tau_\pi^{(r)} - \tilde{z}_\pi\big)^2\Big], \qquad \bar{z}_\pi = \tilde{z}_\pi\, \sigma\Big(\frac{\tilde{z}_\pi - \tau_\pi^{(K)}}{T_b}\Big),
\end{aligned}$$

where z is the target's pre-activation $\hat{a}_f$ at the anchors of the training contexts on the unmodified stream, $\tilde{z}_\pi$ its value under (1) with method π, and the means run over those anchors. Under A and C, training uses evaluation's sparsity-preserving fill (Section 3.3), since circuits trained against a dense mean, a code no Top-K SAE produces, rely on a background that evaluation removes: at each position the members are kept and only the largest-mean non-members fill the rest of the Top-K, with straight-through gradients and evaluation's own means.

The terms. The squared error *reproduces* the pre-activation rather than the activation, which Top-K censors at zero, so the loss keeps a gradient where the mask must learn. Under Z, C, and A it compares *censored* reads $\bar{z}$, which multiply the pre-activation by a soft Top-K gate ($\tau_\pi^{(K)}$ the K-th largest pre-activation of the site's other latents, width $T_b = 0.05\,|z|$), so these terms also see whether the target would survive its site's Top-K. The *rank-keep* term $\mathcal{K}_\pi$ is zero whenever the target holds its natural rank r among its site's latents ($\tau_\pi^{(r)}$ the r-th largest of the others' pre-activations under the circuit). The weights γ_π score one mask under zero and both mean ablations at once, because a mask trained under one method learns whatever that method rewards (Appendix V.3). The *resample* term (R) replaces every latent outside the circuit with the sparse code of a donor, one of the target's

training contrast contexts redrawn at every step, so the mask must also hold its target against real backgrounds; its donors are never used in evaluation. The penalty is flat per latent, and the guard $\lambda\,(1-\sigma(\theta))\,|\alpha-1|$ stops a closed gate from carrying an inflated coefficient. The temperature T decays geometrically from 1 to 0.05 (Savarese et al., 2020), so training converges to the binary gates of the circuit $C = \{(s,i) : \theta_{s,i} > 0\}$ that evaluation scores. Coefficients start at $\alpha = 1$; *unweighted circuit masking* keeps them there. Training runs AdamW for 400 steps on batches of 4; the weights, penalty, learning-rate schedule, and their provenance are in Table 4, and the rationale for each term in Appendix B.1.

3.3 Evaluation

Every reported score is read on held-out contexts and reads the target's activation (after Top-K) in every term of its ratio; the full evaluation family and its settings are in Tables 3 and 4.

Intervention scores. Let A^+ and A^- be the target's mean activation on activating and contrast contexts. **Sufficiency to induce**, $\phi_{\text{ind}} = (A^-_{\text{int}} - A^-)/(A^+ - A^-)$, sets the circuit's nodes in contrast contexts to their mean values at the anchors of the activating contexts, times α for a weighted circuit ($\phi^{\alpha}_{\text{ind}}$, the form reported throughout): near 1, the circuit switches the target on. **Necessity** is one minus the completeness score of Marks et al. (2025), with SAE errors preserved and the nodes zero-ablated: with $a_{M\setminus C}$ the target's activation on activating contexts when the circuit's nodes are set to zero and every other latent is recomputed,

$$\phi_{\text{sup}}(C) = \frac{a_{\text{pos}} - a_{M\setminus C}}{a_{\text{pos}} - a_{\varnothing}}.$$

Both scores intervene on the circuit as one set. All scores are unclipped ratios: success means landing near 1, and a score far above 1 is overshoot, not quality.

Ablation methods. The choice of ablation is load-bearing (Li and Janson, 2024; Miller et al., 2024). Mean ablation over activating contexts leaks target signal into the ablation value increasingly with depth (in the protocol-v1 run, the activating-mean empty circuit already exceeds the circuit's members at natural scale for 32% of targets at layers 8–11, Appendix I; exploratory floors in Appendix S.2), while mean ablation over contrast contexts does not; dense means put the model on streams a Top-K SAE can never emit, so training and evaluation re-impose the per-position Top-K after applying them (sparsity-preserving; Appendix S.1). Patching from real contexts, often preferred for keeping the model on-distribution (Heimersheim and Nanda, 2024), is scored in Appendix L.

Random-circuit baselines. Fitted coefficients could make any set look faithful, so every random circuit, matched to the weighted circuit's size site by site, receives the *same coefficient fitting* on its own nodes before scoring. Its members are latents active near the target's anchor on its training contexts, matched to the circuit's members on corpus frequency (the *anchor-active* baseline) or also on contribution to the target (the *contribution-matched* baseline; Appendix I).

Pass criterion. A circuit passes if, on the held-out strongest contexts and the activation read, its faithfulness under zero ablation and the sparsity-preserving A and C means lies in $[0.75, 1.5]$ and its necessity is at least 0.9; the resample ablation of training is not part of the criterion, and (P) is its idealised form. The band is asymmetric by design: below 0.75 the circuit fails to restore its target, while an overshoot up to 1.5 still restores it. Sufficiency to induce, specificity (Appendix K), the random-circuit baselines, and a *donor-background test* are reported rather than required. In the donor-background test, everything outside the circuit takes a real donor context's code; a circuit passes it when it meets the pass criterion and its recovery of the target under activating and under held-out contrast donors lies in the same band (Appendix L). A target is *vacuous*, and is excluded and counted, when one of the criterion's empty-circuit baselines already reaches 95% of the unmodified activation; this depends only on the target and its contexts, never on the circuit (Appendix B).

4 Experiments and Results

Experimental setup. We run WCM on three models (Table 2). TuringLLM-1.0-254M is a 12-layer decoder-only transformer trained from scratch on a documented synthetic corpus of ∼2 billion tokens generated by Phi-3 Mini (Abdin et al., 2024), a more tractable subject than a model trained on an opaque crawl; its 36 Top-K sparse autoencoders have 40,960 latents with $K = 128$ (Appendices A and A.1). Pythia-70m (Biderman et al., 2023) and GPT-2 small (Radford et al., 2019) use public Top-K dictionaries of 32,768 latents at every site, EleutherAI's ($k = 16$; 18 dictionaries) and OpenAI's ($k = 32$; 48 dictionaries, including the residual stream between attention and MLP; Gao et al., 2024). Only the model and dictionary adapters differ; every other step runs through the same code. The public models read fixed, deduplicated corpora of 4M 64-token windows, and TuringLLM its own interpretability corpus; TuringLLM fits in bfloat16, the precision its pipeline was validated in, and the public models in float32 (Table 4).

Targets are sampled with no prioritisation heuristic. On TuringLLM they are 8,000 latents stratified over layer and site kind, drawn from the targets of an earlier run (Appendix H), which were drawn uniformly per site but excluded 879 dead or context-poor attention-output latents; the list contains the sweep, null-panel, case-study, and gallery targets by construction. On each public model, 512 targets are drawn at random about equally per site, from latents that fire in at least 200 distinct corpus windows and in at most half of them, so rare and hub latents are excluded; disjoint companion lists serve the sparsity-penalty pilot, the random-circuit baselines, the stable core, and an unweighted arm (Table 4). The sparsity penalty was calibrated, and the pass criterion set, on the same TuringLLM frame, so those choices are not independent of the TuringLLM evaluation.

Contexts. A target trains on 32 strongest and 16 mid-band activating contexts and is scored on 16 held-out strongest contexts (primary) and 16 held-out mid-band contexts (reported), with pools deduplicated and split by whole groups of copies, so that no copy of a training context is held out (Appendix B.2). It has 64 contrast contexts (48 train, 16 held out): corpus sequences most similar to its strongest contexts in pooled final-layer representation, kept only if the target stays out of its site's Top-K on them (Appendix R).

Runs. In the *production run* we fit WCM once per target at $\lambda = 10^{-3}$ on every model and score each circuit on its held-out contexts; every circuit is also re-scored with every coefficient set to 1 and under the donor-background test, and companion runs add the random-circuit baselines, a stable-core refit, and, on Pythia-70m, unweighted circuit masking (Appendix G). In a *sweep* we fit weighted circuit masking at eight penalties and unweighted circuit masking at five to 190 TuringLLM targets, plus an arm without the resample term (Section 4.2). Analyses of an earlier TuringLLM run under the first protocol, and of development experiments, appear only in the appendices, marked (Appendices B and H).

4.1 Weighted Circuits Explain Most Latents on Three Models

In the production run, WCM finds a circuit that meets the pass criterion for 88.0% of 7,464 TuringLLM targets (95% bootstrap interval 87.2–88.7), 81.9% of 508 Pythia-70m targets (78.5–85.2), and 82.4% of 511 GPT-2 small targets (79.1–85.5), all at $\lambda = 10^{-3}$ (Table 1). Median scores are close to 1 (Z 0.92–0.96, A 0.94–1.00, C 0.93–1.00; necessity 1.00), and circuits hold a median of 609, 294, and 371 upstream latents. Excluded targets are counted: 214 TuringLLM targets have no usable context record, three Pythia-70m targets were left without a held-out context, and 322, 1, and 1 targets are vacuous; counting them all as fails, 82%, 81%, and 82% of the sampled targets pass (Appendix G).

The deepest band passes least on every model, though the fall is not steady (Figure 2): 82.3% at TuringLLM layers 8–11 against 90.7% and 91.2% above, 78.4% at Pythia-70m layers 4–5 (overlapping layers 2–3), and 65.7% at GPT-2 small layers 8–11; deeper circuits are also larger. Attention outputs pass least on TuringLLM (77.5%) and Pythia-70m (64.8%), but not on GPT-2 small, where MLP outputs do (76.7%; by band and kind in Table 12).

Random circuits of the same size fail. Random-circuit baselines matched to each circuit's size site by site, drawn from latents active near the anchor and matched on frequency or also on contribution,

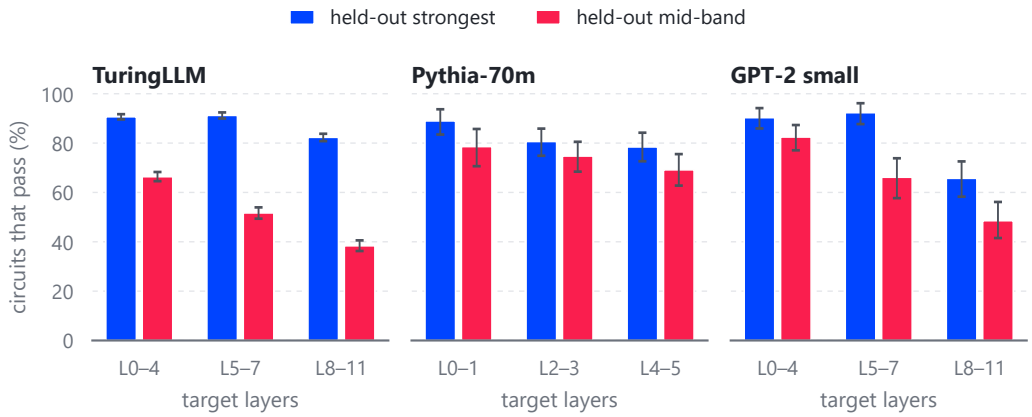


Figure 2: Circuits pass for most targets on all three models, least often in the deepest layers, and transfer only in part to weaker contexts. Pass rates of the production circuits at $\lambda = 10^{-3}$ by target depth band, on held-out strongest (blue) and held-out mid-band (red) contexts, with 95% bootstrap intervals over targets, under the criterion of Section 3.3.

Model	Scored	Pass, strongest (%) [95% CI]	Pass, mid-band (%) [95% CI]	Size, median (IQR)	Induce
TuringLLM	7,464	88.0 [87.2, 88.7]	53.5 [52.3, 54.8]	609 (403–954)	0.78
Pythia-70m	508	81.9 [78.5, 85.2]	73.6 [69.6, 77.4]	294 (132–653)	0.88
GPT-2 small	511	82.4 [79.1, 85.5]	66.8 [62.6, 70.9]	371 (208–666)	0.81

Table 1: Most targets pass on every model. Production run at $\lambda = 10^{-3}$: targets scored, pass rate on held-out strongest and held-out mid-band contexts with 95% bootstrap intervals over targets, median circuit size (interquartile range), and median sufficiency to induce. Scored targets are those sampled (8,000, 512, 512) less those without a record and those that are *vacuous*, where ablating every upstream latent already leaves the target above 95% of its activation and no circuit can be told from none (322, 1, 1; Table 9). Mid-band rates are over their own population: on TuringLLM, 358 targets have no mid-band pool and 1,200 of 7,428 mid-band rows are vacuous.

and given the same coefficient fitting, pass for none of the 366 targets tested on the three models, fitted or at $\alpha = 1$, while the production circuits of the same targets pass for 83–86% (Table 20 and Appendix I).

Robustness. Circuits transfer only in part to weaker, held-out mid-band contexts (53.5%, 73.6%, and 66.8% pass; on TuringLLM 38.4% at layers 8–11). With everything outside the circuit taken from real donor contexts, 69.2% of TuringLLM, 66.1% of Pythia-70m, and 41.1% of GPT-2 small targets still pass the donor-background test, so real backgrounds are harder than means (Appendix L). On the TuringLLM sweep the resample term adds only 5.2 points to the pass rate (not significant, −0.6 to 11.6; Table 23), but without it no circuit passes the donor-background test (65.9% with it). A refit with reordered training contexts shares a median Jaccard of 0.39–0.51 with the original circuit, and the shared core alone passes for 44–72% of targets (Appendix P). Specificity is reported, not required: 29–34% of circuits, and a similar share of passing ones, also restore the target's co-firing siblings to at least 80% of the target's own restoration under some ablation, and are flagged as possible amplifiers (Appendix K).

4.2 Fitted Coefficients Make Faithful Circuits Several Times Smaller

Without their fitted coefficients the same nodes almost never pass: with every coefficient set to 1, the pass rate falls from 88.0% to 1.8% on TuringLLM, from 81.9% to 13.8% on Pythia-70m, and from

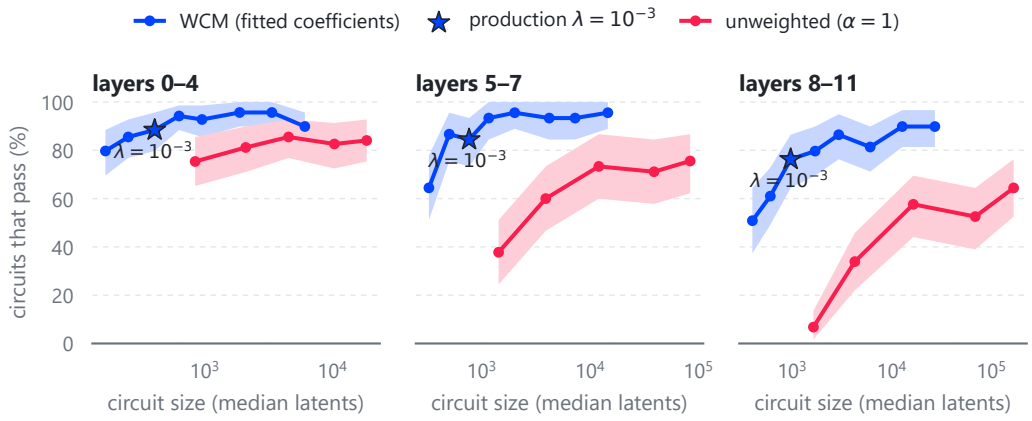


Figure 3: Over the size range both methods cover, weighted circuits pass more often than unweighted ones, and the gap widens with depth; WCM's pass rate plateaus at weak penalties and declines past 5×10^{-4}. Weighted (blue) against unweighted ($\alpha = 1$, red) circuit masking on the 190 TuringLLM sweep targets, by target depth band (columns), one point per sparsity penalty, WCM at eight penalties from 4×10^{-3} to 2.5×10^{-5} and unweighted masking at five from 10^{-3} to 10^{-5}, pass rate (with 95% bootstrap bands) against the median circuit size; the star marks the production penalty, $\lambda = 10^{-3}$.

82.4% to 3.3% on GPT-2 small, and to 0% in the deepest band of every model (Table 13). No circuit passes at $\alpha = 1$ that fails with its coefficients. Since random circuits fail under the same fitting, the coefficients are necessary but not sufficient: they make a selected node set faithful, not an arbitrary one.

Unweighted circuit masking isolates what the coefficients buy: the same objective, fill, resample term, and optimiser with every α fixed at exactly 1, fitted to the 190 TuringLLM sweep targets (173 scored) at five penalties alongside WCM at eight (Figure 3). Target by target, the smallest passing unweighted circuit is a median 6.0× larger than the smallest passing weighted one over the 142 targets both methods pass (interquartile range 4.2–13.8): 5.1×, 7.9×, and 13.8× at layers 0–4, 5–7, and 8–11, where it holds 6,124 latents against 436 (Table 14). This ratio is conservative, since it leaves out the targets unweighted masking never passes. Unweighted circuits never reach the 83% that WCM reaches on the same 173 targets at $\lambda = 10^{-3}$, peaking at 75.1% (68.8–81.5) at a median of 7.6×10^4 latents; at layers 8–11 they peak at 64.4%, against 76.3% for WCM at the production penalty and up to 89.8% at weaker ones (Table 11). Over the size range both methods cover, weighted circuits pass more often, by a margin that grows with depth (at layers 0–4 it lies within the intervals). Under real backgrounds the gap is wider: unweighted circuits pass the donor-background test for at most 8.7% of these targets at any penalty, and weighted ones for 66–83% (Appendix L). On Pythia-70m, unweighted circuit masking at $\lambda = 10^{-3}$ passes for 59.8% of 127 targets against 83.5% for WCM, with circuits a median 2.3× larger, and the gap is widest at depth (37.5% against 87.5% at layers 4–5; Table 10).

4.3 One Sparsity Penalty Transfers across Models

We use $\lambda = 10^{-3}$ throughout; it was set on TuringLLM. On the TuringLLM sweep, the pass rate stays on a plateau of 89–93% from $\lambda = 2.5 \times 10^{-5}$ to 5×10^{-4}, while each doubling of λ shrinks the median circuit by a factor of 1.4–1.9, from 12,040 latents to 990; beyond it the rate declines, to 83.2% at 10^{-3} and 65.9% at 4×10^{-3} (Table 11). The production penalty thus sits just past the plateau's edge: against 5×10^{-4} it gives up 5.8 points of pass rate, spread over every depth band, for circuits 1.7× smaller (573 against 990 latents). On each public model, a size-only rule fixed before a 30-target pilot kept 10^{-3} unless the pooled median circuit size left $[10^2, 10^3]$, and it kept it on both (medians

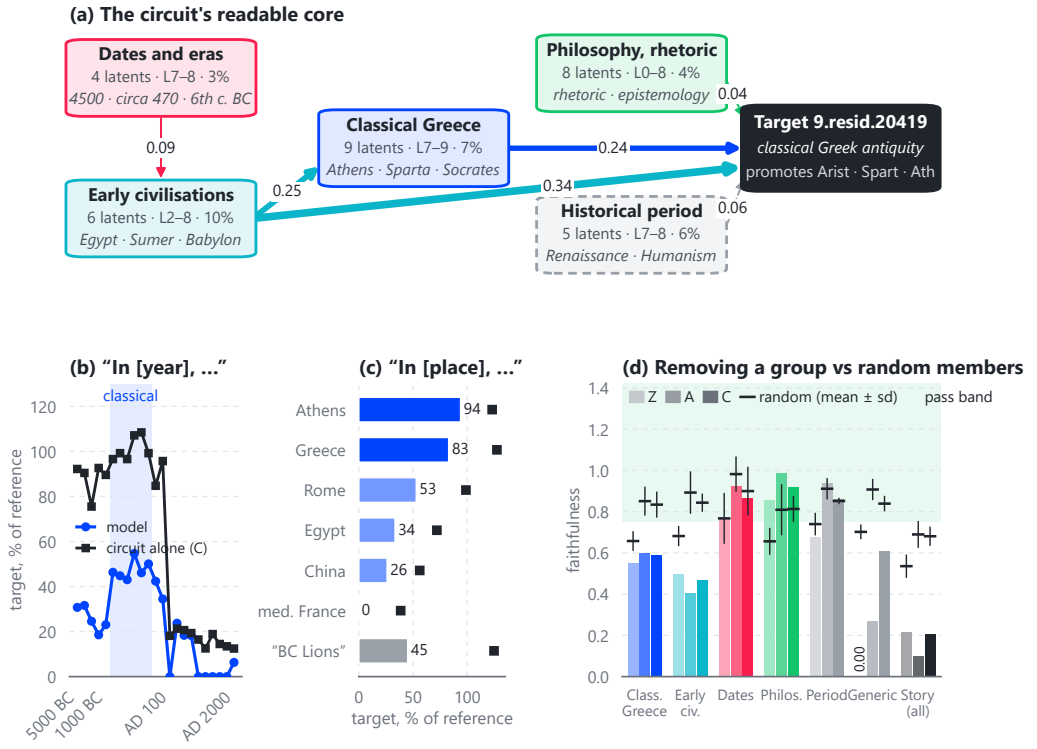


Figure 4: A passing circuit has a readable core: the production-run circuit of a latent for classical Greek antiquity (layer 9, residual stream, 945 latents; Section 4.4). **(a)** Role groups among the circuit's largest contributors, the protocol-v1 run's groups restricted to latents still present, named from their latents' own top contexts, with each group's size, layers, and share of the circuit's contribution at the target's anchor. Arrows give the total drop in the destination (or the target) when the source group is removed, with all other members recomputed (activating-mean ablation, circuit alone, held-out strongest contexts); the dates group's drop in the target (0.07) is not drawn. The dashed grey group fires about as much on contrast contexts. Not drawn: a government and empire group (2 latents, 0.9% of the contribution; removing it gives 0.90/1.03/0.96 against 0.81/0.96/0.91 for random members), a generic core of 7 latents, and 904 further latents. **(b,c)** The target on template sentences ("In [year], . . . " and "In [place], . . . ") as a percentage of its mean activation on its held-out strongest contexts, in the unmodified model (lines, bars) and with the circuit run alone under contrast-mean ablation (black squares; the empty circuit gives 0). **(d)** Faithfulness with each group removed (bars) against removing the same number of random members of similar contribution rank (mean ± sd over 8 draws); shaded, the pass band.

328 and 428 latents; Table 15). This is transfer of one setting, not a claim that it is the best one: on Pythia-70m, 3×10^{-4} passed as often in the pilot with circuits 2.8× larger. Sizes and pass rates are measured at this penalty and a 400-step budget, by which circuits shrink slowly (a median 2.8–5.2% of members leave over the last 100 steps), though in the deepest band of TuringLLM and Pythia-70m the circuits still shrinking pass less often (Appendix G).

4.4 A Circuit for Classical Antiquity Has a Readable Core

We open the running example's circuit to show what a passing circuit contains. Its target, a layer-9 residual-stream latent, fires where a passage places its subject in classical Greek antiquity, and its logit effect promotes "Arist", "Spart", "Ath", and "Greek". It tracks era and place rather than a word:

completing "In [year], the scholar lived in the city of" (a different template from the sentence of Section 3), it reaches 42–55% of its typical activation for 800–200 BC, 19–32% for older dates, and at most 6% from AD 1000 on, and for places it gives ancient Athens 94%, Rome 53%, Egypt 34%, and medieval France 0% (Figure 4b,c).

Its weighted circuit in the production run holds 945 latents and passes on the held-out strongest contexts (Z, A, C 0.91, 1.08, 0.98; necessity 1.00). Its largest contributors still include the readable core of this target's circuit in the earlier run, whose role groups, named from the members' own top contexts, we keep for the 41 of their 46 latents still present (Figure 4a; overlap in Appendix M). Latents for early civilisations (Egypt, Sumer, Babylon; 6 latents, 10% of the contribution) feed latents for classical Greece itself (Athens, Sparta, Socrates; 9 latents, 7%): removing the early-civilisation group lowers the classical group by 25%, and a group for dates and eras feeds the early-civilisation group more weakly (9%). Removing the early-civilisation group takes the target to 0.50, 0.40, and 0.47 under Z, A, and C, and removing the classical group to 0.55, 0.60, and 0.59, against 0.68, 0.89, 0.84 and 0.66, 0.85, 0.83 when as many random members of similar contribution rank are removed (Figure 4d). Removing all 19 story-group latents, 2% of the circuit, takes the target to 0.22, 0.10, and 0.21 (random: 0.54, 0.69, 0.68). Latents for philosophy and rhetoric, the subject of the target's training contexts, matter no more than random ones.

Run alone under contrast-mean ablation, where the empty circuit gives 0, the circuit keeps the target's ordering but not its levels: every BC year reads 75–109% of the typical activation (19–55% in the model), every AD year 12–21%, and medieval France 39% (0% in the model) (Figure 4b,c). The core is a reading, not the whole circuit: the story groups carry 20% of the contribution, "BC" alone partly triggers the target ("The BC Lions" 45%), a generic core of seven latents that fire on most text is also necessary (removing it gives 0.00, 0.27, 0.61), and under activating-mean ablation the circuit restores co-firing sibling latents to 0.84, though under contrast-mean ablation it is specific (siblings 0.44; Appendix K).

A second circuit, for magnetic-confinement fusion, shows a sharper core inside an amplifier: ten of its 1,322 latents carry the computation, but it lifts about 12,000 latents above the Top-K cut (Appendix M). Further cases are in Appendix M; since they were found by searching, Appendix N adds circuits for nine randomly drawn targets of the earlier run.

5 Discussion and Limitations

Circuit size is a measurement at a stated scaling convention. At natural scale ($\alpha = 1$), unweighted circuits for deep TuringLLM targets need far more latents than weighted ones to pass, and pass less often at any size we fitted (Section 4.2). We hypothesise that a latent's activation, reached through an additive residual stream, is a dense function of many small contributions, much of it redundant, which fewer, rescaled latents can carry; a circuit's size is therefore meaningful only alongside whether its coefficients were fitted.

5.1 Using WCM as a Framework

WCM is better read as a framework for solving for circuits than as one fixed algorithm. Its parts are interchangeable: an objective (the quantity reproduced, the ablation methods it is reproduced under, the regularisers, and their weights; Eq. (2)), a parametrisation (a gate and a coefficient per node), data (training, contrast, and held-out contexts), and an optimiser. The production recipe is one setting of these parts, each chosen by an experiment (Table 29), and changing a part changes the circuits found. With every coefficient fixed at 1 the same nodes pass for 2–14% of targets (Section 4.2); in exploratory fits, coefficients trained without zero ablation overshoot it by one to three orders of magnitude (Appendix V.3); circuits trained against close contrast contexts are the most faithful on close backgrounds and the least on any other (Appendix J); and circuits failed a resample test until training included a resample term (Appendix L). When WCM fails an evaluation, we therefore suggest first asking whether training ever saw that evaluation's domain, and if not, adding the term, contexts, or setting that puts it there, while keeping the evaluation's own contexts and donors out of training

so that it stays a held-out test (Appendix U). A pass rate then describes a recipe, a dictionary, and an evaluation together, not WCM alone, which is why the ablation methods used for training and evaluation belong in the method statement (Mueller et al., 2025; Mahale, 2026).

Node sets are samples; pass rates are stable. A refit with reordered training contexts passes about as often on TuringLLM and Pythia-70m but shares a median Jaccard of only 0.39–0.51 with the original circuit (Appendix P), so we treat sizes and scores, not memberships, as properties of the target; composing circuits into chains must then manage classes of node sets.

Latent targets reduce the experimenter's choice of what to explain. Among sampled targets, hand search found variables no task set of ours would have contained, such as names that carry a law and a date read as classical antiquity (Section 4.4 and Appendix M); no latent we examined checks whether a stated fact is correct. The bias moves rather than vanishes: which variables exist depends on the dictionary (Leask et al., 2025), and reading a latent still relies on human interpretation.

Recommendations. For latent-targeted circuit discovery we suggest: report faithfulness under more than one ablation method and name the methods used for discovery, training, and evaluation; compare circuit sizes only at matched faithfulness, stating whether coefficients were fitted; report specificity, since a faithful circuit can be an amplifier; report sizes, scores, and stable cores rather than one membership; and keep near and anchor-prefix copies of training contexts out of held-out sets, not only exact copies (Appendix B.2).

5.2 Limitations

Scope. Three models of at most 254M parameters, Top-K dictionaries only (on dense ReLU dictionaries a random-circuit baseline with fitted coefficients stops discriminating; Appendix T), one sparsity penalty at a 400-step budget, and anchors only: a circuit explains the target where it fires, not at every position (Appendix L). **The SAE basis is not ground truth.** Circuits built from ambiguous latents inherit that ambiguity; our scores measure intervention effects, not interpretability. **The objective is a surrogate.** WCM trains on the pre-activation, softened by the censored read, while evaluation reads the activation after Top-K, so targets near the Top-K cut are ill-conditioned (Appendix B). **Faithfulness does not imply specificity.** Rank-keep limits lifts past the target but does not remove amplifier circuits (Appendix K). **Generality.** TuringLLM's synthetic corpus aids control but limits generality; the public models, with 512 targets each, partly address it, and contrast contexts are one of several defensible constructions (Appendix J).

6 Conclusion

We asked what makes an internal latent fire, and answered with weighted circuit masking. Two results carry the paper. Circuits for arbitrary latents can be found at scale and tested strictly, on three models at one sparsity penalty: 82–88% of the scored sampled targets (81–82% of all sampled targets) get a circuit of a few hundred latents that is faithful under zero and both mean ablations and necessary on held-out text, while random circuits of the same size and fitting never pass; fewer pass at depth and on weaker contexts. And fitted coefficients are what keep those circuits small: without them the same nodes almost never pass, and for TuringLLM targets in the last four layers the smallest passing unweighted circuit is a median 14× larger; unweighted circuits pass for at most 64% of those targets at any size we fitted, against 76% for weighted circuits at the production penalty on the same sweep targets. Because the targets are sampled rather than chosen, the results describe a population of internal variables, as recovered by one dictionary, rather than a few tasks we thought to ask about; and because a circuit's size depends on whether its coefficients were fitted, a claim that some computation is dense or sparse needs its scaling convention stated. Circuits for single latents are a building block: composing them into chains, with propagation tested at every link, is the path from dependencies to a map of the model's internal computation.

Reproducibility

The full system (pipeline stages, evidence stores, the discovery methods of Section 3, the target-intervention evaluation, and the analysis code that produces every figure and table in this paper) is available at `github.com/DanielJamesDavies/turing-explorer-circuit-discovery`. TuringLLM, its synthetic training corpus, and the trained SAE bank are described in Appendix A; every experiment runs from a released configuration under the protocol of Appendix B, and each analysis reads only the run artifacts written by the pipeline stages. The production run used NVIDIA H100 SXM GPUs, with fit times per model in Table 8.

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A Model Cards

Table 2 gives what the results depend on for each model. TuringLLM follows the GPT residual-stream design with SwiGLU MLPs (Shazeer, 2020) and RMSNorm (Zhang and Sennrich, 2019); it is meant as a subject whose training data is fully known, not as a strong model. The public models and dictionaries are used as released, read at the hook points the dictionaries were trained on.

A.1 Sparse Autoencoder Quality

Figure 5 reports, for each TuringLLM site, explained variance, $1 - \mathrm{Var}(x - \hat{x})/\mathrm{Var}(x)$; the fraction of the 40,960 latents that fire at least once over the interpretability corpus; and loss recovered, the share of the cross-entropy gap between the model and zero-ablating the site that splicing in the reconstruction recovers (Marks et al., 2025), over 12,288 tokens. Attention SAEs explain 0.93–0.99 of variance, residual SAEs decline with depth to 0.85, and MLP SAEs reach 0.77 at layer 9. Two sites are weak: only 37% of the layer-1 attention latents ever fire (84% at layer 2, at least 93% elsewhere), and the layer-11 attention SAE recovers 55% of the gap despite explaining 0.96 of variance (layer 10: 84%). Of the public dictionaries, the $k = 16$ Pythia-70m ones are the lossiest per site (Table 2).

	TuringLLM-1.0-254M	**Pythia-70m** (Biderman et al., 2023)	**GPT-2 small** (Radford et al., 2019)
Model			
Parameters	254M	70M	124M
Layers, d_{model}, heads	12, 1,024, 16	6, 512, 8	12, 768, 12
Block	sequential; SwiGLU MLP (4,096); RMSNorm	parallel attention and MLP; GELU MLP (2,048); LayerNorm	sequential; GELU MLP (3,072); LayerNorm
Context	1,024	2,048	1,024
Tokenizer	Phi-3 Mini tokenizer; embedding and unembedding sized 50,304 (padded)	GPT-NeoX; 50,304	GPT-2 BPE; 50,257
Training data	~2B synthetic tokens: Phi-3 Mini (Abdin et al., 2024) educational documents over 18 subject areas, each with 11 paraphrases (Vorobev and Kuznetsov, 2023); a final pass over the originals only	the Pile (Gao et al., 2020), 300B tokens	WebText
Training	trained here: 3,985 steps, minimum cross-entropy 1.62 (train), 1.80 (validation)	released checkpoint	released checkpoint
Dictionaries			
Source	trained here, Top-K	EleutherAI Top-K	OpenAI v5 Top-K (Gao et al., 2024), via jbloom's release
k, latents	128, 40,960	16, 32,768	32, 32,768
Sites	attention out, MLP out, residual; 12 layers (36)	attention out, MLP out, residual; 6 layers (18)	attention out, residual between attention and MLP, MLP out, residual; 12 layers (48)
Input	the raw activation	the raw activation	per-token layer-normalised; edits scaled back by the token's standard deviation
Training data	see the note below	the Pile, 8.2B tokens, context 2,049	OpenWebText (Gokaslan and Cohen, 2019), context 64, no BOS
Quality	explained variance 0.77–0.99; layer-1 attention: 37% of latents fire; layer-11 attention: 55% of loss recovered (Figure 5)	FVU 0.02–0.29; ΔCE +0.02 to +0.47 nats, largest at the residual stream (+0.16 to +0.47)	FVU $\leq$ 0.30; ΔCE $\leq$ +0.30 nats, largest at the two residual sites
Analysis corpus (64-token windows, no BOS for the public models)			
Windows	24.8M, in 3,030 shards	4M, one pinned file of the uncopyrighted Pile	4M, pre-tokenised OpenWebText

Table 2: Model cards. FVU is the fraction of variance unexplained and ΔCE the cross-entropy increase when one site's reconstruction is spliced in, over 1,024 WikiText tokens (16 windows of 64, no BOS); Table 4 gives the sampling frame, corpora and precision. [*pending: TuringLLM SAEs' training data and token count (Daniel to obtain, DAN-185)*]

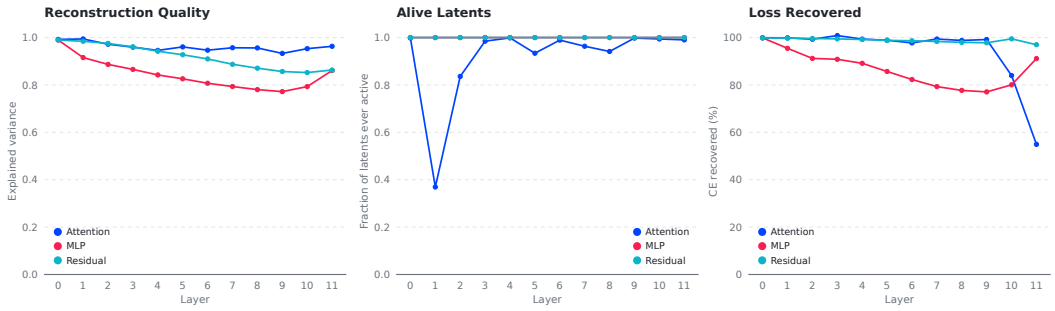


Figure 5: Quality of the 36-SAE bank by site. Left: explained variance of each SAE's reconstruction. Middle: fraction of latents that fired at least once over the full collection pass. Right: percentage of the cross-entropy gap recovered when splicing each SAE's reconstruction into the model, relative to zero-ablating the site.

B Protocol

Every circuit in the main text is discovered and scored under the settings in Table 4, the production protocol, on every model; only the model and its dictionary change. It differs from the first protocol, protocol v1, in five ways, each fixed before any production fit: the A and C terms are trained under the sparsity-preserving fill that evaluation uses, rather than a dense mean; context records are built afresh, deduplicated and split by whole groups of copies (Appendix B.2); training adds the resample term, with a stronger rank-keep weight; the learning rate follows a warm-up and geometric decay, and the Z, A, and C terms read the target through the censored read (Section 3.2); and the pass band's floor is 0.75 rather than 0.8, with vacuous targets defined by every baseline of the rule. Many appendix analyses use the circuits of an earlier TuringLLM run under protocol v1 (Appendix H) and are marked where they appear: the details of that run (Appendix H), the coefficient analyses and harder-null design (Appendix I), the method comparison (Appendix D), the contrast-source comparison (Appendix J), specificity (Appendix K), the resample tests (Appendix L), the case studies and random gallery (Appendices M and N), and the cross-GPU refit and stable core (Appendix P). Analyses marked *exploratory* predate protocol v1, and each states how its settings differ. Table 3 names the evaluation family; every reported number carries its row, is read on the held-out activating contexts (or held-out contrast contexts), and reads the target's activation in every term of its ratio, except the secondary pass rate on the pre-activation read (Table 4, Read).

Score	Contexts	Circuit latents	Other latents
ϕ_{ind}	contrast	set	untouched
ϕ_{sup}	activating	zero	recomputed
$\phi^{\text{free}}_{\text{Z}}$	activating	recomputed (permitted)	zero
$\phi^{\text{free}}_{\text{A}}$	activating	recomputed	activating means, dense
$\phi^{\text{free}}_{\text{A,sp}}$	activating	recomputed	activating means, sp
$\phi^{\text{free}}_{\text{C,sp}}$	activating	recomputed	contrast means, sp
$\phi^{\text{free}}_{\text{Z},\alpha}$ / $\phi^{\text{free}}_{\text{A},\alpha}$	activating	recomputed, $\times\,\alpha$	zero / activating means
$\phi^{\alpha}_{\text{ind}}$	contrast	set to $\alpha\times$ anchor means	untouched

Table 3: The evaluation family. Intervention scores: ϕ_{ind} sufficiency to induce, ϕ_{sup} necessity; "set" means set, at every position, to the node's mean value at the anchors of the activating contexts. Faithfulness ϕ^{free}_{π} measures whether the circuit reproduces the target alone with its nodes recomputed, and each variant names its ablation method π (Z zero, A mean over activating contexts, C mean over contrast contexts; sp sparsity-preserving; Section 3.1). The α rows score weighted circuits with their scaling coefficients applied (Section 4.2). SAE errors are preserved in every row, and every ratio reads the target the same way in all of its terms.

Setting	Value
Models and dictionaries	
TuringLLM-1.0-254M	Top-K, $k = 128$, 40,960 latents; attention output, MLP output, residual stream at 12 layers
Pythia-70m	EleutherAI Top-K, $k = 16$, 32,768 latents; the same three sites at 6 layers
GPT-2 small	OpenAI Top-K, $k = 32$, 32,768 latents, layer-normalised input; attention output, residual stream between attention and MLP, MLP output, residual stream at 12 layers

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Setting	Value
Targets per model	TuringLLM: 8,000, a frozen list drawn with seed 20261026, stratified over the layer and site-kind cells (about 229 per cell; precision about ±1.1 points overall and ±3 per depth band and site kind), from the 15,046 targets of the protocol-v1 run (Appendix H: 455 drawn uniformly at random at every site with an upstream SAE site, restricted to those an earlier run of the pipeline could fit, which dropped 879 dead or context-poor attention-output targets); it contains the 190 sweep targets, the 126-target null panel, and the case-study and gallery targets. 214 have no context record (205 too few distinct strongest contexts, 9 too few contrast contexts), and no fit failed. Pythia-70m and GPT-2 small: 512 each, drawn uniformly at random with seed 20261024, 32 at each of Pythia-70m's 16 sites with an upstream site and 10–11 at each of GPT-2 small's 47, from latents that fire in at least 200 distinct windows of the model's corpus and in at most half of them (hub cap; the same cap would remove 1.75% of the TuringLLM targets); disjoint companion lists of 30 (pilot for λ), 128 (harder nulls), 32 (stable core), and on Pythia-70m 128 (unweighted arm)
Corpora	TuringLLM: its interpretability corpus, 24.8M windows of 64 tokens (Appendix A); Pythia-70m: 4,000,000 windows from one pinned file of the uncopyrighted Pile (Gao et al., 2020); GPT-2 small: 4,000,000 windows of OpenWebText (Gokaslan and Cohen, 2019), as tokenised for its dictionaries, from three pinned files; exact duplicate windows removed from the public corpora before the stores are built
Precision	TuringLLM fits and scores in bf16, its validated precision; Pythia-70m and GPT-2 small fit and score in float32 with TF32 matrix multiplies; the public context stores are built in one float32 pass per model, since a bf16 pass did not reproduce the same top-context sets in a check before the run
Contexts	
Activating contexts	up to 64 strongest and up to 64 mid-band contexts per target, deduplicated (exact, anchor-prefix and near copies, below); a pool with fewer than 8 is treated as absent. Each pool is split 3:1 into training and held-out contexts: ranked by anchor activation in blocks of 4, block b holds out position $(b + 2) \bmod 4$, so ranks 1–2 always train, with no group of copies on both sides. With full pools the target trains on 32 strongest (a stratified two-thirds of the 48 strongest training contexts) and 16 mid-band, and holds out 16 strongest (primary) and 16 mid-band (reported); targets with fewer than 8 mid-band contexts train on all their strongest training contexts (358 TuringLLM targets have no mid-band pool; none on the public models); 383, 240, and 165 TuringLLM, Pythia-70m, and GPT-2 small targets hold fewer than 16 held-out strongest contexts and 262, 11, and 11 fewer than 8 (Table 5). TuringLLM's strongest pools draw candidates beyond the stores' 64 from a corpus scan of each target's 512 strongest contexts, ranked by the stores' own statistic; mid-band contexts are held out only when the pool has at least 32
Held-out rule	every held-out set excludes each training-side context (strongest and mid-band training contexts, all 64 contrast contexts, and every corpus copy of them) and its exact, anchor-prefix (identical up to the anchor) and near copies (a shared run of $\geq$ 32 tokens, or $\geq$ 50% of its 8-grams shared), and any copy of an already accepted held-out context; held-out activating contexts must fire at the anchor. Pools are split by whole groups of copies, so no group spans training and held-out contexts, and the rule is applied once more as a final check; rows are only removed, so a held-out set may hold fewer than 16 (floor 8, reported); a target with no held-out strongest context left is not fitted (3 Pythia-70m targets; Appendix B.2)

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Setting	Value
Contrast contexts	64 per target: candidates ranked by highest cosine similarity to any of the target's strongest stored contexts (pooled final-layer representations), its activating contexts excluded; the $128n$ most similar (8,192 for $n = 64$; 24,576 in the deduplicated records) are scanned in order and kept only if the target is outside its Top-K at every position, until 64 are kept (Appendix R); 48 train, 16 held out, stratified by similarity; the held-out 16 then go through the held-out rule against the contrast training contexts and every activating context, must stay silent at every position, and are refilled from the selector's next candidates; contrast contexts are deduplicated by the exact and near rule (they have no anchor). TuringLLM targets with too few contrast contexts get no record (9, with 0–2), and 6 fitted TuringLLM targets have fewer than 64 (at least 6); every Pythia-70m and GPT-2 small target has 64
Anchor	position of the target's peak activation in each context
Weighted circuit masking, WCM[Z+C+A] with the resample term	
Ablation methods	Z zero ablation (weight 1); C mean over contrast contexts ($\gamma_{\mathrm{C}} = 0.25$); A mean over activating contexts ($\gamma_{\mathrm{A}} = 0.25$), at every layer; A and C are trained under the evaluation's sparsity-preserving fill, bit for bit (Appendix S.1), with straight-through gradients through the fill's selection, and with the evaluation's means (A over the reference contexts below, C over the 48 training contrast contexts); R resample ablation ($\gamma_{\mathrm{R}} = 1$), below
Resample term	every non-member takes the sparse code of a donor at the same position, the donors being the target's 48 training contrast contexts, one drawn for each training context at every step (seeded); members kept, and the donor's largest active non-members fill the remaining Top-K budget; SAE error live; no rank-keep or censored read; unweighted circuit masking takes the same term at $\alpha = 1$
Objective	MSE to the target's natural pre-activation at the anchors, each term divided by one shared $\nu = \mathrm{mean}(\mathrm{target}^2)$ (uncensored); under Z, C, and A both sides pass through the censored read $v\,\sigma((v - \tau^{(K)})/T_b)$, $\tau^{(K)}$ the K-th largest pre-activation of the site's other latents (under the ablation for the circuit, in the unmodified model for the target), $T_b = 0.05\,\lvert z_b\rvert$ at anchor b, at least 10^{-4}
Rank-keep	$\gamma_{\mathrm{K}} = 10^{-1}$ on $\mathrm{relu}(\tau^{(r)} - \hat{a}_f)^2$ inside the Z, C, and A terms (so weighted $\gamma_\pi \gamma_{\mathrm{K}}/\nu$), r the target's rank among its site's other latents at the training anchor in the unmodified model, capped at K
Gates	$m = \sigma(\theta/T)$, $\theta_0 = 4$; T annealed geometrically from 1 to 0.05; nodes are $\{m > \frac{1}{2}\}$
Scaling coefficients	$\alpha = \mathrm{softplus}(\psi)$, $\alpha_0 = 1$; closed gates charged $\lambda(1 - \sigma(\theta))\,\lvert\alpha - 1\rvert$ (no temperature). Unweighted circuit masking fixes every α at exactly 1 and is otherwise identical, including the sparsity-preserving training fill and the resample term
Sparsity penalty	$\lambda = 10^{-3}$, flat per latent, on every model. TuringLLM: calibrated there; the sweep fits WCM at $\lambda \in \{0.25, 0.5, 1, 2.5, 5, 10, 20, 40\} \times 10^{-4}$ and unweighted circuit masking at $\{0.1, 0.3, 1, 3, 10\} \times 10^{-4}$ on 190 targets. Public models: a 30-target pilot at three values (Pythia-70m $\{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, GPT-2 small $\{10^{-3}, 3 \times 10^{-3}, 10^{-2}\}$) under a rule stated before it: keep 10^{-3} unless the pooled median circuit size leaves $[100, 1000]$, else take the value whose median is nearest 494 (a protocol-v1 TuringLLM median) on a log scale; size only, never pass rate. It kept 10^{-3} on both models (Section 4.3)
Optimiser	AdamW over θ and ψ, weight decay 0.05, 400 steps, batch 4 in a fixed order; learning rate warmed up linearly over the first 40 steps from 0.0079 to a peak of 0.1575 (3.15×0.05), then decayed geometrically to 0.0079 at the last step, so the summed rate is about that of a constant 0.05

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Setting	Value
Execution	nodes recomputed from the edited stream; SAE errors preserved; every upstream site edited
Provenance	$\gamma_C = \gamma_A = 0.25$ at every layer is a protocol decision; γ_K and γ_R come from a selection on a 72-target TuringLLM check panel under a rule stated before it; the learning-rate schedule and the censored read were validated on TuringLLM targets; λ is calibrated on TuringLLM; the 400-step schedule is examined in the binarisation sweep (Appendix V). TuringLLM: the calibration targets are not disjoint from the evaluation set; the pass rule was set on a 1,416-target subset of the same frame. Pythia-70m and GPT-2 small: λ is checked on the 30-target pilot, disjoint from the 512 main targets, and every other setting carries over from TuringLLM unchanged
Evaluation (Section 3.3), on the held-out contexts	
Reference contexts	the evaluation's A means, the anchor means set by sufficiency to induce, the specificity siblings, and the random-circuit pool are computed on the strongest training contexts (48 with a full pool) for every target, one reference whether or not the target trains on mid-band contexts; C means use the 48 training contrast contexts, as in training
Read	the pass criterion reads the target's activation (after Top-K), in every term of every ratio; a secondary pass rate applies the same rule to the pre-activation, asking whether the circuit reproduces the target's input apart from whether the target then survives the Top-K competition. On that read Z and necessity are measured from a baseline of 0, latent off ($\phi_Z^{\text{free}} = \hat{a}_C/\hat{a}_{\text{pos}}$, $\phi_{\text{sup}} = (\hat{a}_{\text{pos}} - \hat{a}_{M\setminus C})/\hat{a}_{\text{pos}}$, hats for pre-activations), because with every upstream latent zeroed a residual-stream target's pre-activation leaves its natural range and the empty-circuit form is undefined for most of them; A and C keep their mean-ablation baselines; for attention and MLP targets, where the empty circuit is well defined, the empty-circuit baseline is reported as a sensitivity check. The two reads are never mixed inside one rule. Training uses the pre-activation, which keeps a gradient where Top-K censors the activation
Faithfulness	ϕ_π^{free} under Z, dense A, sparsity-preserving A and C; each of a_C, $a_\varnothing$, a_{pos} is the target's mean over the held-out anchors, so the score is a ratio of means, not a mean of per-context ratios, and its spread over contexts within a target is not reported
Necessity	$\phi_{\text{sup}}(C) = (a_{\text{pos}} - a_{M\setminus C})/(a_{\text{pos}} - a_\varnothing)$, the circuit as one set, its nodes set to zero; $a_\varnothing$ the zero-ablation empty circuit. Two details differ from the completeness score of Marks et al. (2025): SAE errors are always preserved, where they ablate the circuit's error nodes, and the nodes are zero-ablated rather than mean-ablated, so the score needs no reference distribution
Sufficiency to induce	ϕ_{ind} and ϕ_{ind}^α on the held-out contrast contexts, nodes set at every position to (α times) their mean at the anchors; ϕ_{ind}^α is the one reported
Random-circuit baseline (weak)	size-matched site by site, drawn from latents active on the target's strongest training contexts; coefficients fitted with the production objective for 400 steps, gates started fully open on the random nodes ($\theta_0 = 40$) and held closed elsewhere, no sparsity penalty; in the protocol-v1 run one draw on each of 747 targets, stratified by depth and site kind; in the production run on the 112 scored TuringLLM null-panel targets only

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Setting	Value
Harder random circuits	fitted as above, at the production circuit's per-site counts; anchor-active: latents active at the anchor or up to 8 positions before it on some training context, each member's frequency decile matched, production members allowed; contribution-matched: also matched on contribution to the target (activation × gradient of the pre-activation, binned by deciles of the production members' own contributions), production members excluded; every set also scored at $\alpha = 1$; 126 TuringLLM sweep targets (14 per depth band and site kind) and 128 per public model, one draw each and a second on 27–32 (Appendix I)
Donor-background test (reported)	bases: the 16 held-out strongest and 16 held-out contrast contexts; every non-member at every upstream site takes a donor's natural sparse code at the same position (members at $\alpha\times$ live, the donor's active non-members filling the Top-K budget, SAE error live); donor pools A (the 48 strongest training contexts), C (the 16 held-out contrast contexts, never a training donor) and T (512 random text windows from the corpus, filtered against every record); 8 paired draws per pool, the empty circuit under the same donors; recovery = $(\bar{a}_C - \bar{a}_\varnothing)/(a_{\text{pos}} - \bar{a}_\varnothing)$, pooled over draws; a circuit passes the test when it meets the pass criterion and its recovery under A and C each lies in $[0.75, 1.5]$; also scored at $\alpha = 1$ and with members clamped
Specificity (reported)	the target's rank and Top-K share; latents lifted above the unmodified Top-K cut, against the empty circuit; sibling faithfulness against the target's (Appendix K)
Near-threshold targets	median rank among all the site's latents at the held-out strongest anchors $\geq K/2$ (uncapped, unlike the rank-keep r), identified from the unmodified model; kept in the headline pass rate and also reported as a stratum on both reads, with the split of its targets into passing both rules, the pre-activation rule only, the activation rule only, and neither
Pass criterion	faithfulness under Z, sparsity-preserving A, and C all in $[0.75, 1.5]$, and necessity ≥ 0.9 (held-out strongest contexts, activation read); targets with a vacuous denominator, $a_{\text{pos}} \leq 0$ or $a_{\text{pos}} - e < 0.05\, a_{\text{pos}}$ for any of the empty-circuit baselines e of Z, A, and C, excluded and counted, with the all-targets rate (vacuous counted as fails) reported as a lower bound; a score still undefined on a non-vacuous target is a fail; the largest deviation $\max_\pi \lvert\phi_\pi^{\text{free}} - 1\rvert$ reported alongside. Secondary: the same rule on the pre-activation read (Read, above), reported separately over all targets and the near-threshold stratum

Table 4: The protocol behind every reported circuit. Values marked pending are fixed before the corresponding runs are reported.

B.1 The Loss Terms in Detail

Censored read. Under Z, C, and A, both the circuit's pre-activation and the unmodified one pass through the soft Top-K gate $\nu\,\sigma((\nu - \tau^{(K)})/T_b)$, with $\tau^{(K)}$ the K-th largest pre-activation of the site's other latents (under the ablation for the circuit, in the unmodified model for the target) and $T_b = 0.05\,|z_b|$ at anchor b. The terms therefore also see whether the target would survive its site's Top-K, which is what evaluation reads, while keeping a gradient below the cut; the resample term and the normaliser ν use the uncensored pre-activation.

Rank-keep. Reproducing the pre-activation does not keep the target in its site's Top-K, since the loss constrains the stream along one direction of d_{model} and other latents at the site can rise past the target. The rank-keep term, with r the target's rank among its site's other latents at an anchor in the unmodified model (capped at K), is zero whenever the target holds that rank, so it never pulls a circuit towards a state the unmodified model does not have (rejected alternatives in Appendix V.1).

Weights, guard, and normaliser. Under the contrast-context mean alone a mask learns the difference from that mean, never a set that suffices alone, and under zero ablation alone it is never asked what separates firing from a near-identical context that does not fire (Appendix V.3); the weights γ_π therefore score one mask under all of them. The guard $\lambda\,(1-\sigma(\theta))\,|\alpha-1|$ is needed because, with coefficients free, the optimiser could hold a gate just below the keep threshold while inflating its coefficient, emptying the node set while keeping its effect. All terms measure the same quantity on the same target, so one shared, bounded normaliser ν suffices (Appendix V). The gates are optimised directly rather than sampled (Louizos et al., 2018; Li and Janson, 2024); the circuit is the set of gates above $\frac{1}{2}$, which is the same at any temperature. Coefficients start at $\alpha = 1$, so training starts exactly at unweighted behaviour, and the zero-ablation term anchors them (Appendix V.3).

Learning-rate schedule. The learning rate warms up over the first tenth of the steps and then decays geometrically, which keeps its ratio to the gate temperature roughly fixed while the gates anneal; its peak restores the total step size of a constant rate.

Specificity, near-threshold targets, and the pre-activation read. Every score reads the target alone, so a circuit that restores the target by lifting its whole neighbourhood at the site passes them all; specificity (the target's rank, the latents lifted above the Top-K cut, and the faithfulness of co-firing *sibling* latents) is a measured property, not a criterion (Appendix K). A *near-threshold* target, whose median rank at the held-out anchors of the unmodified model is at least $K/2$, has an ill-conditioned activation read, since small shifts near the cut push it out of the Top-K. A secondary pass rate applies the pass criterion to the pre-activation read, with Z and necessity measured from 0 (Read, in Table 4). In the production run it is 75.7% (74.7–76.7), 73.4% (69.5–77.2), and 81.2% (77.9–84.5) on TuringLLM, Pythia-70m, and GPT-2 small, against 88.0%, 81.9%, and 82.4% on the activation read (Table 17). The near-threshold stratum holds 597 TuringLLM targets (8.0%, all of them attention outputs), 38 Pythia-70m targets (7.5%, 30 of them attention outputs), and 3 GPT-2 small targets; on TuringLLM it passes for 59.6% on the activation read, against 90.4% for the other targets, and for only 33.5% on the pre-activation read (28.9% and 5.3% on Pythia-70m). Of the TuringLLM stratum, 21% pass both rules, 12% the pre-activation rule only, 38% the activation rule only, and 28% neither.

B.2 Held-out Contexts

The protocol-v1 run (Appendix H) built its held-out sets by the 3:1 split of each stored pool, and an audit of every record found that they were not clean: 66.8% of held-out strongest contexts repeated a training-side context, 35.6% exactly, 50.2% up to the anchor, and 55.1% as a near-duplicate. A context identical up to its anchor is the same input for every score, since each score reads the target at the anchor of a causal model. The cause is the corpus construction: about half of all 64-token windows (52.9%) were written to the interpretability corpus twice, the two copies have identical activations and so sit at adjacent ranks of a pool, and the split, which holds out one rank in every four, places most such pairs on both sides. The mid-band held-out sets were nearly clean (3.2% of contexts identical up to the anchor, 0.4% near-duplicates).

The production run therefore fits every model from records built so that this cannot occur. For TuringLLM they are built afresh (protocol.md §12.2): each strongest pool is deduplicated before the split, removing exact, anchor-prefix and near copies and corpus twins; candidates beyond the stores' 64 slots come from a scan of the corpus's 18.2M distinct windows for each target's 512 strongest contexts, ranked by the stores' own statistic; the split is group-aware, so no group of copies straddles training and held-out contexts; and contrast contexts come from the same close, silence-checked selector, deduplicated by the exact and near rule, since they have no anchor. For Pythia-70m and GPT-2 small each pool is split the same way, by whole groups of copies (exact, anchor-prefix, and near-duplicate), and unstable-firing rows are dropped. On every model the held-out rule of Table 4 is then applied once more as a final check; it only removes rows, so a held-out set can hold fewer than 16 (floor 8, reported), and an admissible activating context must fire at its anchor. Three Pythia-70m targets were left with no held-out strongest context and were not fitted (Table 9). Every set's size is in Table 5. The median target holds 16 held-out strongest contexts on every model; 383 TuringLLM,

Model	Fitted	Strongest 16	Strongest < 16	Strongest < 8	Strongest min	No mid-band	Mid-band < 16	Contrast < 64	Contrast min
TuringLLM	7,786	7,403	383	262	2	358	63	6	6
Pythia-70m	512	272	240	11	0	1	103	0	64
GPT-2 small	512	347	165	11	3	0	45	0	64

Table 5: Held-out set sizes in the production run, per fitted target: held-out strongest sets that are full (16), below 16, below the floor of 8, and the smallest; targets with no held-out mid-band pool or fewer than 16 held-out mid-band contexts; and targets with fewer than 64 contrast contexts (training and held-out) and the fewest. Pythia-70m counts include the three targets left with no held-out strongest context, which were not fitted.

240 Pythia-70m, and 165 GPT-2 small targets hold fewer than 16, and 262, 11, and 11 fewer than the floor of 8 (the Pythia-70m counts include the three unfitted targets). Targets below the floor are scored and counted in the pass rates. On TuringLLM the short sets are mostly attention-output targets (329 of the 383 below 16, 234 of the 262 below 8), spread evenly over depth. TuringLLM's strongest pools come almost entirely from the corpus scan: all but 4 of their contexts (99.999%).

C Attribution Baselines and Position-Aware Node Sets

WCM is compared against attribution methods that score every upstream latent and keep the top of a ranking. All of them run on each target's training contexts and score a latent's effect on the target's *encoder pre-activation* at the anchor,

$$\hat{a}_f(x,t) = \left\langle W_e^{(f)},\, z(x,t) \right\rangle + b_e^{(f)},$$

with $W_e^{(f)}$ the target's encoder row, $b_e^{(f)}$ its effective bias, and $z(x,t)$ the pre-SAE dense activation: unlike the activation, which Top-K censors to zero, it is defined on every forward pass, and gradients reach every upstream latent through identity passthroughs at each SAE site (Marks et al., 2025), with the SAE errors at their clean values.

C.1 Estimators

The two gradient estimators share one path, from every upstream site's mean over the target's activating training contexts (its A ablation value, $\bar{c}^s$) to its natural code c^s, all sites interpolated together and held at their interpolated codes, and both multiply a gradient by $(c^s - \bar{c}^s)$:

- **Attribution patching** (AtP) takes the gradient at the clean point, giving the first-order estimate of mean-ablating each latent (Nanda, 2023; Syed et al., 2023; Kramár et al., 2024).
- **Integrated gradients from the mean** (IG) (Sundararajan et al., 2017) averages the gradient over 10 points of the path. It follows the IE_{ig} node estimator of Marks et al. (2025), the node-level, activation-space counterpart of EAP-IG (Hanna et al., 2024), in form but not in detail: theirs interpolates one submodule at a time with the rest of the model live (as in their released code), starts from a zero patch when no counterfactual exists, and scores the error terms as nodes.

The two differ only in where they linearise, with target, contexts, path, and selection held fixed. Two further baselines use them:

- **Hierarchical attribution** applies the gating algorithm of Ge et al. (2024) to our SAE sites. Without transcoders our graph is not linear, so its scores are first-order gradients rather than their exact linear attributions, aggregated over contexts rather than per prompt; its threshold is bisected towards the weighted circuit's size, and every bisection point is scored.
- **Ranked members with fitted coefficients** fill the weighted circuit's per-site counts from the top of a ranking at each site, then fit coefficients exactly as for random circuits (Section 3.3: only the coefficients move, with no sparsity penalty), and are scored at $\alpha = 1$ and fitted. The rankings are IG, the latent's mean activation at the target's anchor over the training contexts (no

gradient), and the correlation of that activation with the target's across contexts (co-activation). Against WCM, these rows separate what the selection contributes from what the coefficients contribute.

IG makes the attribution baselines comparable in kind to the published estimator, but not identical to it, so the output-targeted comparison of Appendix E uses its own reimplementation of the published recipe; a version of that recipe with the target latent as its metric, which would isolate the choice of target from everything else, has not been run [*pending: DAN-48*].

Exact activation patching. Attribution patching tracks exact activation patching more closely than integrated gradients do. On ten targets (one per depth band and site kind, and a tenth at layers 8–11), we measured the exact effect of each candidate latent: the change in the target's pre-activation at the anchor when that latent alone is set to its training-context mean at every position and everything else stays live, averaged over the training contexts, which is the quantity AtP linearises. The candidates, 3,200–3,900 per target out of 8k–760k active latents, are the 2,000 top-ranked by each of AtP and IG and 1,000 other active latents drawn at random, so the comparison covers the heads of the two rankings and a random sample of the rest. Under the aggregate ranking defined below, the median rank correlation with the exact effect is 0.86 for AtP and 0.71 for IG (0.81 and 0.62 on magnitudes), and of the exact top-n latents at the weighted circuit's size, AtP's top-n holds a median 79% and IG's 54%; ranked by their largest score at any context and position, both track the exact effect less closely (0.72 and 0.62). These measurements use training contexts only. Members ranked by exact effect, given fitted coefficients, give a ground-truth row for the comparison [*pending: exact-ranked circuits scored on the rebuilt held-out sets, DAN-159*].

C.2 Rankings and Node Sets

Attribution returns a score for every latent at every context and position, so the baselines must decide how the scores form one ranking and which node set a ranking defines. Three rankings collapse contexts and positions: a latent's largest |score| at any context and position; the mean over contexts of its summed |score| over positions; and the magnitude of the mean over contexts of its summed signed score, the aggregate ranking of prior work (Marks et al., 2025). A truncated baseline keeps the top n latents of one ranking, with n the target's weighted-circuit size.

At natural size the Top-K constraint shapes the node set: the code is k-sparse *per position* while a target reads its whole causal prefix, so a position-collapsed circuit must cover the union of every position's active set, thousands of distinct latents per site, or recomputed faithfulness collapses. The baselines' node set is therefore a set of latents, the same kind of object (P) optimises over, collected across positions. Latents outside the circuit are ablated everywhere; nodes are not forced to any value: they are recomputed from the modified stream at every position, so positions instantiate naturally (allow, don't force), exactly the circuit-only execution of Section 3.1. Discovery is position-aware to match: attribution is computed per position, a percentile cut on |attribution| (pooled over sites and positions, resolved per batch of four contexts) keeps individual (latent, position) entries, and the node set is the union over the positions of the target's causal prefix. With a mean-ablation path nearly every entry is nonzero, so the median cut keeps about half of them, which is why these unions are large (the natural-size rows of Table 6). No template or cross-example alignment is required. Both attribution signs are kept, as a design principle rather than an implementation detail: the node set is *stream reconstruction*, not a selection of what drives the target, and a negative-attribution latent still carries part of the residual stream that nodes are recomputed from, so removing it corrupts recomputed execution.

A node set's size at a stated faithfulness is a measured quantity, and Appendix F argues it measures the structure of the computation rather than a defect of the method.

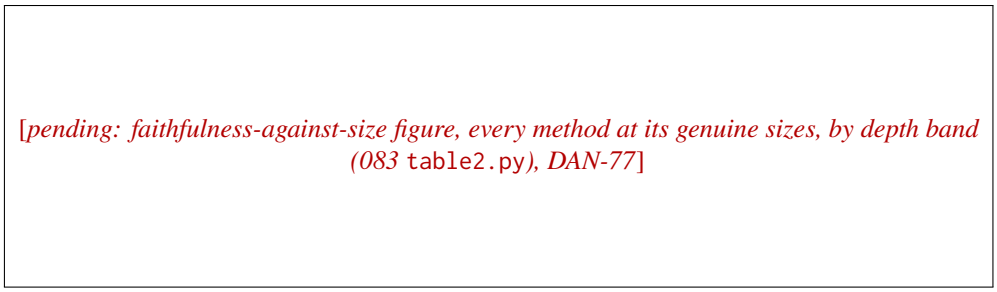


Figure 6: Pass rate (top) and median activating-mean faithfulness (bottom) against median circuit size, by target depth band, on the 190 sweep targets: weighted and unweighted circuit masking at each sparsity penalty, AtP and IG truncated to the weighted circuit's size at each WCM penalty (under the largest-score and aggregate rankings) and at their natural size, every bisection point of the hierarchical attribution, and the ranked-member baselines at $\alpha = 1$ and fitted.

D Method Comparison across Circuit Sizes

Status: the results of this comparison are pending. The attribution baselines did not run with the production run, so this appendix states the design and the questions; no claim in the main text rests on it [*pending: attribution baselines on the production sweep (pod track T7)*]. Every method will be compared on the 190 TuringLLM sweep targets, scored on their held-out contexts (Appendix B.2) with the pass criterion of Section 3.3, by depth band. The comparison will be a curve first (Figure 6): pass rate and median activating-mean faithfulness against median circuit size, with every method at the sizes it actually produces. WCM will appear at its eight sparsity penalties and unweighted circuit masking at its five (Figure 3); AtP and IG truncated to the weighted circuit's size at every WCM penalty and also at their natural size; every bisection point of the hierarchical attribution; and the ranked-member baselines at the weighted circuit's per-site counts (Appendix C). Table 6 will summarise the curve at one point per method: each method at its natural operating point, with WCM and unweighted circuit masking at the same sparsity penalty, $\lambda = 10^{-3}$, rather than at matched size, since no unweighted penalty on the grid gives circuits as small as WCM's (in the protocol-v1 sweep the nearest was $\lambda = 10^{-3}$ for 99% of targets, at a median 1.85× the WCM size); then AtP and IG truncated to the WCM size at $\lambda = 10^{-3}$ under each ranking; then the ranked members at $\alpha = 1$ and fitted. An exploratory version of this comparison, on a 22-target panel and with older attribution code, is retired; only its attribution-mass measurement, below, is kept.

Three questions will structure the comparison. First, at natural size, does attribution reconstruct the target with its full per-position union, and do the methods' node sets coincide there, so that method identity matters only under pressure to be compact? [*pending: natural-size rows and node-set overlap, DAN-159*] Second, at the weighted circuit's size, does any attribution ranking pass, the aggregate ranking and the hierarchical attribution included, and how does unweighted circuit masking at the same penalty compare? On the exploratory panel every truncation failed under mean ablation and the weighted circuit passed on most targets. [*pending: truncated and unweighted rows by depth band, DAN-159*] Third, do fitted coefficients rescue members chosen by attribution, by activation, or by exact effect, which separates what the selection contributes from what the coefficients contribute? [*pending: ranked-member rows at $\alpha = 1$ and fitted, DAN-159 and DAN-43*] In earlier, exploratory experiments, a method tended to win the scores whose ablation method it was trained or ranked under, which we call the *home-turf effect* (also in the binarisation sweep, Appendix V), and under the prior work's own first-order evaluation the exploratory panel's ranking inverts (below); we therefore plan to report rankings as a matrix of discovery method against ablation method, never a single column.

Attribution-mass recovery (exploratory). Ge et al. (2024) score circuits by attribution recovery inside their linearised graph, with no intervention; our first-order analogue, measured on the retired exploratory panel, is *attribution-mass recovery*, the share of total activation-times-gradient mass a node set captures (a surrogate: the exact leaf-sum identity exists only in their substitute model). Under

Method	n (med)	$\phi_{\mathrm{Z}}^{\mathrm{free}}$	$\phi_{\mathrm{A}}^{\mathrm{free}}$	$\phi_{\mathrm{C}}^{\mathrm{free}}$	ϕ_{sup}	pass (%) 0–4	5–7	8–11
Natural operating point								
weighted circuit masking, $\lambda = 10^{-3}$	?	?	?	?	?	?	?	?
unweighted circuit masking, $\lambda = 10^{-3}$	?	?	?	?	?	?	?	?
AtP, position union	?	?	?	?	?	?	?	?
IG, position union	?	?	?	?	?	?	?	?
hierarchical attribution, bisected	?	?	?	?	?	?	?	?
Truncated to the weighted circuit's size								
AtP, largest score	?	?	?	?	?	?	?	?
AtP, mean |score|	?	?	?	?	?	?	?	?
AtP, aggregate	?	?	?	?	?	?	?	?
IG, largest score	?	?	?	?	?	?	?	?
IG, mean |score|	?	?	?	?	?	?	?	?
IG, aggregate	?	?	?	?	?	?	?	?
Ranked members at the weighted circuit's per-site counts: $\alpha = 1$ / fitted								
IG-ranked, $\alpha = 1$	?	?	?	?	?	?	?	?
IG-ranked, fitted	?	?	?	?	?	?	?	?
activation-ranked, $\alpha = 1$	?	?	?	?	?	?	?	?
activation-ranked, fitted	?	?	?	?	?	?	?	?
co-activation-ranked, $\alpha = 1$	?	?	?	?	?	?	?	?
co-activation-ranked, fitted	?	?	?	?	?	?	?	?

Note. Weighted and unweighted circuit masking share $\lambda = 10^{-3}$ rather than a matched size, because no unweighted penalty on the grid gives circuits as small as WCM's. Weighted-circuit and fitted rows apply their coefficients. A dash marks a cell with no defined value, such as a band in which every circuit of the row has a vacuous denominator.

Table 6: [*pending: the finding, first: which methods pass at the weighted circuit's size and by how much, DAN-159*] The method comparison at one point per method on the 190 sweep targets: medians of held-out faithfulness under zero (Z), activating-mean (A), and contrast-mean (C) ablation and of necessity (ϕ_{sup}), median size (n), and the pass rate under the criterion of Section 3.3 by target depth band. Rows (Appendix C): *weighted* and *unweighted circuit masking* at the same sparsity penalty; *position union*, every latent kept by the per-position attribution cut, collected over the target's causal prefix (the natural size); *hierarchical attribution*, the gating algorithm of Ge et al. (2024) with its threshold bisected towards the weighted circuit's size; *largest score*, *mean |score|*, and *aggregate*, the top-n latents (n the weighted circuit's size) by a latent's largest |attribution| at any context and position, by its mean summed |attribution|, and by the magnitude of its mean summed signed attribution (the ranking of Marks et al. (2025)); *ranked members*, the weighted circuit's per-site counts filled from the top of an IG, mean-anchor-activation, or co-activation ranking, at $\alpha = 1$ or with fitted coefficients. Bold marks the best and underlining the second-best entry of each column (faithfulness: closest to 1; ϕ_{sup} and pass rate: highest). [*pending: every cell, from experiments/083-v1-comparison* `table2.py`*, DAN-159; the exact-patching-ranked rows on ten targets as a separate block or a note, DAN-159*]

it that panel's intervention ranking inverts: plain top-n attribution leads (median signed recovery 0.76), the replication follows (0.72), and the weighted circuit trails (0.52) while being the only arm that passed the intervention test there. The home-turf effect therefore reaches evaluation frameworks themselves: a circuit that reproduces the target on held-out intervention while holding a third of the absolute attribution mass (0.37 median) shows that first-order attribution mass is not what faithfulness is made of.

E Output-Targeted and Latent-Targeted Circuits

On three deep targets, output-targeted and latent circuits fail each other's objective. A reimplementation of the sparse-feature-circuits recipe (Marks et al., 2025) (integrated gradients one submodule at a time from a zero patch, as in its released code, error nodes, an output metric) on the same model and SAE bank scores $\phi_{\text{ind}} = \phi_{\text{sup}} = 0.000$ on each of the three deep target latents it was run on (layers 8–10), even at its most permissive threshold, and our circuits score near zero on logit faithfulness (Appendix R.1): "which latents support this output?" and "what makes this latent fire?" are answered by circuits that fail each other's objective. The output-targeted arm ran at its ceiling threshold, in-sample, and before protocol v1 and its evaluation fixes [*pending: re-run under protocol v1, DAN-48*].

F Individual Contributions Undercount Faithfulness

This appendix is exploratory, and so are all of its numbers: it uses node sets from an earlier version of the attribution baselines (Appendix C) at natural scale ($\alpha = 1$), on one target each at layers 2, 8, 9, and 10, under the evaluation that preceded protocol v1. Leave-one-out is blind to how these node sets work: removing any single node changes the target by 0.000, the noise level. Yet cutting nodes in bulk by their own attribution ranking craters function: dropping the bottom half by score moves $\phi_{\text{Z}}^{\text{free}}$ from 0.975 to 0.895 at layer 2 but from 1.119 to 0.595 at layer 8; dropping the bottom 90% yields 0.635, 0.000, 0.222, and 0.005 at layers 2, 8, 9, and 10. The scores being cut are minuscule, $2\text{–}5 \times 10^{-4}$ in attribution units. Nodes individually at the noise level collectively carry the entire deep reconstruction. Individual-contribution thresholding, the standard selection rule across circuit discovery, therefore undercounts what faithfulness needs, which is why circuit size is reported here as a measurement at a stated faithfulness.

G The Production Run in Detail

Execution and exclusions. TuringLLM's main fit began with a gate of 64 targets, whose circuits and scores were checked in full before the remaining targets ran; targets whose fit failed would have been quarantined and counted, and none did. Of the 8,000 targets, 214 have no context record (205 with 1–7 distinct strongest contexts on the deduplicated corpus, 9 with 0–2 contrast contexts) and 322 are vacuous, leaving 7,464 scored (Table 9). Pythia-70m and GPT-2 small fitted their 512 main targets after the sparsity-penalty pilot (Section 4.3). On Pythia-70m three fits stopped because the target had no held-out strongest context: its strongest contexts are runs of near-adjacent windows, which the grouped split placed wholly in training. On GPT-2 small 86 fits ran out of GPU memory and all succeeded when retried. Each public model has one vacuous target. Vacuous targets come mostly from the activating-mean baseline (Table 7). Of the 322 TuringLLM targets vacuous on held-out strongest contexts, 219 are vacuous under A (217 under A alone), 103 under the zero-ablation baseline that Z and necessity share, and 2 under C; on held-out mid-band contexts, 1,063 of 1,200 are vacuous under A. The A-vacuous targets are almost all attention outputs (199 of 219, 133 of them at layers 8–11), whose activating-mean empty circuit already restores them, while the Z-vacuous ones are residual-stream and MLP targets (102 of 103), mostly at layers 0–4 (67). The public models' one strongest and five or six mid-band vacuous targets all come from A.

Model	Held-out set	Rows	Vacuous	Z / necessity	A	C	$a_{\text{pos}} \leq 0$
TuringLLM	strongest	7,786	322	103	219	2	0
	mid-band	7,428	1,200	160	1,063	36	0
Pythia-70m	strongest	509	1	0	1	0	0
	mid-band	509	5	0	5	0	0
GPT-2 small	strongest	512	1	0	1	0	0
	mid-band	512	6	0	6	0	0

Term columns overlap: a row counts under every denominator that is within 5% of a_{pos}.

Table 7: Vacuous targets in the production run, by held-out set and by the empty-circuit baseline that makes them vacuous: zero ablation (shared by Z and necessity), the sparsity-preserving activating mean (A), and the contrast mean (C). A target counts under every baseline that is within 5% of its unmodified activation, so the term columns overlap.

Model	Fits	Fit time, s: median (p10–p90)	Median by depth band, s	Summed fit time (h)	Main-stage wall clock (h)	GPU layout
TuringLLM	7,786	149 (54–259)	82 / 156 / 238	336.1	12.9	1949 fits: 4 GPUs × 6; 5837 fits: 5 GPUs × 6
Pythia-70m	509	78 (37–130)	38 / 74 / 121	11.5	0.6	509 fits: 4 GPUs × 6
GPT-2 small	512	174 (77–343)	104 / 198 / 308	27.5	1.5	156 fits: 5 GPUs × 4; 356 fits: 5 GPUs × 6

Fit times are per-process wall seconds on NVIDIA H100 SXM GPUs shared by several fits under CUDA MPS.

Table 8: Fit time of the production run's main stage: per-process wall time of each fit, with several fits sharing a GPU under CUDA MPS, its median by depth band, the summed fit time, and the stage's wall clock, with the number of GPUs and fits per GPU.

Hardware and fit time. Every fit ran on NVIDIA H100 SXM GPUs, four or five at a time, with six fits sharing each GPU under CUDA MPS (four for 156 of the GPT-2 small fits; Table 8). A fit takes a median of 149 s, 78 s, and 174 s of process wall time on TuringLLM, Pythia-70m, and GPT-2 small, rising with depth (on TuringLLM from 82 s at layers 0–4 to 238 s at layers 8–11); the main stage summed 336, 11.5, and 27.5 process-hours and took 12.9, 0.6, and 1.5 hours of wall clock.

Depth, site kind and transfer. By depth band and site kind (Table 12), the lowest TuringLLM cell is attention outputs at layers 8–11 (68.1%) and the highest residual-stream targets at layers 5–7 (99.1%); on Pythia-70m the lowest is attention outputs at layers 4–5 (55.6%), and on GPT-2 small MLP outputs at layers 8–11 (58.1%), while every GPT-2 small attention target at layers 0–4 passes (44 of 44). On held-out mid-band contexts the TuringLLM pass rate falls with depth from 66.4% to 51.7% and 38.4%.

How circuits fail. Each scored circuit is counted once, in the order: passes, fails necessity, fails Z while necessity holds, or fails only A and/or C (Table 16 and Figure 7); no score was undefined. On TuringLLM, 4.7% of circuits fail only under the mean ablations, 4.7% fail Z, and 2.6% fail necessity; with depth the mean-only share grows from 1.6% at layers 0–4 to 3.3% and 9.3%, and the Z share from 1.9% to 4.9% and 7.7%. On Pythia-70m and GPT-2 small, failing Z is the largest class (9.3% and 8.6% of circuits), against 1.4% and 5.5% mean-only and 7.5% and 3.5% necessity. Failures are almost all on the low side: only 4% of failing TuringLLM circuits, and none on the public models, have any faithfulness above 1.5. On TuringLLM and GPT-2 small, necessity failures are mostly at layer 0 (123 of 197 and 14 of 18), where 29.6% and 45.2% of circuits fail necessity, consistent with the token embedding reaching layer-0 targets directly without being ablated; on Pythia-70m they

Model	Frame	Targets	No record	Quarantined	Skipped	Fit error	No circuit	Vacuous	Scored	No mid pool	Mid vacuous	Mid scored
TuringLLM	—	8,000	214	0	0	0	0	322	7,464	358	1,200	6,228
Pythia-70m	486,698	512	0	0	0	3	0	1	508	0	5	504
GPT-2 small	1,527,910	512	0	0	0	0	0	1	511	0	6	506

Table 9: Targets excluded from the production pass rates, by model and reason; vacuous targets are excluded by a property of the target and its contexts, never of its circuit (Section 3.3).

Depth	Targets	Pass, WCM (%)	Pass, unweighted (%)	WCM − unweighted (pp)	Size WCM / unweighted	Size ratio U/W (IQR)
all layers	127	83.5 [77.2, 89.8]	59.8 [51.2, 68.5]	+23.6 [+15.7, +31.5]	318 / 826	2.28 (1.70–3.04)
layers 0-1	31	87.1 [74.2, 96.8]	71.0 [54.8, 87.1]	+16.1 [+0.0, +32.3]	107 / 186	1.62 (1.28–2.20)
layers 2-3	48	77.1 [64.6, 89.6]	75.0 [62.5, 87.5]	+2.1 [-4.2, +8.3]	279 / 721	2.45 (1.94–3.30)
layers 4-5	48	87.5 [77.1, 95.8]	37.5 [25.0, 52.1]	+50.0 [+35.4, +62.5]	670 / 1,608	2.35 (1.89–3.36)

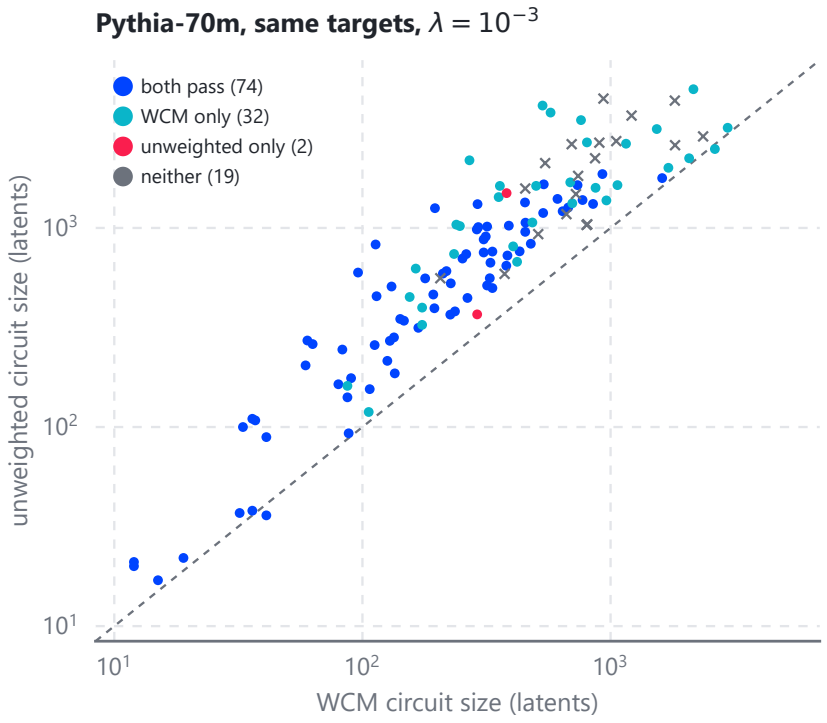


Table 10: On a second model too, unweighted circuits at the same penalty pass less often and are larger, most at depth. Weighted against unweighted circuit masking on Pythia-70m at $\lambda = 10^{-3}$, on the same 127 scored targets: pass rate with 95% bootstrap interval, the paired difference, and median circuit size and per-target size ratio, overall and by depth band.

Method	λ	Targets	Pass (%) [95% CI]	Size, median (IQR)	Z / A / C	Worst dev.
WCM	2.5e-05	185	91.3 [87.3, 95.4]	12,040 (6,389–22,519)	0.96 / 0.97 / 0.98	0.06
WCM	5e-05	185	93.1 [89.0, 96.5]	6,250 (3,368–11,663)	0.95 / 0.97 / 0.99	0.07
WCM	0.0001	185	90.2 [85.5, 94.2]	3,542 (1,875–5,530)	0.95 / 0.97 / 0.98	0.08
WCM	0.00025	185	91.3 [87.3, 95.4]	1,594 (936–2,662)	0.93 / 0.98 / 0.98	0.09
WCM	0.0005	185	89.0 [84.4, 93.1]	990 (615–1,504)	0.92 / 0.99 / 0.98	0.10
WCM	0.001 (production)	185	83.2 [77.5, 88.4]	573 (395–961)	0.92 / 0.99 / 0.98	0.11
WCM	0.002	185	77.5 [71.1, 83.2]	401 (264–591)	0.91 / 0.99 / 0.94	0.14
WCM	0.004	185	65.9 [59.0, 72.8]	263 (173–413)	0.87 / 0.99 / 0.94	0.18
unweighted	1e-05	185	75.1 [68.8, 81.5]	75,874 (27,669–135,836)	0.93 / 0.93 / 1.00	0.11
unweighted	3e-05	185	69.4 [62.4, 76.3]	31,575 (14,668–56,762)	0.90 / 0.91 / 0.97	0.13
unweighted	0.0001	185	72.8 [66.5, 79.2]	9,004 (5,858–15,944)	0.89 / 0.90 / 0.95	0.15
unweighted	0.0003	185	59.5 [52.0, 67.1]	3,041 (2,172–4,408)	0.84 / 0.88 / 0.90	0.19
unweighted	0.001	185	42.2 [34.7, 49.7]	1,201 (907–1,617)	0.74 / 0.84 / 0.84	0.29

Depth	Method	λ	Targets	Pass (%) [95% CI]	Size (median)
layers 0-4	WCM	2.5e-05	73	89.9 [82.6, 95.7]	5,943
layers 0-4	WCM	5e-05	73	95.7 [89.9, 100.0]	3,280
layers 0-4	WCM	0.0001	73	95.7 [89.9, 100.0]	1,834
layers 0-4	WCM	0.00025	73	92.8 [85.5, 98.6]	930
layers 0-4	WCM	0.0005	73	94.2 [88.4, 98.6]	615
layers 0-4	WCM	0.001	73	88.4 [79.7, 95.7]	396
layers 0-4	WCM	0.002	73	85.5 [76.8, 92.8]	246
layers 0-4	WCM	0.004	73	79.7 [69.6, 88.4]	163
layers 0-4	unweighted	1e-05	73	84.1 [75.4, 92.8]	18,140
layers 0-4	unweighted	3e-05	73	82.6 [72.5, 91.3]	10,130
layers 0-4	unweighted	0.0001	73	85.5 [76.8, 92.8]	4,431
layers 0-4	unweighted	0.0003	73	81.2 [71.0, 89.9]	2,047
layers 0-4	unweighted	0.001	73	75.4 [65.2, 85.5]	822
layers 5-7	WCM	2.5e-05	48	95.6 [88.9, 100.0]	14,195
layers 5-7	WCM	5e-05	48	93.3 [84.4, 100.0]	7,071
layers 5-7	WCM	0.0001	48	93.3 [84.4, 100.0]	4,082
layers 5-7	WCM	0.00025	48	95.6 [88.9, 100.0]	1,958
layers 5-7	WCM	0.0005	48	93.3 [84.4, 100.0]	1,128
layers 5-7	WCM	0.001	48	84.4 [73.3, 93.3]	743
layers 5-7	WCM	0.002	48	86.7 [75.6, 95.6]	487
layers 5-7	WCM	0.004	48	64.4 [51.1, 77.8]	316
layers 5-7	unweighted	1e-05	48	75.6 [62.2, 86.7]	81,714
layers 5-7	unweighted	3e-05	48	71.1 [57.8, 84.4]	37,909
layers 5-7	unweighted	0.0001	48	73.3 [60.0, 86.7]	11,719
layers 5-7	unweighted	0.0003	48	60.0 [46.7, 73.3]	3,791
layers 5-7	unweighted	0.001	48	37.8 [24.4, 51.1]	1,391
layers 8-11	WCM	2.5e-05	64	89.8 [81.4, 96.6]	26,246
layers 8-11	WCM	5e-05	64	89.8 [81.4, 96.6]	12,352
layers 8-11	WCM	0.0001	64	81.4 [71.2, 89.8]	5,888
layers 8-11	WCM	0.00025	64	86.4 [76.3, 94.9]	2,847
layers 8-11	WCM	0.0005	64	79.7 [69.5, 89.8]	1,648
layers 8-11	WCM	0.001	64	76.3 [64.4, 86.4]	937
layers 8-11	WCM	0.002	64	61.0 [49.2, 72.9]	586
layers 8-11	WCM	0.004	64	50.8 [37.3, 64.4]	387
layers 8-11	unweighted	1e-05	64	64.4 [52.5, 76.3]	161,940
layers 8-11	unweighted	3e-05	64	52.5 [39.0, 64.4]	66,720
layers 8-11	unweighted	0.0001	64	57.6 [44.1, 69.5]	15,944
layers 8-11	unweighted	0.0003	64	33.9 [22.0, 45.8]	4,130
layers 8-11	unweighted	0.001	64	6.8 [1.7, 13.6]	1,586

Table 11: Weighted circuits pass at 89–93% from $\lambda = 2.5 \times 10^{-5}$ to 5×10^{-4} and less at stronger penalties; unweighted circuits pass less at every penalty. The TuringLLM sparsity-penalty sweep, weighted circuit masking at eight penalties and unweighted at five, on 190 targets (185 with a record, 12 of them vacuous), pooled (top) and by depth band (bottom): pass rate with 95% bootstrap interval and median circuit size; the production arm, $\lambda = 10^{-3}$, is the main circuit of each target.

Model	Group	Circuits	Pass, strongest (%) [95% CI]	Pass, mid-band (%) [95% CI]	Size, median (IQR)	Induce
TuringLLM	layers 0-4	2,931	90.7 [89.6, 91.7]	66.4 [64.6, 68.3]	393 (262–541)	0.81
	layers 5-7	1,981	91.2 [90.0, 92.4]	51.7 [49.4, 53.9]	721 (508–914)	0.73
	layers 8-11	2,552	82.3 [80.8, 83.8]	38.4 [36.3, 40.6]	989 (637–1,321)	0.78
	residual	2,670	95.7 [94.8, 96.4]	58.2 [56.2, 60.1]	786 (458–1,163)	0.96
	MLP	2,711	88.4 [87.2, 89.6]	45.2 [43.3, 47.2]	471 (340–666)	0.66
	attention	2,083	77.5 [75.8, 79.3]	60.2 [57.4, 62.8]	671 (500–1,008)	0.35
Pythia-70m	layers 0-1	127	89.0 [83.5, 93.7]	78.6 [70.6, 85.7]	87 (35–176)	1.01
	layers 2-3	191	80.6 [74.9, 85.9]	74.7 [68.4, 80.5]	250 (133–481)	0.85
	layers 4-5	190	78.4 [72.6, 84.2]	69.1 [62.8, 75.5]	670 (372–1,138)	0.71
	residual	191	92.7 [89.0, 96.3]	83.5 [78.2, 88.3]	199 (61–474)	1.03
	MLP	158	86.1 [80.4, 91.1]	76.6 [69.6, 82.9]	248 (130–536)	0.78
	attention	159	64.8 [57.2, 72.3]	58.9 [51.3, 66.5]	485 (261–1,095)	0.16
GPT-2 small	layers 0-4	206	90.3 [85.9, 94.2]	82.4 [77.1, 87.3]	258 (111–448)	0.92
	layers 5-7	130	92.3 [87.7, 96.2]	66.2 [57.7, 73.8]	398 (218–584)	0.77
	layers 8-11	175	65.7 [58.3, 72.6]	48.5 [41.5, 56.1]	587 (328–899)	0.69
	residual	131	82.4 [75.6, 88.5]	64.3 [55.8, 72.1]	298 (174–528)	1.12
	residual (mid)	131	84.7 [78.6, 90.1]	63.6 [55.0, 72.1]	300 (164–549)	1.08
	MLP	129	76.7 [69.0, 83.7]	68.2 [60.5, 76.0]	329 (217–623)	0.60
	attention	120	85.8 [79.2, 91.7]	71.4 [63.0, 79.0]	592 (392–894)	0.00

Table 12: Pass rates of the production circuits by depth band and site kind, on held-out strongest and held-out mid-band contexts, with 95% bootstrap intervals, median circuit size, and median sufficiency to induce.

Model	Group	Circuits	Pass, fitted α (%)	Pass, $\alpha = 1$ (%)	Difference (pp)	Z / A / C, fitted	Z / A / C, $\alpha = 1$
TuringLLM	all layers	7,464	88.0 [87.2, 88.7]	1.8 [1.5, 2.1]	-86.1 [-86.9, -85.4]	0.93 / 0.99 / 0.98	0.00 / 0.04 / 0.04
	layers 0-4	2,931	90.7 [89.6, 91.7]	4.6 [3.9, 5.4]	-86.0 [-87.2, -84.7]	0.97 / 1.03 / 1.03	0.09 / 0.22 / 0.25
	layers 5-7	1,981	91.2 [90.0, 92.4]	0.0 [0.0, 0.0]	-91.2 [-92.5, -90.0]	0.91 / 0.98 / 0.97	0.00 / 0.02 / 0.01
	layers 8-11	2,552	82.3 [80.8, 83.8]	0.0 [0.0, 0.0]	-82.3 [-83.8, -80.8]	0.88 / 0.93 / 0.91	0.00 / 0.00 / 0.00
Pythia-70m	all layers	508	81.9 [78.5, 85.2]	13.8 [10.8, 16.9]	-68.1 [-72.0, -64.0]	0.96 / 1.00 / 1.00	0.17 / 0.27 / 0.26
	layers 0-1	127	89.0 [83.5, 93.7]	46.5 [37.8, 55.1]	-42.5 [-51.2, -33.9]	1.01 / 1.03 / 1.04	0.77 / 0.81 / 0.82
	layers 2-3	191	80.6 [74.9, 85.9]	5.8 [2.6, 8.9]	-74.9 [-81.2, -68.6]	0.96 / 0.99 / 0.99	0.21 / 0.27 / 0.28
	layers 4-5	190	78.4 [72.6, 84.2]	0.0 [0.0, 0.0]	-78.4 [-84.2, -72.6]	0.92 / 0.98 / 0.96	0.00 / 0.04 / 0.03
GPT-2 small	all layers	511	82.4 [79.1, 85.5]	3.3 [1.8, 4.9]	-79.1 [-82.4, -75.5]	0.92 / 0.94 / 0.93	0.00 / 0.00 / 0.00
	layers 0-4	206	90.3 [85.9, 94.2]	8.3 [4.8, 12.1]	-82.0 [-87.4, -76.7]	0.97 / 1.00 / 1.00	0.09 / 0.10 / 0.12
	layers 5-7	130	92.3 [87.7, 96.2]	0.0 [0.0, 0.0]	-92.3 [-96.9, -87.7]	0.90 / 0.91 / 0.91	0.00 / 0.00 / 0.00
	layers 8-11	175	65.7 [58.3, 72.6]	0.0 [0.0, 0.0]	-65.7 [-72.6, -58.9]	0.86 / 0.88 / 0.86	0.00 / 0.00 / 0.00

Table 13: Without their fitted coefficients the same nodes almost never pass. The production circuits with their fitted coefficients and with every coefficient set to 1, on the same targets and held-out strongest contexts, by model and depth band: pass rates with 95% bootstrap intervals, the paired difference, and median Z / A / C. On Pythia-70m, 59 of the 70 circuits that pass at $\alpha = 1$ are at layers 0–1.

Depth	Matched on	Targets (both)	WCM size	Unweighted size	Ratio
layers 0–4	50% pass rate	—	≤163	≤822	—
layers 0–4	75% pass rate	—	≤163	≤822	—
layers 0–4	WCM's pass rate at $\lambda = 10^{-3}$ (88.4%)	—	396	—	—
layers 0–4	smallest passing circuit, per target	61	163	822	5.1×
layers 0–4	smallest with Z, A, C within 0.3 of 1, per target	64	161	836	5.1×
layers 5–7	50% pass rate	—	≤316	2,414	≥7.6×
layers 5–7	75% pass rate†	—	388	74,234	191.3×
layers 5–7	WCM's pass rate at $\lambda = 10^{-3}$ (84.4%)	—	466	—	—
layers 5–7	smallest passing circuit, per target	39	386	2,315	7.9×
layers 5–7	smallest with Z, A, C within 0.3 of 1, per target	41	330	1,582	5.0×
layers 8–11	50% pass rate	—	≤387	10,328	≥26.7×
layers 8–11	75% pass rate	—	901	—	—
layers 8–11	WCM's pass rate at $\lambda = 10^{-3}$ (76.3%)	—	937	—	—
layers 8–11	smallest passing circuit, per target	42	436	6,124	13.8×
layers 8–11	smallest with Z, A, C within 0.3 of 1, per target	46	422	4,360	11.7×
all layers	50% pass rate	—	≤263	1,824	≥6.9×
all layers	75% pass rate†	—	367	74,229	202.5×
all layers	WCM's pass rate at $\lambda = 10^{-3}$ (83.2%)	—	573	—	—
all layers	smallest passing circuit, per target	142	284	1,793	6.0×
all layers	smallest with Z, A, C within 0.3 of 1, per target	151	273	1,410	5.6×

† layers 5–7, 75% pass rate: unweighted maximum 75.6% [62.2, 86.7] (at $\lambda = 10^{-5}$); non-monotone curve.

† all layers, 75% pass rate: unweighted maximum 75.1% [68.8, 81.5] (at $\lambda = 10^{-5}$); non-monotone curve.

≤: the method's sparsest arm already reaches the level, so the size is an upper bound.

Table 14: At matched faithfulness, unweighted circuits are several times larger, and the ratio grows with depth. Weighted against unweighted circuit masking on the 190 TuringLLM sweep targets. Rows by pass-rate level: the median circuit size at which each method's pooled pass rate reaches 50%, 75%, or WCM's production rate, interpolated in log size along its penalty curve; ≤ marks a level already reached at the method's smallest circuits (so the size is an upper bound and the ratio a bound, ≥), and a dash a level never reached. The 75% rows are fragile, since the unweighted curve is not monotone and peaks at 75.1% (68.8–81.5). Rows by target: the median of each target's smallest passing circuit (or smallest circuit with every faithfulness within 0.3 of 1) over the penalties fitted, and the median per-target ratio, over targets both methods reach; these are the conservative comparison.

Model	λ	Circuits	Size, median (IQR)	In $[10^2, 10^3]$ (%)	Median size by depth band	Pass (%, reported)
Pythia-70m	0.0003	30	924 (304–1,641)	40	90 / 963 / 3,032 (L0-1, L2-3, L4-5)	87
	0.001 (kept)	30	328 (128–641)	63	34 / 382 / 684 (L0-1, L2-3, L4-5)	87
	0.003	30	169 (62–258)	67	16 / 189 / 320 (L0-1, L2-3, L4-5)	80
GPT-2 small	0.001 (kept)	30	428 (208–788)	77	430 / 254 / 451 (L0-4, L5-7, L8-11)	77
	0.003	30	190 (85–286)	70	182 / 137 / 214 (L0-4, L5-7, L8-11)	63
	0.01	30	70 (40–98)	23	56 / 82 / 82 (L0-4, L5-7, L8-11)	43

Table 15: The sparsity-penalty pilot on the public models (30 targets each, disjoint from the main targets). The rule, fixed before the pilot, reads the pooled median circuit size only; pass rates are shown for information.

Model	Group	n	Pass (%) [95% CI]	A/C only (%)	Z (%)	Necessity (%)	Undefined (%)	Fails above 1.5 (% of fails)	Largest deviation
TuringLLM	L0-4	2,931	90.7 [89.6, 91.7]	1.6	1.9	5.8	0.0	7	0.08
	L5-7	1,981	91.2 [90.0, 92.4]	3.3	4.9	0.6	0.0	5	0.11
	L8-11	2,552	82.3 [80.8, 83.8]	9.3	7.7	0.6	0.0	2	0.16
	all	7,464	88.0 [87.2, 88.7]	4.7	4.7	2.6	0.0	4	0.11
	residual	2,670	95.7 [94.8, 96.4]	0.7	1.0	2.6	0.0	0	0.10
	MLP	2,711	88.4 [87.2, 89.6]	3.3	5.2	3.2	0.0	2	0.11
	attention	2,083	77.5 [75.8, 79.3]	11.6	8.9	2.0	0.0	6	0.14
Pythia-70m	L0-1	127	89.0 [83.5, 93.7]	0.0	3.1	7.9	0.0	0	0.06
	L2-3	191	80.6 [74.9, 85.9]	1.0	11.0	7.3	0.0	0	0.06
	L4-5	190	78.4 [72.6, 84.2]	2.6	11.6	7.4	0.0	0	0.12
	all	508	81.9 [78.5, 85.2]	1.4	9.3	7.5	0.0	0	0.07
	residual	191	92.7 [89.0, 96.3]	0.5	2.1	4.7	0.0	0	0.06
	MLP	158	86.1 [80.4, 91.1]	0.6	8.9	4.4	0.0	0	0.06
	attention	159	64.8 [57.2, 72.3]	3.1	18.2	13.8	0.0	0	0.15
GPT-2 small	L0-4	206	90.3 [85.9, 94.2]	0.0	1.0	8.7	0.0	0	0.06
	L5-7	130	92.3 [87.7, 96.2]	2.3	5.4	0.0	0.0	0	0.13
	L8-11	175	65.7 [58.3, 72.6]	14.3	20.0	0.0	0.0	0	0.20
	all	511	82.4 [79.1, 85.5]	5.5	8.6	3.5	0.0	0	0.11
	residual	131	82.4 [75.6, 88.5]	7.6	4.6	5.3	0.0	0	0.09
	residual (mid)	131	84.7 [78.6, 90.1]	4.6	6.1	4.6	0.0	0	0.07
	MLP	129	76.7 [69.0, 83.7]	2.3	17.1	3.9	0.0	0	0.15
	attention	120	85.8 [79.2, 91.7]	7.5	6.7	0.0	0.0	0	0.13

Outcome classes are exclusive, in the order pass, undefined, necessity, Z, A/C only; held-out strongest, non-vacuous.

Table 16: How the production circuits fail. Each non-vacuous circuit on held-out strongest contexts is counted once, in the order pass, undefined score, necessity below 0.9, Z outside the band, and A and/or C only, by model, depth band, and site kind, with the pass rate and its 95% bootstrap interval, the share of failing circuits with any of Z, A, C above 1.5, and the median largest deviation $\max_\pi |\phi_\pi^{\text{free}} - 1|$.

Model	n	Pass, activation (%)	Pass, pre-activation (%)	Near-threshold	Near: activation (%)	Near: pre-activation (%)	Near: both / pre only / act. only / neither (%)	Others: activation (%)
TuringLLM	7,464	88.0 [87.2, 88.7]	75.7 [74.7, 76.7]	597 (8.0%)	59.6 [55.6, 63.5]	33.5 [29.6, 37.4]	21 / 12 / 38 / 28	90.4 [89.8, 91.1]
Pythia-70m	508	81.9 [78.5, 85.2]	73.4 [69.5, 77.2]	38 (7.5%)	28.9 [15.8, 44.7]	5.3 [0.0, 13.2]	5 / 0 / 24 / 71	86.2 [83.0, 89.1]
GPT-2 small	511	82.4 [79.1, 85.5]	81.2 [77.9, 84.5]	3 (0.6%)	66.7 [0.0, 100.0]	66.7 [0.0, 100.0]	67 / 0 / 0 / 33	82.5 [78.9, 85.8]

Table 17: The pre-activation pass rate and the near-threshold stratum (Appendix B.1). Pass rates on the activation read (the criterion) and on the pre-activation read (Z and necessity from 0), with 95% bootstrap intervals; near-threshold targets (median unmodified rank at the held-out strongest anchors at least $K/2$), their pass rates on both reads, their split into passing both rules, the pre-activation rule only, the activation rule only, and neither; and the activation-read pass rate of the other targets.

are spread over depth (6 of 38 at layer 0). Near-threshold targets (Appendix B.1) are all attention outputs on TuringLLM and pass less often (59.6% against 90.4%). On held-out mid-band contexts the median largest deviation $\max_\pi |\phi_\pi^{\text{free}} - 1|$ is 0.36, 0.15, and 0.26, against 0.11, 0.07, and 0.11 on held-out strongest contexts, and on TuringLLM it grows with depth from 0.24 to 0.57 (strongest: 0.08 to 0.16).

Circuit sizes, sufficiency to induce, training dynamics. Circuit sizes have interquartile ranges of 403–954, 132–653, and 208–666 latents on TuringLLM, Pythia-70m, and GPT-2 small. Sufficiency to induce, which no circuit is trained on, has a median of 0.78, 0.88, and 0.81 on held-out contrast contexts; on TuringLLM it is lowest for attention-output targets (median 0.35, against 0.96 for the residual stream and 0.66 for MLP outputs). 73.9% (72.9–74.9), 72.0% (68.1–75.8), and 66.1% (62.0–70.3) of circuits switch the target on to at least half its natural level (76.7%, 75.7%, and 67.0% of passing circuits), from over 90% of residual-stream targets on every model to 43.8%, 34.0%, and 11.7% of attention-output targets (Table 18).

Every fit records its per-step curve (Figure 8). All gates start open, so no latent falls below the keep threshold before a median step of 37 on every model; membership then collapses from a median of 2.9×10^5–7.9×10^5 candidates, and a circuit is within twice its final size by a median step of 95, 96, and 125 on TuringLLM, Pythia-70m, and GPT-2 small. Over the last 100 of the 400 steps the median circuit loses 3.2%, 2.8%, and 5.2% of its members, more at depth, and 24%, 24%, and 51% are still shrinking by more than 5%. In the deepest band, the half of circuits still shrinking most pass less often than the settled half: 76.8% against 87.9% at TuringLLM layers 8–11 (difference 8.1–13.9 points) and 61.1% against 95.8% at Pythia-70m layers 4–5; at GPT-2 small layers 8–11 the gap (59.8% against 71.6%) lies within its interval, and at layers 0–4 of TuringLLM and GPT-2 small

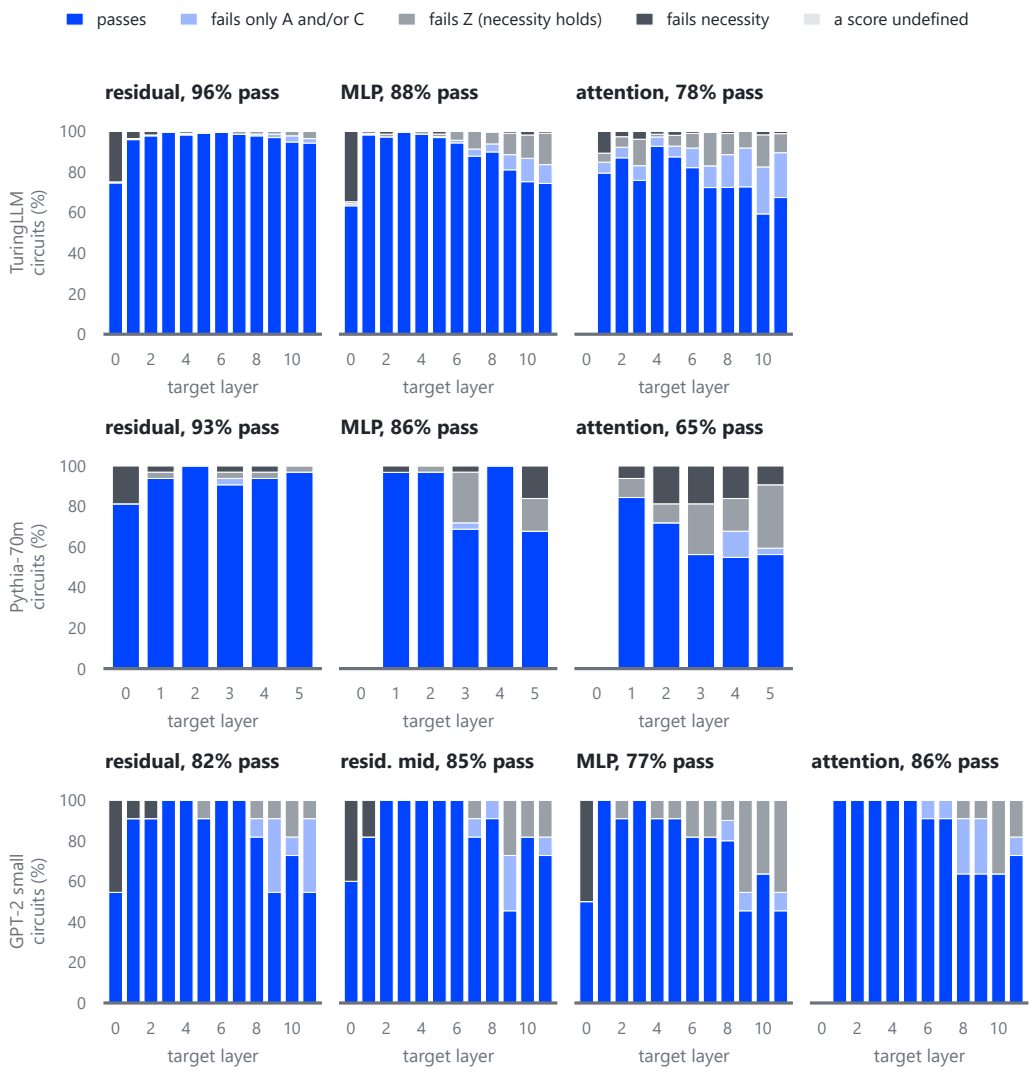


Figure 7: Failures grow with depth on TuringLLM and GPT-2 small, are commonest at attention outputs on TuringLLM and Pythia-70m, and at layer 0 are mostly necessity failures. Outcomes of the production circuits on held-out strongest contexts by target layer (bars) and site kind (columns), one row per model; each circuit is counted once: it passes, fails only the mean ablations (A, C, or both) while zero ablation and necessity hold, fails zero ablation while necessity holds, or fails necessity. Panel titles give each site kind's pass rate; layer-0 attention has no upstream site.

Model	n	All (%)	Passing (%)	residual (%)	residual (mid) (%)	MLP (%)	attention (%)
TuringLLM	7,464	73.9 [72.9, 74.9]	76.7 [75.6, 77.7]	96.2 [95.4, 96.9]	—	75.1 [73.5, 76.7]	43.8 [41.8, 46.0]
Pythia-70m	508	72.0 [68.1, 75.8]	75.7 [71.6, 79.8]	97.9 [95.8, 99.5]	—	79.1 [72.8, 85.4]	34.0 [26.4, 41.5]
GPT-2 small	511	66.1 [62.0, 70.3]	67.0 [62.2, 71.5]	91.6 [86.3, 96.2]	93.9 [89.3, 97.7]	62.8 [54.3, 71.3]	11.7 [5.8, 17.5]

Table 18: The share of production circuits whose sufficiency to induce, $\phi^{\alpha}_{\text{ind}}$ on held-out contrast contexts, is at least 0.5, with 95% bootstrap intervals, over all non-vacuous circuits, passing circuits, and by site kind; undefined scores count as below. Reported for every circuit; not part of the pass criterion.

the shrinking half passes more often. Reported sizes are therefore sizes at a fixed budget.

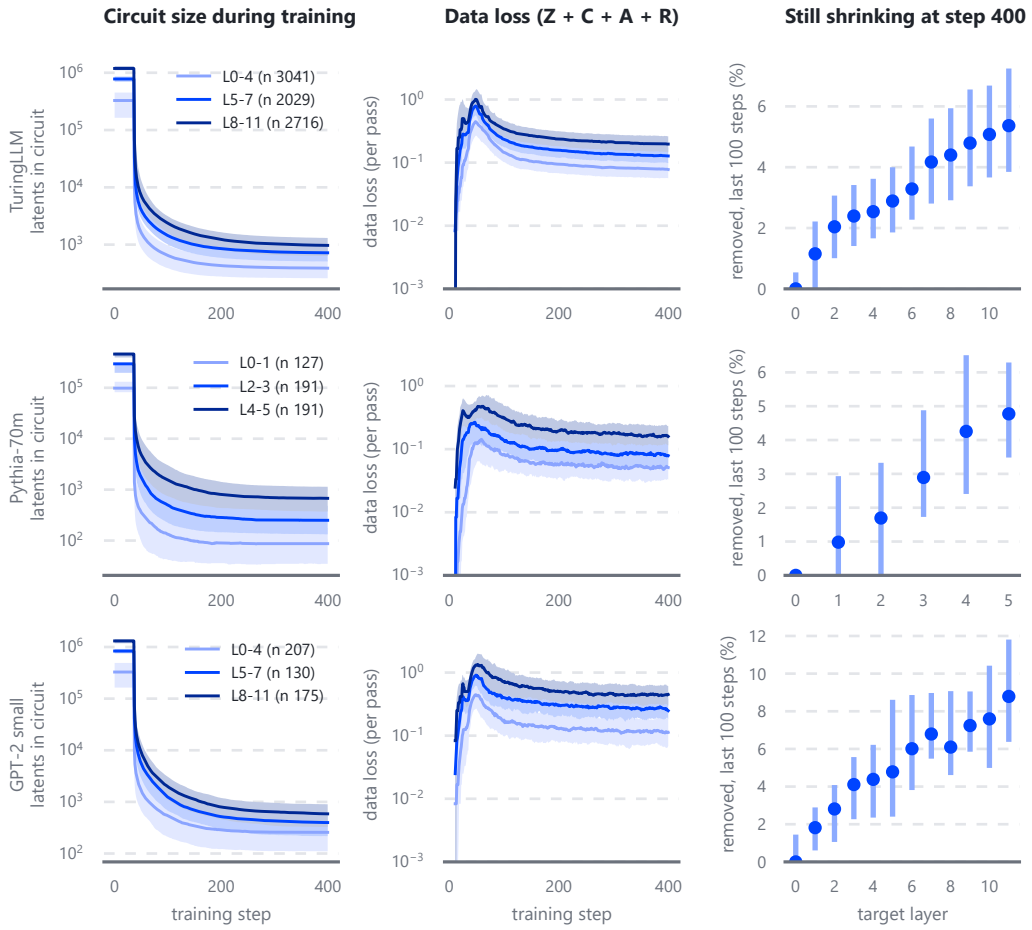


Figure 8: Production circuits shrink slowly by the 400-step budget, more at depth. Training dynamics of the production fits by target depth band, one row per model (median and interquartile range). Left: circuit size (latents above the keep threshold) per step. Middle: the data loss (the Z, C, A, and R terms, with the rank-keep parts, penalty excluded), averaged over each pass through the training contexts. Right: the share of members removed over the last 100 steps, by target layer (median and interquartile range).

H The Protocol-v1 TuringLLM Run

Before the production run, WCM was run once at scale on TuringLLM under the first protocol, protocol v1: A and C trained under a dense mean, rank-keep $\gamma_K = 3 \times 10^{-3}$, a constant learning rate of 0.05, and no resample term or censored read. It fitted 15,046 sampled targets (455 per site with an upstream site, less dead or context-poor attention targets) at $\lambda = 10^{-3}$, with a pass band floor of 0.8, and its held-out sets repeated training text (Appendix B.2), so its numbers are not comparable with the production run's and are reported here only because several appendix analyses use its circuits. Of its 14,994 non-vacuous circuits 46% passed (residual stream 65%, MLP 52%, attention output 14%; layers 0–4, 5–7, 8–11: 69%, 45%, 22%), at a median of 494 latents (386 for passing circuits; 91% within 10^2–10^3); on held-out mid-band contexts 12% passed (Figure 9). Random circuits of the same size, drawn from latents active anywhere on the strongest training contexts and given the same coefficient fitting, passed on none of 747 targets, but their coefficients barely moved, so that null shows only that unrelated latents cannot be fitted (Appendix I). Re-scored with every coefficient at 1, 1.5% of 1,746 sampled circuits passed, against 44.7% as fitted (Figure 16). On 190 sweep targets, weighted circuits explained more targets than unweighted ones at every circuit budget (Figure 10): where both passed, the smallest unweighted circuit was a median 3.7×, 9.6×, and 10.5× larger at layers 0–4, 5–7, and 8–11 (13,075 against 906 latents at depth), and over the sweep 80% of deep targets passed with coefficients against 41% without; the running example passed with 728 latents, while unweighted masking failed at every penalty up to 24,837. On ten targets with exact patching, the median rank correlation with the exact effect was 0.86 for attribution patching (AtP) and 0.71 for integrated gradients (IG) (Sundararajan et al., 2017) (Appendix D).

Where the fall with depth comes from. Zero ablation and necessity both hold for three circuits in four or more at every depth (79%, 80%, and 76% at layers 0–4, 5–7, and 8–11; Figure 9); what grows with depth is the share that fail only under the two mean ablations, from 10% of circuits at layers 0–4 to 53% at 8–11 (over all layer-11 circuits, median A and C faithfulness are 0.64 and 0.48), and these make up 59% of all failing circuits. Attention outputs are the exception: 40% of their circuits fail even zero ablation, and 63% of those belong to near-threshold targets, which barely enter their site's Top-K in the unmodified model and read 0 once they slip out of it (1,142 targets, 7.6% of the run, almost all attention outputs; 0.8% pass, against 50% for the remaining targets). Layer 0 departs from the trend (55%): its circuits are faithful (90%), but over a third (37%) fail necessity, because the token embedding reaches layer-0 targets directly and is never ablated.

Transfer to weaker contexts. On the held-out mid-band contexts, where targets fire more weakly, 12% pass (median largest deviation 0.77, against 0.31 on the strongest contexts), so circuits fitted mostly to strong contexts transfer only in part to weak ones: a circuit is a circuit for the regime it was fitted in.

Why attention outputs fail. Three local tests on samples of this run's attention targets, under protocol v1, locate the cost in source positions. About 85% of an attention target's clean contribution comes from source positions other than its own and the first token, for passing and failing targets alike. Failing targets read from more positions and more heads than passing ones at the same circuit size (layers 0–4: 4.5 against 2.4 effective positions, 7.1 against 4.5 effective heads). Under activating-mean ablation the dominant head of a failing circuit keeps 0.42 of its contribution when only its attention pattern is ablated and 0.60 when only its values are (passing circuits: 1.04 and 0.80), so the circuit loses both where the head attends and what it reads. Holding attention patterns at their clean values does not rescue them (5 of 120 failing circuits), and raising the A and C weights or the rank-keep weight barely helps (21% to 29% on 24 targets); at a weaker penalty about half pass, at ~9k latents (10 of 22 attention targets at $\lambda = 2.5 \times 10^{-5}$). Near-threshold targets are a separate case, largely a Top-K censoring effect: their clean margin above the first excluded latent is 25%, against 81–91% for other targets, and over all 1,142 of them the pre-activation read gives median faithfulness 0.93 under zero ablation but only 0.50 and 0.42 under the two mean ablations, so the read explains their zero-ablation failures but not all of their mean-ablation ones.

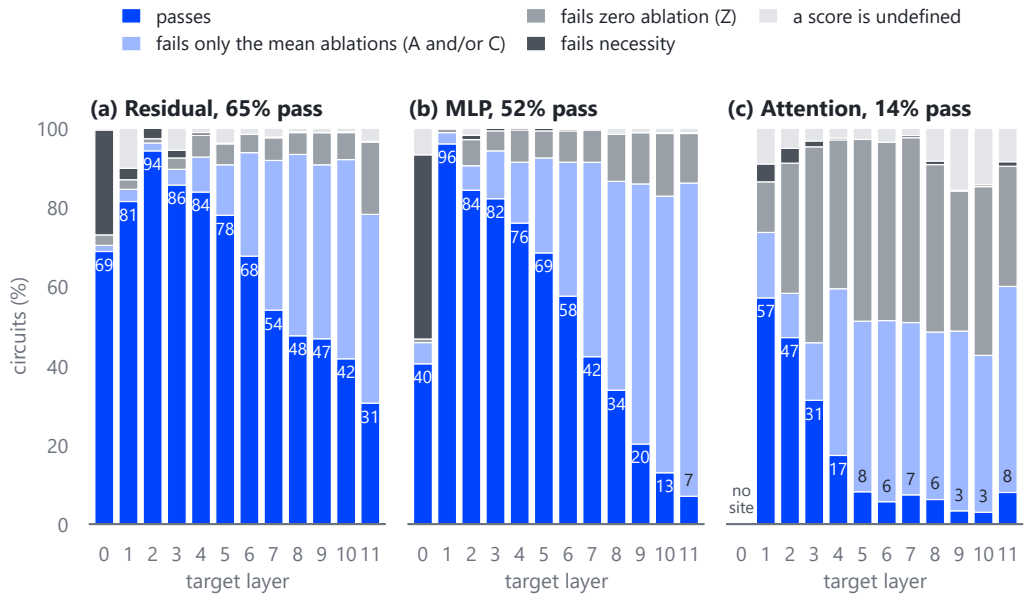


Figure 9: Protocol-v1 run: pass rates fell with depth and were lowest at attention outputs; with depth, failing circuits increasingly failed only the two mean ablations. Outcomes of the weighted circuits of 14,994 TuringLLM latents (the run's non-vacuous circuits) at $\lambda = 10^{-3}$, by target layer and site kind, scored with the pass band's earlier floor of 0.8. Each circuit is counted once: it passes; fails only the mean ablations (A, C, or both) while zero ablation and necessity hold; fails zero ablation while necessity holds; fails necessity; or has an undefined score (3%, counted as a fail). Bar numbers are pass rates (%), and panel titles give each site kind's overall pass rate; layer-0 attention has no upstream site.

Sufficiency to induce by site kind. Sufficiency to induce, unlike reconstruction, is structured by the target's site kind (Figure 12). It asks a question no circuit is trained on: setting the circuit's latents to α times their mean values at the anchors of the activating contexts, does the target switch on in held-out contrast contexts where it is silent? Across the protocol-v1 run the median is 0.84 (0.98 among passing circuits, of which 88% switch the target on to at least half its natural level). Residual-stream targets are induced almost everywhere (96% at half or more; median 1.19, rising with depth to 1.35 at layers 8–11), MLP-output targets partly (70%; median 0.69), and attention-output targets rarely beyond layer 4 (28% and 29% at layers 5–7 and 8–11; median 0.20 overall). Which latents *can* be induced from upstream is therefore itself a per-kind property, and characterising it is open.

Training dynamics. Every fit of the protocol-v1 run records its per-step curve (Figures 13 and 14). All gates start open, so no latent falls below the keep threshold for the first 57 steps; membership then collapses from 10^5–10^6 candidates to a median of about 1,700 by step 100, and a circuit is within twice its final size by a median step of 137. It is not converged at the protocol's 400 steps: the median circuit loses 12% of its members over the last 100 steps (16–17% at layers 10 and 11), and 88% are still shrinking by more than 5%. The data loss stays nearly flat over that stretch while the penalty falls, so the late pruning costs little training fit, but at depth the circuits still shrinking pass less often: 30% against 67% for the settled half at layers 4–7, and 9% against 36% at layers 8–11, with no difference at layers 0–3. Reported sizes are therefore sizes at a fixed budget. The sparsity penalty dominates the loss throughout: its tail of nearly closed gates outweighs the members, and the data terms end at 0.3% of the loss. Before their 0.25 weights, the contrast-mean and activating-mean errors are about twice the zero-ablation error, the same ordering as held-out faithfulness.

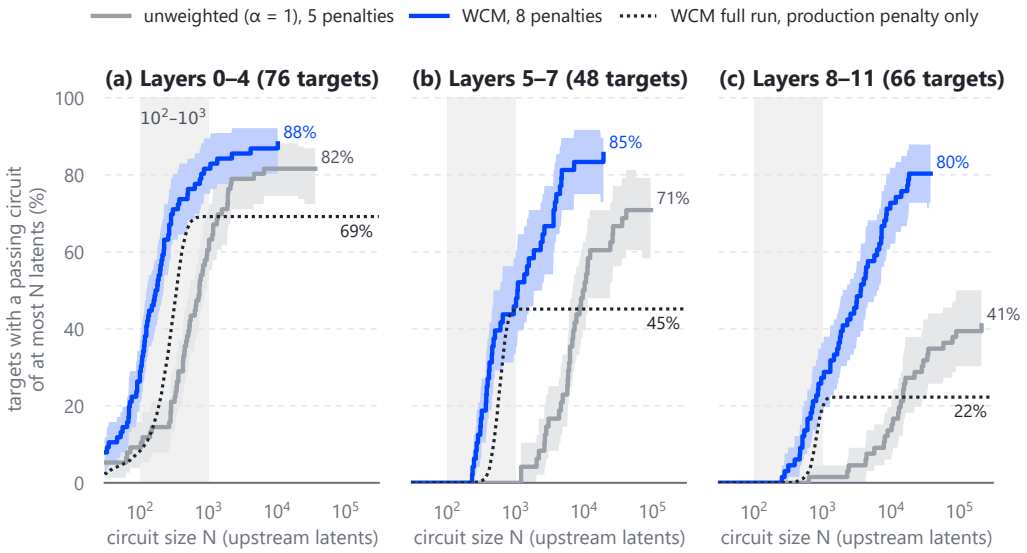


Figure 10: Protocol-v1 sweep: fitted coefficients explained more targets at every circuit budget, and under both methods deeper targets needed larger circuits. For each target depth band, the share of 190 sweep targets with a circuit of at most N upstream latents that passes the criterion (floor 0.8), taking each target's smallest passing circuit over the sparsity penalties it was fitted at: weighted circuit masking (blue) at eight penalties, λ from 4×10^{-3} to 2.5×10^{-5}, and unweighted circuit masking ($\alpha = 1$, grey) at five, 10^{-3} to 10^{-5}. Targets that never pass count as unexplained; shading is a 90% bootstrap interval over targets, and each curve ends at the largest circuit fitted. Dotted: the full run (~15,000 targets) at $\lambda = 10^{-3}$ alone, which levels off at that penalty's pass rate.

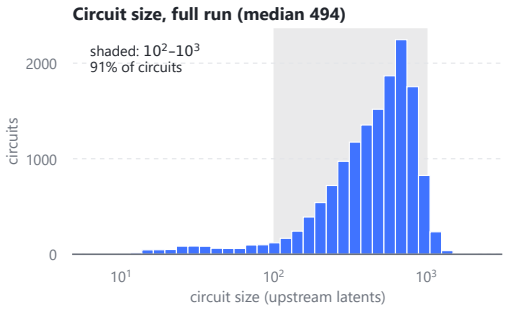


Figure 11: Circuit sizes in the protocol-v1 run at $\lambda = 10^{-3}$: 91% of circuits hold 10^2–10^3 upstream latents (shaded); median 494.

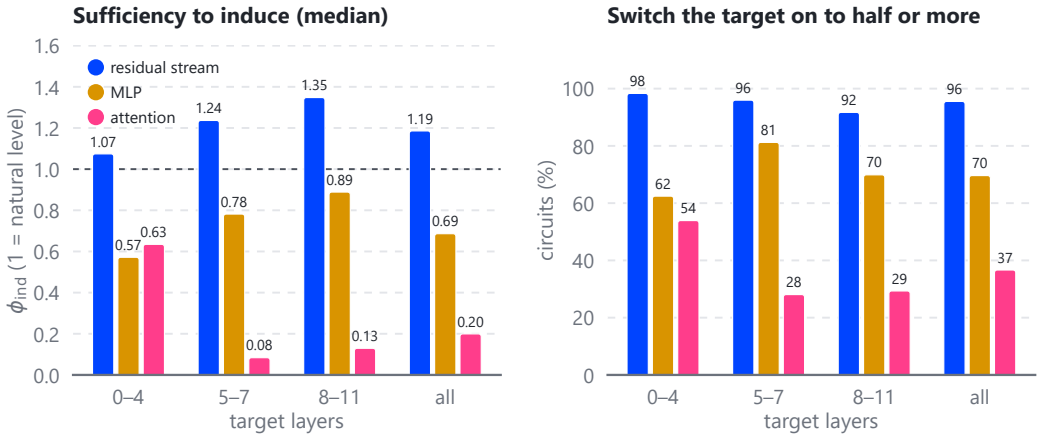


Figure 12: Sufficiency to induce is set by site kind: residual-stream targets are switched on most often and attention outputs least often, especially beyond layer 4. Sufficiency to induce across the protocol-v1 run (14,994 non-vacuous TuringLLM circuits), on held-out contrast contexts where the target is silent. Left: median $\phi^{\alpha}_{\text{ind}}$ by target depth band and site kind (1 is the target's natural level, dashed). Right: the share of circuits that switch the target on to at least half its natural level. Reported for every circuit; not part of the pass criterion.

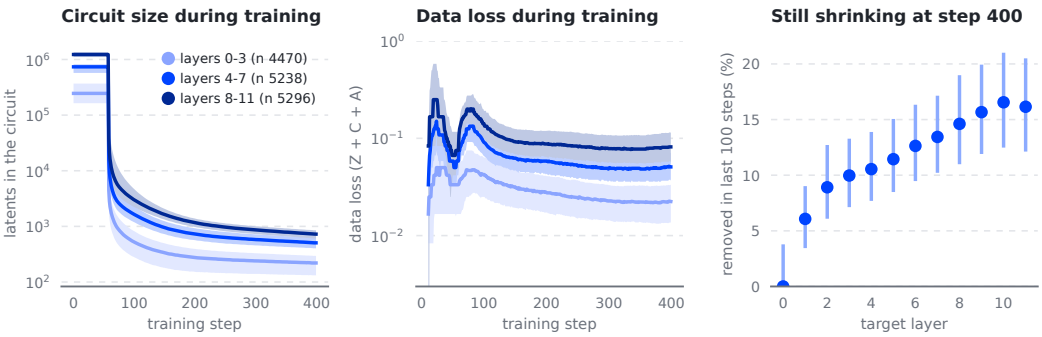


Figure 13: Circuits are still shrinking at the 400-step budget, most at depth, while the data loss stays nearly flat. Training dynamics of the 15,004 protocol-v1 fits, by target depth band (layers 0–3, 4–7, and 8–11; median and interquartile range). Left: circuit size (latents above the keep threshold) per step. Middle: the data loss (the Z, C, and A terms, penalty excluded), averaged over each pass through the 48 training contexts (12 steps), since every step sees one batch of 4. Right: the share of members removed over the last 100 steps, by target layer (median and interquartile range).

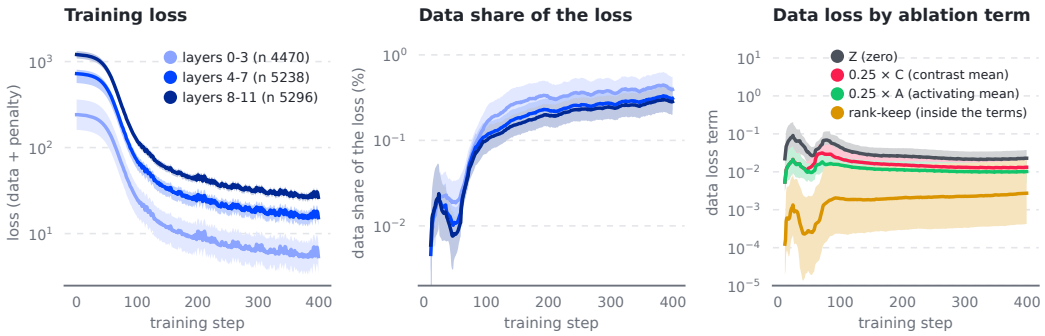


Figure 14: The training loss of the protocol-v1 fits (median and interquartile range; per-pass averages as in Figure 13). Left: the total loss, data terms plus sparsity penalty, by target depth band. Middle: the data terms' share of it; the rest is the penalty on open gates. Right: the data loss split into its ablation terms as weighted in the objective (Z, 0.25× C, 0.25× A), with the rank-keep contribution that each term contains.

I Coefficients and the Weighted-versus-Unweighted Sweep

The weighted-against-unweighted sweep, the member analysis, and the protocol-v1 details of the triad and the weak null below use circuits of the protocol-v1 run (Appendix H), scored with the earlier pass floor of 0.8; the coefficient triad figure and the harder nulls report the production run, whose sweep and $\alpha = 1$ re-score are in Section 4.2.

Weighted against unweighted circuits across the sparsity penalty. The comparison at a fixed penalty (Figure 15) uses the one value both grids share, $\lambda = 10^{-3}$; the rest comes from the full sweep, which fits WCM at $\lambda \in \{2.5, 5, 10, 20, 40\} \times 10^{-4}$ and unweighted circuit masking (every coefficient fixed at exactly 1, all else including the training fill as WCM) at $\lambda \in \{0.1, 0.3, 1, 3, 10\} \times 10^{-4}$ to 191 targets (five per layer and site kind, plus the pilot targets; 190 with usable contexts) and scores every circuit on the held-out strongest contexts. Figure 10 adds WCM at $\lambda \in \{0.25, 0.5, 1\} \times 10^{-4}$ for every sweep target, so that both methods reach circuits of $\sim 10^4$ latents and beyond. At the shared penalty unweighted circuits keep 1.85× as many latents and pass 54% against 64% at layers 0–4, but 2.1% against 38% at 5–7 and 1.5% against 18% at 8–11 (Figure 15): their scores are lower one by one, and the criterion asks for all of them at once. On the budget curves of Figure 10, weighted circuits within 10^3 latents pass for 82%, 48%, and 27% of targets at layers 0–4, 5–7, and 8–11, against 63%, 0%, and 2% for unweighted circuits; the full run, fitted at the production penalty alone, levels off at 69%, 45%, and 22% (dotted), while weaker penalties carry the same targets much further.

Unweighted masking needs a weaker penalty to reach comparable faithfulness, so the two grids differ. Figure 19 separates the three ablation methods: the gap is smallest under zero ablation and largest under the two mean ablations, where unweighted circuits of $\sim 10^3$ latents score 0.18–0.37 at layers 5–11 (zero ablation 0.58–0.72). On the five-penalty grid WCM stops near 10^3 latents, so the size ratios of Appendix H come from the eight-penalty comparison of Figure 10, which adds three weaker WCM penalties: per target, the smallest passing unweighted circuit is a median 3.7× (62 targets that pass under both methods), 9.6× (34), and 10.5× (26; median 906 against 13,075 latents) larger than the weighted one at layers 0–4, 5–7, and 8–11. Pooled over depth bands, WCM's pass rate rises monotonically as the penalty weakens: 31%, 35%, 42%, 47%, and 58% at $\lambda = 40, 20, 10, 5, 2.5 \times 10^{-4}$, where the median circuit holds 218, 327, 472, 747, and 1,103 latents; the production setting is one point on this curve, which has not levelled off. At the production penalty the per-score medians are all lower unweighted but not dramatically so at layers 8–11 (zero ablation 0.58 against 0.89, activating mean 0.27 against 0.69, contrast mean 0.18 against 0.51; Figure 18); the pass rates differ far more because the criterion needs every score at once.

The coefficient triad. Figure 16 sets the versions of a production circuit side by side on all three models: the circuits as fitted, the same nodes with every coefficient set to 1, and the anchor-active and contribution-matched random circuits given the same coefficient fitting: 88.0%, 1.8%, 0%, and 0% pass on TuringLLM, 81.9%, 13.8%, 0%, and 0% on Pythia-70m, and 82.4%, 3.3%, 0%, and 0% on GPT-2 small (the nulls on 112, 126, and 128 targets). In the protocol-v1 run (Appendix H), on 1,746 circuits, median faithfulness falls from 0.94, 0.95, and 0.89 with fitted coefficients to 0.00, 0.09, and 0.06 at natural scale, under Z, A, and C. At natural scale the nodes seldom overshoot the target: 4 of the 1,746 circuits exceed 1.5 under any ablation method, all at attention-output targets. More often they fall short of the empty circuit itself under activating-mean ablation (18%; 7% at layers 0–4, 32% at 8–11, and 29% of attention-output targets). There the activating-context means that the members replace already carry more of the target (a median 1.16 of a natural 3.06) than the members do at their natural scale (0.61), and the fitted coefficients make up the difference; these circuits pass less often with fitted coefficients (16%, against 51% for the other targets).

Harder random circuits. The protocol-v1 run's random circuits (DAN-17; the "dan17" arm of Table 20, rerun on the TuringLLM null panel) are the weakest null that fits coefficients. On the 747 targets their median faithfulness is 0.00 under every ablation method at every depth, only one has even a single score in the pass band, and none induces the target to half its level in contrast

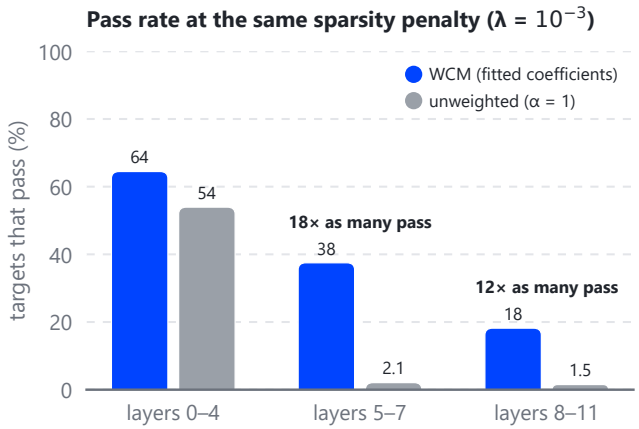


Figure 15: At the shared sparsity penalty, unweighted circuits pass almost as often as weighted ones at layers 0–4 but rarely beyond. Weighted against unweighted circuit masking at $\lambda = 10^{-3}$ (the production setting): the share of targets whose circuit passes the criterion of Section 3.3 (faithfulness under Z, A, and C in $[0.8, 1.5]$ and necessity ≥ 0.9, on held-out contexts) for weighted circuit masking (blue) and unweighted circuit masking ($\alpha = 1$, grey), by target depth band; bold labels give the ratio of pass rates. 190 targets; the per-score medians are in Figure 18.

contexts. Their members are latents active anywhere on the target's strongest training contexts, and most are silent at the positions that feed the anchor, so in that run their coefficients received no gradient: the median coefficient is exactly 0.798, the value weight decay alone reaches from 1 in 400 steps at that run's constant learning rate, and on the circuits' own training contexts the median zero-ablation faithfulness is 0.00 (0.4% reach 0.8). The null therefore shows that unrelated latents cannot be fitted, not that WCM's selection matters. Two harder nulls keep everything else and change only where the members come from. The *anchor-active* null draws, at each site, the production circuit's member count from latents active at the anchor or up to 8 positions before it on at least one training context, matching each member's corpus-frequency decile; production members may be drawn. The *contribution-matched* null also matches each member's contribution to the target, the mean over training contexts of its activation times the gradient of the target's pre-activation, binned by deciles of the production members' own contributions, and excludes the production members: it is a substitution test, asking whether other latents of the same strength can stand in for the selected ones, as overlap-penalised search does for output circuits (Chen et al., 2026b). Each set is fitted exactly as above and scored on held-out and on training contexts, with fitted coefficients and with every coefficient at 1; the latter is the size-matched, nothing-fitted null of prior work (Shi et al., 2024; Seth et al., 2026), and once circuits carry fitted coefficients the fair control fits them on the random members too. In the production run both run on 126 TuringLLM sweep targets (14 per depth band and site kind; 112 scored, as 3 have no record and 11 are vacuous) and on 128 targets per public model (126 scored on Pythia-70m), one draw each and a second draw on 27–32 targets per model, so that each target's WCM score can also be placed within its own null draws (Table 20). Both nulls pass for none of the targets in any depth band on any model, fitted or at $\alpha = 1$, while the production circuits of the same targets pass for 80–89% per band on TuringLLM, 79–90% on Pythia-70m, and 65–96% on GPT-2 small. The anchor-active nulls score a median 0.00 under every ablation method and keep little necessity (0.04, 0.28, and 0.12); the contribution-matched nulls also score 0.00 under zero ablation, but on TuringLLM keep some activating- and contrast-mean faithfulness (0.34 and 0.26), and their necessity is high on TuringLLM and GPT-2 small (0.66 and 0.91), so latents of matched contribution are needed for the target without sufficing for it. On the targets with two draws, the production circuit's lowest faithfulness score exceeds every draw's on every target, and it passes while every draw fails on 77–82% of them. The nulls also fail on the contexts they were fitted on: none passes on the training contexts of any target, where the production circuits pass for 90, 97, and

Model	Circuit	Targets	Pass held / train (%)	Held Z / A / C	Held nec.	Train Z / A / C	Train nec.	α moved (%)	WCM > both draws (%)	Jaccard w/ prod.	Exact bin (%)
TuringLLM	production (WCM)	112	85.7 / 90.1	0.92 / 0.98 / 0.98	1.00	0.94 / 0.99 / 0.98	1.00	100 (99–100)	—	—	—
	weak (protocol v1)	112	0.0 / 0.0	0.00 / 0.00 / 0.00	0.01	0.00 / 0.00 / 0.00	0.01	92 (88–94)	—	—	—
	anchor-active	112	0.0 / 0.0	0.00 / 0.00 / 0.00	0.04	0.00 / 0.01 / 0.00	0.04	99 (98–99)	100	0.007	100
	contribution-matched	112	0.0 / 0.0	0.00 / 0.34 / 0.26	0.66	0.00 / 0.42 / 0.30	0.64	99 (99–99)	100	0.000	68
Pythia-70m	production (WCM)	126	82.5 / 96.8	0.97 / 1.00 / 0.99	1.00	0.98 / 1.01 / 1.01	1.00	99 (99–100)	—	—	—
	anchor-active	126	0.0 / 0.0	0.00 / 0.00 / 0.00	0.28	0.00 / 0.00 / 0.00	0.27	93 (86–96)	100	0.022	100
	contribution-matched	126	0.0 / 0.0	0.00 / 0.00 / 0.00	0.29	0.00 / 0.00 / 0.00	0.30	96 (91–97)	100	0.000	58
GPT-2 small	production (WCM)	128	82.8 / 87.5	0.93 / 0.94 / 0.94	1.00	0.94 / 0.95 / 0.94	1.00	99 (99–100)	—	—	—
	anchor-active	128	0.0 / 0.0	0.00 / 0.00 / 0.00	0.12	0.00 / 0.00 / 0.00	0.13	92 (85–95)	100	0.006	100
	contribution-matched	128	0.0 / 0.0	0.00 / 0.00 / 0.00	0.91	0.00 / 0.00 / 0.00	0.92	94 (90–96)	100	0.000	71

Medians, draw 0, fitted coefficients; α moved: $|\alpha - 0.7901| > 0.01$, median (IQR) over every fitted draw.

Table 19: The harder random circuits fail on their own training contexts too, although their coefficients move as much as the production circuits'. Per model and circuit: pass rate on held-out and training strongest contexts, median Z/A/C and necessity on each, the median share of coefficients moved more than 0.01 from the weight-decay-only value (0.790; IQR in brackets), the share of two-draw targets where the production circuit's lowest score beats both draws', the median Jaccard overlap with the production members, and the share of members matched within their own decile bin (Appendix I).

88% (TuringLLM, Pythia-70m, GPT-2 small), and their median zero-ablation faithfulness is 0.00 on training contexts as on held-out ones (Table 19). With 0 passes of 112–128 targets, the exact one-sided 95% upper bound on each null's pass rate is 2.6% on TuringLLM and 2.3% on the public models (6–9% within a depth band). Unlike the protocol-v1 null, these fits are not inert: under the production schedule the coefficient that weight decay alone reaches is 0.790, and the median circuit moves 92–99% of its coefficients more than 0.01 from it, about as many as the production circuits (99–100%). The weak null of the protocol-v1 run, refitted in the production run on the TuringLLM panel, moves 92% of its coefficients too, yet still scores 0.00, so on this schedule it fails for want of usable members rather than for want of gradient. The anchor-active draws barely overlap the production circuits although production members could be drawn: a median 1.4%, 4.5%, and 1.3% of a draw's members are production members (Jaccard 0.007, 0.022, and 0.006; at most 0.10). The contribution-matched null falls short of the production members' strength where it matters most. It finds a same-site non-member in the member's own contribution decile for a median 68%, 58%, and 71% of members, but for only 22%, 11%, and 22% of the members in the top decile; its drawn set's median contribution reaches 0.87, 0.71, and 0.94 of the members' median but its mean only 0.51, 0.29, and 0.39 of theirs, since the strongest contributors are already in the circuit. That shortfall does not explain the failure: the contribution-matched sets are needed for the target without sufficing for it. Their necessity grows with depth (TuringLLM 0.37, 0.72, 0.74 by band; GPT-2 small 0.55, 1.00, 1.00; Pythia-70m 0.05, 0.21, 0.53) and with how much of the members' contribution the draw recovered (Spearman $\rho = 0.54$–0.69 across draws). On GPT-2 small, removing the set from the running model switches the target off outright in 36% of draws (52–56% from layer 5 on), as removing the production members does in 87%, because a Top-K latent falls to zero once its pre-activation drops below the k-th largest; yet no draw reproduces the target alone, with $Z \geq 0.75$ in none of GPT-2 small's draws and 2% of TuringLLM's. Necessity is coefficient-blind (it removes the members from the live model), so the fitted and $\alpha = 1$ rows share it; once contribution is matched, high necessity says the target depends on many strong upstream latents, not that these latents are its circuit.

What the coefficients do to members. Comparing each member's value inside the circuit-only run with its value in the unmodified model, at the target's anchors on held-out contexts, shows that the coefficients do not return members to their natural values (Figure 17a; 600 circuits of the protocol-v1 run, 200 per depth band). Before the coefficients act, the ablated environment leaves 14% of members unchanged, pushes 8–10% out of their site's Top-K, and raises 11–14% to three or more times their natural value; 28–30% lie within 25% of it. After the coefficients fewer do (22–23%), with the spread widening on both sides, and weighting members by their gradient-times-activation contribution to the target gives the same picture. A member's coefficient is unrelated both to how far the ablation moved it and to how much it contributes (rank correlations within ±0.1). What the coefficients change is the target. With every coefficient set to 1, the members alone bring the target to a median 0%, 19%, and 7% of its natural activation under Z, A, and C (21%, 24%, and 12% of its pre-activation), against 94%, 97%, and 90% with the fitted coefficients (Figure 17b). To first order the coefficients raise the

Model	Circuit	Targets	Pass (%) [95% CI]	Production − arm (pp)	Z / A / C (median)	Necessity
TuringLLM	production circuit (WCM)	112	85.7 [78.6, 92.0]	—	0.92 / 0.98 / 0.98	1.00
	same members, $\alpha = 1$	112	2.7 [0.0, 6.2]	+83.0 [+75.9, +89.3]	0.00 / 0.04 / 0.04	1.00
	random members (weak baseline)	112	0.0 [0.0, 0.0]	+85.7 [+78.6, +92.0]	0.00 / 0.00 / 0.00	0.01
	random members, $\alpha = 1$	112	0.0 [0.0, 0.0]	+85.7 [+78.6, +92.0]	0.00 / 0.00 / 0.00	0.01
	anchor-active random	112	0.0 [0.0, 0.0]	+85.7 [+78.6, +92.0]	0.00 / 0.00 / 0.00	0.04
	anchor-active random, $\alpha = 1$	112	0.0 [0.0, 0.0]	+85.7 [+78.6, +92.0]	0.00 / 0.00 / 0.00	0.04
	contribution-matched random	112	0.0 [0.0, 0.0]	+85.7 [+78.6, +92.0]	0.00 / 0.34 / 0.26	0.66
	contribution-matched random, $\alpha = 1$	112	0.0 [0.0, 0.0]	+85.7 [+78.6, +92.0]	0.00 / 0.00 / 0.00	0.66
Pythia-70m	production circuit (WCM)	126	82.5 [75.4, 88.9]	—	0.97 / 1.00 / 0.99	1.00
	same members, $\alpha = 1$	126	14.3 [8.7, 20.6]	+68.3 [+60.3, +76.2]	0.14 / 0.24 / 0.25	1.00
	anchor-active random	126	0.0 [0.0, 0.0]	+82.5 [+76.2, +88.9]	0.00 / 0.00 / 0.00	0.28
	anchor-active random, $\alpha = 1$	126	0.0 [0.0, 0.0]	+82.5 [+76.2, +88.9]	0.00 / 0.00 / 0.00	0.28
	contribution-matched random	126	0.0 [0.0, 0.0]	+82.5 [+76.2, +88.9]	0.00 / 0.00 / 0.00	0.29
	contribution-matched random, $\alpha = 1$	126	0.0 [0.0, 0.0]	+82.5 [+76.2, +88.9]	0.00 / 0.00 / 0.00	0.29
GPT-2 small	production circuit (WCM)	128	82.8 [75.8, 89.1]	—	0.93 / 0.94 / 0.94	1.00
	same members, $\alpha = 1$	128	5.5 [2.3, 10.2]	+77.3 [+69.5, +84.4]	0.00 / 0.00 / 0.00	1.00
	anchor-active random	128	0.0 [0.0, 0.0]	+82.8 [+75.8, +89.1]	0.00 / 0.00 / 0.00	0.12
	anchor-active random, $\alpha = 1$	128	0.0 [0.0, 0.0]	+82.8 [+75.8, +89.1]	0.00 / 0.00 / 0.00	0.12
	contribution-matched random	128	0.0 [0.0, 0.0]	+82.8 [+75.8, +89.1]	0.00 / 0.00 / 0.00	0.91
	contribution-matched random, $\alpha = 1$	128	0.0 [0.0, 0.0]	+82.8 [+75.8, +89.1]	0.00 / 0.00 / 0.00	0.91

Table 20: Random circuits of the same size, given the same coefficient fitting, never pass; the production circuits with every coefficient at 1 rarely do. Held-out strongest contexts, one draw per target: the production circuit (WCM) and the same members at $\alpha = 1$; random members drawn from latents active anywhere on the strongest training contexts (the weak baseline of the protocol-v1 run, TuringLLM only); and anchor-active and contribution-matched random members (Appendix I), fitted and at $\alpha = 1$; pass rate with 95% bootstrap interval, paired difference from the production circuit, and median scores. TuringLLM: 112 of the 126 panel targets (3 without a record, 11 vacuous).

members' summed contribution only about 1.6×, so most of their effect on the target arises in the layers between the members and the target; we report this as an observation, not a mechanism. Deep and failing circuits also lose members to the ablated environment itself: weighted by contribution, 10% of members of layer 8–11 circuits are pushed out of Top-K under contrast-mean ablation, against 1% at layers 0–4, and 9% in failing circuits against 1% in passing ones. At natural scale ($\alpha = 1$), unweighted circuit masking needs $\sim 10^4$ latents before a deep target passes (Figure 10); we interpret the coefficients as locating most of that cost in redundancy, many latents each carrying a small share of a signal that fewer, rescaled nodes carry alone.

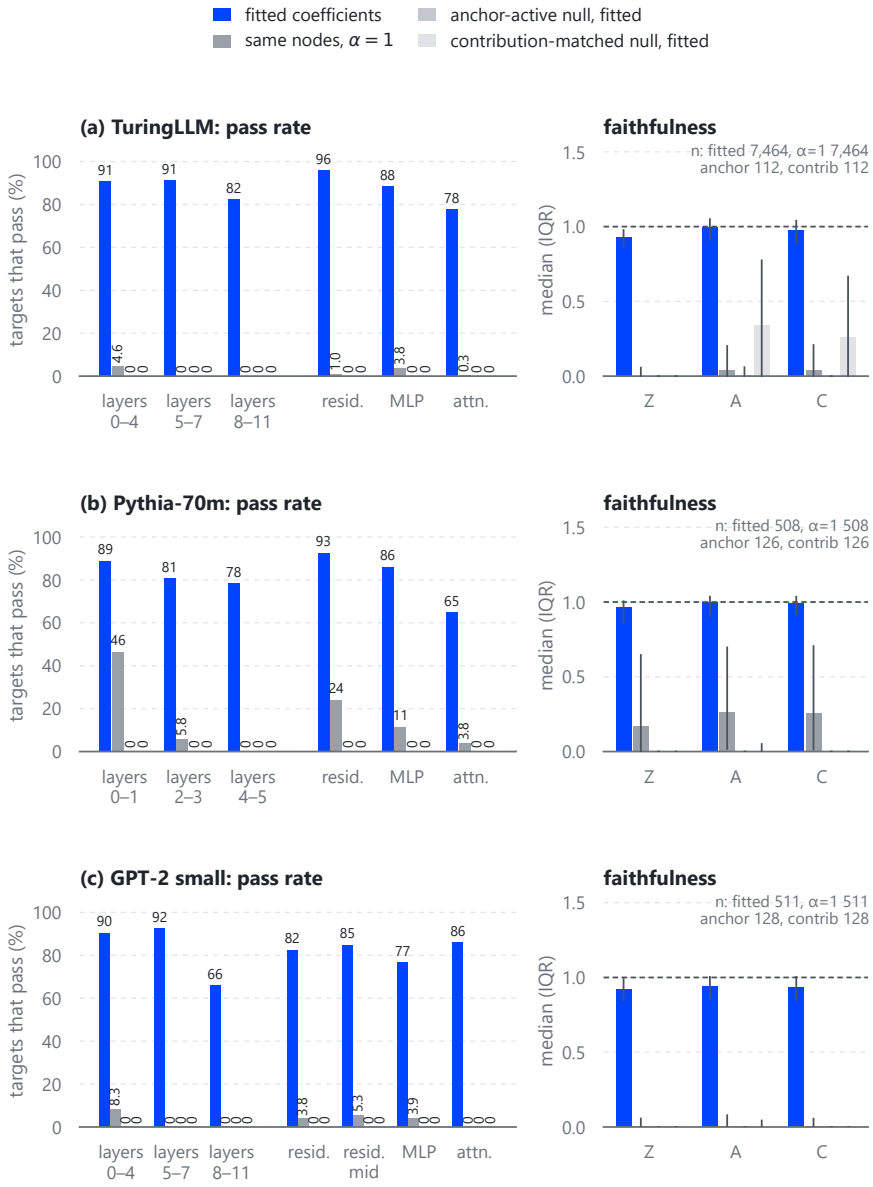


Figure 16: Only the selected nodes with their fitted coefficients pass: the same nodes at natural scale, and random nodes given the same fitting, almost never do. The coefficient triad on the production circuits of the three models at $\lambda = 10^{-3}$. **(a)** The share of targets that pass the criterion of Section 3.3 on held-out strongest contexts, by target depth band and site kind, for the circuits as fitted (blue), the same nodes with every coefficient set to 1 (grey), and the anchor-active and contribution-matched random circuits given the same coefficient fitting (light greys; Appendix I). **(b)** Median faithfulness under zero (Z), activating-mean (A), and contrast-mean (C) ablation, with interquartile ranges; 1 is perfect (dashed). Fitted circuits pass for 88.0%, 81.9%, and 82.4% of TuringLLM, Pythia-70m, and GPT-2 small targets, the same nodes at $\alpha = 1$ for 1.8%, 13.8%, and 3.3%, and both random circuits for none (112, 126, and 128 targets).

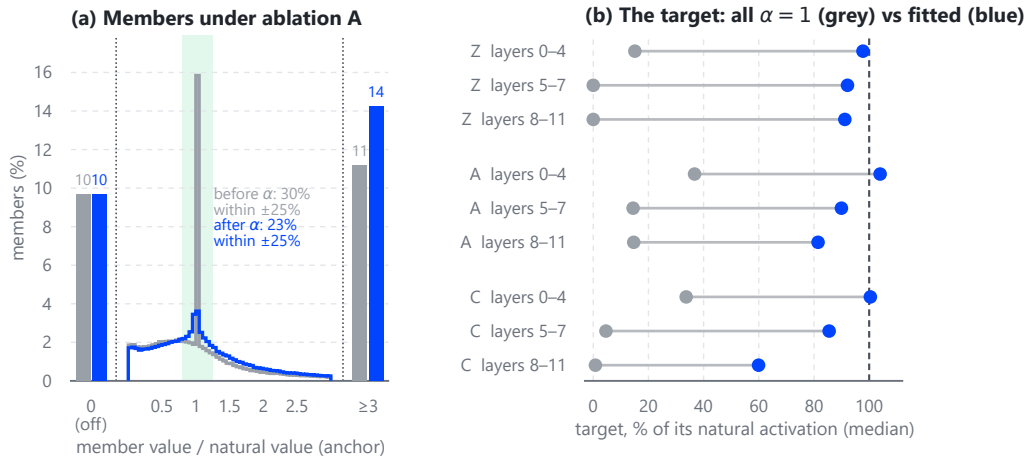


Figure 17: What the coefficients do to circuit members (600 circuits of the protocol-v1 run). **(a)** Each member's value in the circuit-only run divided by its value in the unmodified model, at the target's anchors on held-out contexts, under activating-mean ablation, before (grey) and after (blue) its coefficient; members active in the unmodified model, pooled over circuits. The bars at either side collect members that left their site's Top-K ("0, off") and members at three or more times their natural value; both groups are mostly members that fire weakly in the unmodified model, and together they carry under a tenth of the circuits' contribution to their targets. **(b)** The target's activation in the circuit-only run as a share of its natural activation, median over circuits, with every coefficient set to 1 (grey) and with the fitted coefficients (blue), by ablation method and target depth band.

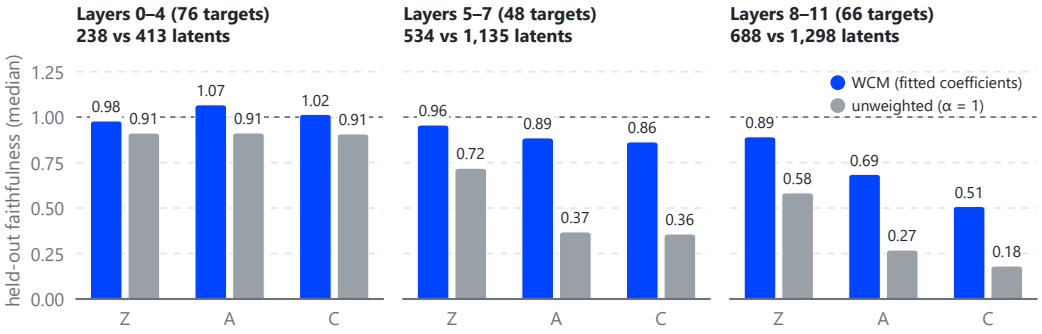


Figure 18: Per-score medians at the production sparsity penalty, $\lambda = 10^{-3}$: held-out faithfulness of weighted (blue) and unweighted (grey) circuit masking under zero ablation (Z) and mean ablation over activating (A) and contrast (C) contexts, by target depth band; panel titles give the median circuit sizes, weighted against unweighted. 1 is perfect (dashed).

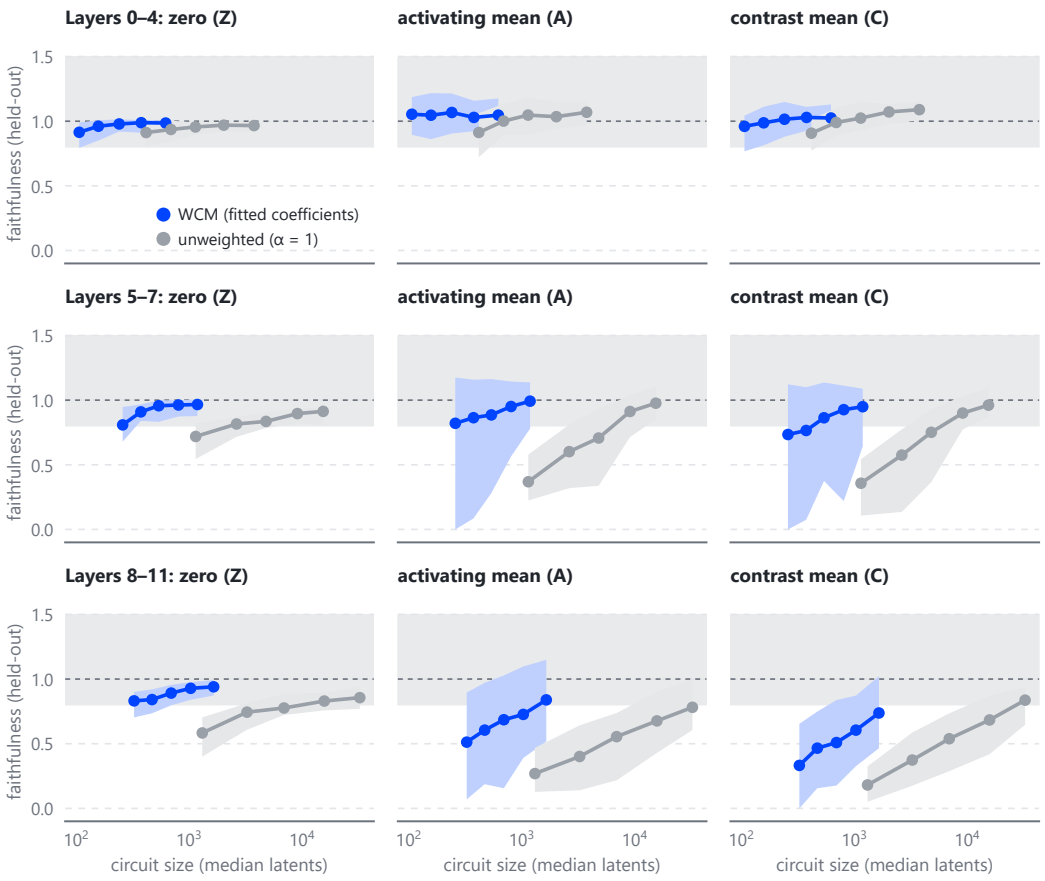


Figure 19: Weighted against unweighted circuit masking by ablation method (rows: target depth band; columns: zero ablation, and mean ablation over activating and over contrast contexts). Median held-out faithfulness and interquartile range against median circuit size, one point per sparsity penalty; the grey band is the pass band $[0.8, 1.5]$. The gap between the methods is smallest under zero ablation, largest under the two mean ablations, and widens with depth.

J Weighted Circuit Masking under Each Contrast Source

The C term of WCM trains against the mean over contrast contexts, so the choice of contrast contexts could shape what a circuit learns. This analysis ran under protocol v1: we refitted 280 targets of the protocol-v1 run (Appendix H; 8 per layer and site kind) under four contrast sources, all checked for silence after Top-K: *close* (the production set, Appendix R), *random*, *distant*, and *fused*, a third of each. For every target the four sets are disjoint, and every circuit, fitted at $\lambda \in \{3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$, is scored on the held-out contexts of the close, random and distant sets alike, so the training source is the only difference between arms and all arms are scored on the same contexts. Scores are interpolated to a matched circuit size of 490 latents, the median close-trained circuit at the production sparsity penalty.

Circuits trained against close contexts are the most faithful on close backgrounds and the least on any other (Figure 20). Their median C faithfulness falls from 0.90 on close to 0.58 on distant backgrounds, and on distant backgrounds they pass 0.27, against 0.37 and 0.38 for circuits trained on random or distant contexts (paired differences +0.099 $[0.052, 0.149]$ and +0.106 $[0.065, 0.152]$, 95% bootstrap intervals over targets). Averaged over the three backgrounds, distant-trained circuits pass more often than close-trained ones (+0.046 $[0.014, 0.082]$), random-trained ones by a similar margin that the interval does not resolve (+0.035 $[-0.002, 0.074]$), and random and distant do not differ (-0.011 $[-0.045, 0.023]$). The source acts mainly on the C term: zero-ablation faithfulness and necessity are the same across sources (activating-mean faithfulness is somewhat higher for close-trained circuits, median 1.00 against 0.92–0.94), and for each source the share of circuits that pass or fail only under C is the same on every background (0.48 to 0.51), so a different background only moves circuits between passing and failing on C.

Why close contexts generalise worst. Variety alone does not explain it. Fused contexts span the widest range of similarity to the target's activating contexts (10th to 90th percentile 0.25 to 0.87, against 0.82 to 0.90 for close, 0.47 to 0.77 for random, and 0.19 to 0.27 for distant), yet fused-trained circuits do no better than random-trained ones (-0.019 $[-0.054, 0.017]$). What close contexts have that the others lack is the target's topic: they sit nearest the activating contexts and nearest to firing (median pre-activation 3% of the activating level, against 0 for random and distant). A circuit trained against their mean keeps the topic's features in its background and learns only what separates the target from near misses; against a background without those features it falls short. The same circuits, re-scored with the C mean taken from the nearest, middle or farthest third of each target's random set, show this within a single source: close-trained circuits pass 0.40, 0.40 and 0.33 as the background moves away from the target's topic (nearest minus farthest +0.068 $[0.039, 0.100]$), while random-trained circuits pass 0.43, 0.44 and 0.41 (+0.014 $[-0.007, 0.036]$).

The production run uses close contrast contexts on all three models, so that the model is the only change between them. For new work, random or distant contrast contexts give circuits whose contrast faithfulness depends less on the background they are evaluated against. An earlier 16-target pilot, with the unchecked retrieval store as a fourth source, found that node sets move no more between contrast sources than between two refits with the same source.

K Specificity

Faithfulness, necessity, and the random-circuit baseline all read the target alone. A circuit can therefore restore its target by lifting the target's whole neighbourhood, its concept, topic, or context type, and pass all three: a concept amplifier rather than a mechanism for that latent. The loss permits it, since reproducing the target's pre-activation constrains the site's stream along one direction of d_{model}, and it leaks into the activation read only when the lifted latents overtake the target in its site's Top-K.

The diagnostic reads the whole code at the target's site, every latent's pre-activation and the SAE's Top-K set at the anchor, in the unmodified model, the empty circuit, and the fitted circuit under each ablation method, on the held-out contexts. It reports the target's rank and Top-K membership; the

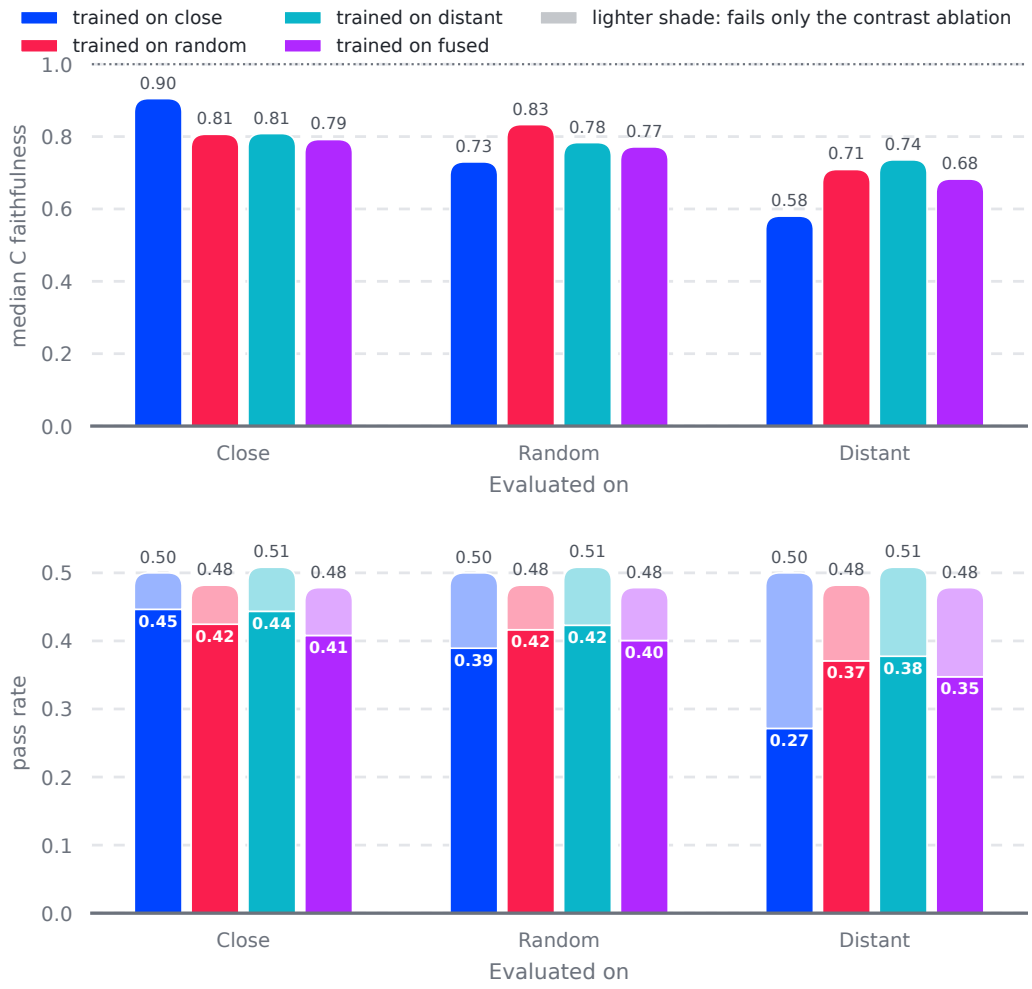


Figure 20: Weighted circuits trained against close contrast contexts generalise worst to other contrast backgrounds. Groups: the contrast set a circuit is evaluated against; bars: the contrast set it was trained against. Top: median faithfulness under the contrast-context mean ablation (C). Bottom: pass rate under the protocol-v1 criterion, each evaluation set supplying its own C; the lighter segment counts circuits that fail only the contrast ablation (zero-ablation and activating-mean faithfulness in band, necessity ≥ 0.9), and its top, the share that would pass with a perfect C term, is the same for each training source on every background. 280 targets at a matched size of 490 latents; the paired differences are in the text.

number of latents lifted above the unmodified model's Top-K cut, against the empty circuit; the overlap of the circuit's Top-K set with the unmodified one; and the faithfulness of *siblings*, latents in the unmodified Top-K at the anchor on at least half of the target's strongest training contexts, computed exactly as the target's, against a control group of the site's other active latents. A circuit is flagged as an amplifier when, under at least one ablation method and on the pre-activation read, its target is restored to at least 0.5 and its median sibling to at least 80% of the target's faithfulness.

The rank-keep term (Section 3.2) keeps the target in its Top-K more often but does not reliably remove amplifiers. In the production run (Table 21), under Z, A, and C the target is restored to a median 0.92, 1.00, and 0.98 of its pre-activation on TuringLLM, its siblings to 0.36, 0.46, and 0.46, and the site's other active latents to 0.17, 0.23, and 0.24; on Pythia-70m the figures are 0.97–1.00

Model	π	n	Target / siblings / control (median)	Lifted: circuit / empty (median)	Target in Top-K (%)	Amplifier under π (%)	Amplifier, any π (%)	Amplifier among passing (%)
TuringLLM	Z	7,464	0.92 / 0.36 / 0.17	742 / 10,977	98	11.9	29.0 [28.0, 30.0]	30.4 [29.3, 31.5]
	A	7,464	1.00 / 0.46 / 0.23	501 / 4	99	23.8		
	C	7,464	0.98 / 0.46 / 0.24	491 / 4	98	23.2		
Pythia-70m	Z	508	0.97 / 0.45 / 0.11	53 / 86	95	24.4	33.7 [29.5, 37.8]	35.1 [30.5, 39.7]
	A	508	1.00 / 0.53 / 0.11	48 / 10	96	26.4		
	C	508	1.00 / 0.52 / 0.12	50 / 10	96	25.4		
GPT-2 small	Z	511	0.92 / 0.32 / 0.00	58 / 9	97	24.7	31.3 [27.4, 35.4]	30.6 [26.4, 35.2]
	A	511	0.94 / 0.32 / 0.00	61 / 29	96	24.5		
	C	511	0.93 / 0.30 / 0.00	59 / 29	96	22.9		

Table 21: Specificity of the production circuits on held-out strongest anchors, by model and ablation method: median pre-activation restoration of the target, its siblings, and a control group of the site's other active latents; median latents lifted above the unmodified Top-K cut by the circuit and by the empty circuit; the share of anchors where the target stays in its Top-K; and the share of circuits flagged as amplifiers (target restored to at least 0.5, median sibling to at least 0.8 of the target) under that method, under any method, and among passing circuits, with 95% bootstrap intervals.

for the target, 0.45–0.53 for siblings, and 0.11–0.12 for the others, and on GPT-2 small 0.92–0.94, 0.30–0.32, and 0.00. Under the two mean ablations, TuringLLM circuits lift a median of 501 and 491 latents above the unmodified Top-K cut, against 4 for the empty circuit (Pythia-70m 48 and 50 against 10; GPT-2 small 61 and 59 against 29), and the target stays in its Top-K at 96–99% of held-out anchors on average, on every model. 29.0% (28.0–30.0), 33.7% (29.5–37.8), and 31.3% (27.4–35.4) of circuits are flagged as amplifiers under at least one ablation method, including 30.4%, 35.1%, and 30.6% of passing circuits; on TuringLLM the share is 44.8% of MLP, 26.8% of residual-stream, and 11.3% of attention-output circuits. 61.2%, 53.1%, and 57.1% of all circuits pass and carry no flag.

In the protocol-v1 run (Appendix H), most circuits were specific (15,004 circuits): under Z, A, and C the target is restored to a median 0.94, 0.99, and 0.92 of its pre-activation, its siblings to 0.26, 0.46, and 0.41, and the site's other active latents to 0.14, 0.28, and 0.24. Under the two mean ablations, circuits lift a median of 578 and 517 latents above the unmodified Top-K cut, against 5 and 4 for the empty circuit, and the target stays in its Top-K on 84% and 80% of held-out anchors. 30% of circuits are flagged as amplifiers under at least one ablation method (44% of MLP, 30% of residual-stream, and 11% of attention-output targets), including 38% of passing circuits; 29% of all circuits pass and carry no flag.

The loss terms we tried for rank and specificity before adopting rank-keep are in Appendix V.1.

L Resample Ablation and Reads at Every Position

The three ablation methods of the pass criterion replace a removed latent with zero or a mean. Causal scrubbing (Chan et al., 2022) sets a stricter test: everything outside the circuit takes its value on another real context. We scored the circuits of the protocol-v1 run (Appendix H), trained without any real background, for the 747 targets that have a random-circuit baseline this way. At each position a removed latent takes a donor context's sparse code (so the donor's sparsity is kept), with donors drawn from the target's activating training contexts (A), its contrast contexts (C), or random corpus text (T); the scores and the pass criterion are otherwise unchanged. With A and C donors, 0.1% of circuits pass, against 43.1% under the pass criterion, with median faithfulness 0.18 (A), 0.04 (C), and 0.01 (T); random circuits pass on none. As for output circuits under node ablation (Miller et al., 2024), resample ablation scores far below mean ablation.

The failure is one of closure, not of membership, for residual-stream targets. In a 45-target diagnostic with C donors, holding the members at their own values in the base context restores residual-stream targets to a median 0.92 (MLP 0.39, attention 0.11), while members recomputed from the mixed stream do not; the node set carries the signal, but the members lose the inputs they receive from outside the circuit. The coefficients show the same dependence: with the base context as its own donor, so that nothing is ablated, residual-stream circuits overshoot to a median 2.16, because coefficients fitted to stand in for an ablated complement over-drive the target when the complement is present.

Before the production protocol, nothing in training asked a circuit to survive a real background,

so we tried adding one: a resample ablation whose replacement values are the sparse codes of the target's training contrast contexts, one donor per context, drawn afresh at every step, with the donor's sparsity kept as in the test; the test's donors are held-out contrast contexts and random corpus text, never seen in training. On 72 targets (8 per depth band and site kind), paired against a development configuration (sparsity-preserving fill, no resample term, rank-keep 3×10^{-3}), adding it as a fourth ablation method raises the resample pass rate from 0% to 24% (95% CI of the difference +14 to +33 points; 33%, 33%, and 4% by depth band), while the rate under the pass criterion moves from 69% to 62% (CI −17 to +3 points), at 1.18× the circuit size and 1.45× the fitting time. Using it in place of the contrast-mean ablation gains as much on the resample test (25%) but costs the pass criterion (51%; CI −29 to −7 points), most at layers 8–11 (46% to 21%), where the median contrast-mean faithfulness falls from 0.83 to 0.48. Under our earlier training protocol, whose dense mean fill left more circuits failing the pass criterion (44% passed), the same addition raised both rates on 99 targets, the resample pass rate to 30% and the pass rate under the criterion to 57%; with the sparsity-preserving training fill (Section 3.2) only the resample gain remains. Deep targets gain least on the resample test under either protocol (4% here, 15% before).

Every score above, like the pass rule, reads the target at the anchor. Read instead at every position of 16 held-out activating and 16 held-out contrast contexts, the circuits track the target only in part. Under zero ablation the median correlation with the target's activation is 0.59 (0.75, 0.58, and 0.50 by depth band), and beyond layer 4 the variance explained is negative (−1.06 at layers 5–7, −2.14 at layers 8–11). The cause is firing where the target is silent: on 31% and 74% of contrast-context positions in those bands, against 1% at layers 0–4. Under the contrast-mean ablation the reads are closer (median correlation 0.66; 3% of contrast-context positions), and passing circuits track the target far better than failing ones (correlation 0.88 against 0.35; variance explained 0.46 against −0.25). A circuit is thus a mechanism for the target's activation where the target fires, not a simulation of the target at every position, and deep circuits under zero ablation overfire away from the anchor.

Faithfulness is therefore relative to the ablation methods a circuit is trained and tested against, and robustness to a stricter test can be trained for by giving training the backgrounds the circuit must survive.

The production run. The experiments above led to the production protocol: the resample term, over the target's training contrast contexts, is a fourth training ablation at weight 1, with the rank-keep weight raised to 10^{-1} (on a 72-target check panel, chosen by a rule stated before the comparison, these two changes raised the number of targets passing the resample test from 19 to 39 of 72 and the pass criterion from 48 to 54, at 1.33× the circuit size). Every production circuit is scored with the donor-background test of Table 4, eight paired draws per donor pool, and passes it when it meets the pass criterion and its recovery under activating and held-out contrast donors lies in $[0.75, 1.5]$; the test's donors are never training donors. On TuringLLM, Pythia-70m, and GPT-2 small, 69.2% (68.1–70.2), 66.1% (62.0–70.3), and 41.1% (36.8–45.4) of all scored targets pass the donor-background test, 78.6%, 80.8%, and 49.9% of the production passes (Table 22). Median recovery is 0.95 and 0.86 under activating and held-out contrast donors on TuringLLM, 0.91 and 0.86 on Pythia-70m, and 0.79 and 0.74 on GPT-2 small, and lower under random-text donors (pool T; 0.75, 0.69, and 0.64), which the test does not require. Recovery depends on the fitted coefficients: at $\alpha = 1$ it falls to 0.04 under both donor pools on Pythia-70m and to 0.00 on GPT-2 small, and with members clamped at their natural values it is about half (0.52–0.55 and 0.44–0.46). Pythia-70m's unweighted circuits pass the donor test for 6.3% of targets. On TuringLLM the donor test falls with depth and is hardest for attention targets: 81.3%, 77.4%, and 48.9% of targets pass it at layers 0–4, 5–7, and 8–11 (89.6%, 84.8%, and 59.4% of the production passes), and 53.2%, 77.2%, and 73.4% of attention, MLP, and residual targets. On the 173 scored sweep targets, the donor pass rate falls as the penalty grows, from 82–83% at $\lambda \leq 10^{-4}$ to 65.9% at the production 10^{-3} and 43.4% at 4×10^{-3}, while unweighted circuits pass it for at most 8.7% at any penalty, as on Pythia-70m. The strict resample test of the first paragraph was rerun on the public models' null panels only: 66.7% and 43.0% of the targets pass it, against 82.5% and 82.8% under the pass criterion, 4.8% and 1.6% at $\alpha = 1$, and none for random circuits; for TuringLLM it is not available in this run.

An arm trained without the resample term, at the production penalty and otherwise identical,

Model	Test	Circuits	Production pass (%)	Test pass (%) [95% CI]	Test pass | production pass (%)	A / C / T (median)
TuringLLM	donor test (WCM)	7,464	88.0 [87.2, 88.7]	69.2 [68.1, 70.2]	78.6 [77.7, 79.6]	0.95 / 0.86 / 0.75
	donor test (sweep: WCM, $\lambda = 2.5 \times 10^{-5}$)	173	91.3 [87.3, 95.4]	82.1 [76.3, 87.3]	89.9 [84.8, 94.3]	0.94 / 0.89 / 0.76
	donor test (sweep: WCM, $\lambda = 5 \times 10^{-5}$)	173	93.1 [89.0, 96.5]	82.7 [76.9, 87.9]	88.8 [83.9, 93.2]	0.96 / 0.90 / 0.78
	donor test (sweep: WCM, $\lambda = 10^{-4}$)	173	90.2 [85.5, 94.2]	82.1 [76.3, 87.3]	91.0 [85.9, 94.9]	0.97 / 0.90 / 0.78
	donor test (sweep: WCM, $\lambda = 2.5 \times 10^{-4}$)	173	91.3 [87.3, 95.4]	78.0 [71.7, 83.8]	85.4 [79.7, 90.5]	0.97 / 0.90 / 0.78
	donor test (sweep: WCM, $\lambda = 5 \times 10^{-4}$)	173	89.0 [84.4, 93.1]	73.4 [67.1, 79.8]	82.5 [76.0, 88.3]	0.96 / 0.88 / 0.76
	donor test (sweep: WCM, $\lambda = 10^{-3}$)	173	83.2 [77.5, 88.4]	65.9 [59.0, 72.8]	79.2 [72.9, 85.4]	0.96 / 0.86 / 0.74
	donor test (sweep: WCM, $\lambda = 2 \times 10^{-3}$)	173	77.5 [71.1, 83.2]	57.2 [49.7, 64.7]	73.9 [66.4, 81.3]	0.95 / 0.83 / 0.71
	donor test (sweep: WCM, $\lambda = 4 \times 10^{-3}$)	173	65.9 [59.0, 72.8]	43.4 [35.8, 50.9]	65.8 [57.0, 73.7]	0.91 / 0.79 / 0.67
	donor test (sweep: WCM without R, $\lambda = 10^{-3}$)	173	78.0 [71.7, 83.8]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.12 / 0.02 / 0.01
	donor test (sweep: unweighted, $\lambda = 10^{-5}$)	173	75.1 [68.8, 81.5]	8.7 [4.6, 13.3]	11.5 [6.2, 16.9]	0.36 / 0.43 / 0.18
	donor test (sweep: unweighted, $\lambda = 3 \times 10^{-5}$)	173	69.4 [62.4, 76.3]	6.9 [3.5, 11.0]	10.0 [5.0, 15.8]	0.24 / 0.31 / 0.13
	donor test (sweep: unweighted, $\lambda = 10^{-4}$)	173	72.8 [66.5, 79.2]	5.2 [2.3, 8.7]	7.1 [3.2, 11.9]	0.14 / 0.18 / 0.06
	donor test (sweep: unweighted, $\lambda = 3 \times 10^{-4}$)	173	59.5 [52.0, 67.1]	3.5 [1.2, 6.4]	5.8 [1.9, 10.7]	0.04 / 0.06 / 0.03
	donor test (sweep: unweighted, $\lambda = 10^{-3}$)	173	42.2 [34.7, 49.7]	1.7 [0.0, 4.0]	4.1 [0.0, 9.6]	0.01 / 0.01 / 0.00
Pythia-70m	strict resample (A, C)	126	82.5 [75.4, 88.9]	66.7 [57.9, 74.6]	80.8 [73.1, 87.5]	0.90 / 0.88 / 0.71
	strict resample, random null	126	—	0.0 [0.0, 0.0]	—	—
	donor test (WCM)	508	81.9 [78.5, 85.2]	66.1 [62.0, 70.3]	80.8 [76.9, 84.4]	0.91 / 0.86 / 0.69
	donor test (unweighted, $\lambda = 10^{-3}$)	127	59.8 [51.2, 68.5]	6.3 [2.4, 11.0]	10.5 [3.9, 18.4]	0.20 / 0.24 / 0.06
GPT-2 small	strict resample (A, C)	128	82.8 [75.8, 89.1]	43.0 [34.4, 51.6]	51.9 [42.5, 61.3]	0.79 / 0.75 / 0.66
	strict resample, random null	128	—	0.0 [0.0, 0.0]	—	—
	donor test (WCM)	511	82.4 [79.1, 85.5]	41.1 [36.8, 45.4]	49.9 [45.1, 54.9]	0.79 / 0.74 / 0.64

Table 22: Real backgrounds are harder than means, more so on GPT-2 small; random circuits and the same nodes at $\alpha = 1$ almost never survive them. The production circuits under real backgrounds: the strict resample test on the public null panels and the donor-background test on every main circuit and, for TuringLLM, on every arm of the sweep (Table 4), with 95% bootstrap intervals; median recovery under activating (A), held-out contrast (C), and random-text (T) donors. TuringLLM's strict test was not run.

Depth	Targets	Pass, v4 (%)	Pass, without R (%)	Difference (pp)	Size v4 / without R
all	173	83.2 [77.5, 88.4]	78.0 [71.7, 83.8]	+5.2 [-0.6, +11.6]	573 / 493
layers 0-4	69	88.4 [79.7, 95.7]	84.1 [75.4, 92.8]	+4.3 [-4.3, +14.5]	396 / 249
layers 5-7	45	84.4 [73.3, 93.3]	82.2 [71.1, 93.3]	+2.2 [-11.1, +15.6]	743 / 553
layers 8-11	59	76.3 [64.4, 86.4]	67.8 [55.9, 79.7]	+8.5 [-1.7, +18.6]	937 / 686

Table 23: On the pass criterion the resample term may help: the point estimate is largest at depth, but no band's interval excludes zero on 173 targets. Under real backgrounds it is decisive (none of the arm without it passes the donor-background test; Table 22). The production arm against an arm trained without the resample term, at $\lambda = 10^{-3}$ and otherwise identical, paired on the scored TuringLLM sweep targets: pass rate with 95% bootstrap interval, the paired difference, and median circuit sizes, overall and by depth band.

isolates it on the 190 sweep targets (173 scored): with the term, 83.2% meet the pass criterion (77.5–88.4), and without it 78.0% (71.7–83.8), a paired difference of 5.2 points (−0.6 to 11.6), with the largest point estimate at layers 8–11 (8.5 points, −1.7 to 18.6); no band's interval excludes zero; circuits trained with the term are larger (573 against 493 latents) (Table 23). Under real backgrounds the term is decisive: with it 65.9% of these targets pass the donor-background test (59.0–72.8), and without it none do, with median recovery 0.12 and 0.02 under activating and contrast donors against 0.96 and 0.86 (Table 22).

The pass rule of the main text uses zero ablation and the two mean ablations only.

M Case Studies

How the cases were found. In the protocol-v1 run, we read passing circuits for targets at layer 6 or deeper (2,238 circuits) with tools that list, for each circuit, its members ranked by contribution at the target's anchor (α times the member's mean activation there times its decoder norm), each member's own top contexts and peak tokens, its engagement on the target's activating against contrast contexts, and its first-order logit effect, and that run hand-written sentences through the clean model. Candidates were kept when a minimal pair of sentences separated the target cleanly. In the production run, a token-driven topic scan read the strongest and contrast contexts of all 7,786 fitted TuringLLM

targets and counted physics, AI, and computing terms; passing targets at layer 5 or deeper, with most strongest contexts on one domain and an anchor that is not simply a domain word, were read by hand (about 75), and 11 were probed with in-domain, near-domain, and same-word-other-sense sentences, which found the fusion circuit. Group names come from the members' own contexts and logit effects, not from automatic interpretation.

Validation. Each case is tested three ways. *Knock-outs*: the circuit is re-scored with one named group removed (its members ablated like any non-member), on the held-out strongest contexts with the production scorer, against the same number of members of similar contribution rank (±15 ranks, excluding every named member and a generic core of latents that fire on most text), over 8 random draws. *Probes*: the target's activation in the clean model on hand-written sentences, as a percentage of its mean activation on its held-out strongest contexts. *Specificity*: the diagnostic of Appendix K. Table 24 summarises seven cases; the remaining tables are in the repository (`experiments/063-case-studies` and `experiments/110-case-studies-v4`).

More on the antiquity circuit. The production circuit shares only 390 latents with this target's protocol-v1 circuit (Jaccard 0.30), but these carry 70% of its contribution, and 34 of the earlier circuit's 40 largest contributors are still members. Run alone under contrast-mean ablation, the production circuit also keeps ancient Greece and Athens (127%, 123%) above Rome (99%) and Egypt (72%), while the 19th century and 1950 read 20% and 15%, and "The BC Lions" reads 125%. Two further measurements qualify Section 4.4. First, the earlier protocol-v1 circuit of this target (728 latents) put more weight on the classical-Greece group, whose removal alone took it to 0.29, 0.22, and 0.32; in the production circuit the same latents share that role with the early-civilisation group, which now feeds them. The readable core is stable as a set of latents across the two fits, but how the fitted coefficients divide the work among them is not. Second, the generic core was needed only under zero ablation in the earlier circuit; in the production circuit it is also needed under activating-mean ablation (A 0.27).

A circuit for magnetic-confinement fusion. The layer-9 residual-stream latent 9.resid.13945 marks magnetic-confinement fusion: on hand-written sentences it reaches 69–103% of its typical activation for tokamaks and stellarators, 68% for inertial confinement, 22% for a fission reactor, and 13% for fusion in the Sun, and 0–10% for MRI scanners, motors, and transformers, which also use strong magnets; sentences with the same wording about a bakery or an architect ("uses a combination of", "employs a complex arrangement of") give 0–6%. Its production circuit holds 1,322 latents and passes (Z, A, C 0.92, 1.11, 0.99; necessity 1.00; 0.91, 1.09, 0.96 when re-scored on the second GPU used for the knock-outs). Its readable core is a physics concept graph (Figure 21): ten latents at layers 5–9 for plasma confinement (tokamaks, stellarators, magnetic confinement for fusion), 9.1% of the contribution and 0.8% of the circuit, feed six for nuclear and power engineering (reactor types and coolants, nuclear reactions, turbines and heat-engine cycles, the power grid; edge 0.28) and three for stars and particles (edge 0.13), alongside five for quantum physics and magnetism. Removing the ten plasma-confinement latents takes the target to 0.17, −0.01, and 0.12 under Z, A, and C (random members of similar rank: 0.79, 0.99, 0.89), and all four physics groups (24 latents) to 0.07, −0.05, and 0.05 (0.73, 0.93, 0.85); the nuclear and power group alone matters less (0.69, 0.71, 0.71 against 0.82, 1.02, 0.90), and the plasma and nuclear groups lower the target, in total, by 0.59 and 0.21. The star and quantum groups are no more necessary than random members, nor is a control group of seven off-topic members (databases, intercultural communication, traffic, oceanography).

The circuit is also an amplifier, more strongly than any other case here. Under activating-mean ablation it lifts 12,080 latents above the Top-K cut, about 24 times the production run's median of 497, and 7,151 under contrast-mean ablation (median 488), and the site's other active latents are restored to 2.98 and 1.80 times their activation (siblings 2.40 and 1.44; Appendix K). Run alone under contrast-mean ablation, where the empty circuit gives 0, it keeps the order of the clean model but not its sharpness: tokamak and stellarator sentences reach 104–115%, inertial confinement 94%, the Sun 66%, and a fission reactor 45%, but MRI, motor, and transformer sentences also reach 22–53% and

the same-wording bakery and architect sentences 49–56%. So the ten plasma-confinement latents carry what makes this target fire, while the circuit as fitted also raises much of the site around it, as the law circuit in Table 24 does under protocol v1 (and the pivot circuit 9.resid.7640, which lifts 15–21k latents).

What the hunt did not find. No latent we examined tracked whether a stated fact was correct: knowledge appears as latents for entities, eras, and domains, and in the sentences we tried, a wrong date or a false definition activated them as much as the true one, or more.

N Circuits for Randomly Drawn Targets

The case studies were chosen; the circuits in Table 25 were not. We drew one target uniformly at random from each depth band and site kind of the protocol-v1 run (Appendix H; seed 20261012, non-vacuous circuits, any outcome) and list each circuit's scores and its three largest members by contribution share at the target's anchor, excluding generic hub latents. Four of the nine pass, close to the run's 46%. The labels are hypotheses written by a language model from each latent's own strongest contexts and logit effect, not verified readings: the target labels come from the run-wide circuit descriptions, and each member was labelled from its own evidence alone.

The shallow token detector `0.mlp.12557` ("control") rests largely on two members that detect the same token, with 8.5% and 7.5% of the contribution, while the deep circuits spread over many members with 1–4% each that carry topic and construction (`8.mlp.2453`, `9.resid.10786`). The failing attention target `5.attn.10168` is near-threshold and reads 0 under every ablation method, as Appendix H describes. Five of the six circuits at layers 5–11 are flagged as amplifiers, against 30% over the run (Appendix K); nine draws are too few to read this as a trend with depth.

Target	What it computes	Minimal pair (clean model)	Story groups removed: Z / A / C (random)	Caveat
9.resid.20419 (Section 4.4)	a date or place read as classical Greek antiquity	"In the 5th century BC, the philosopher . . . " 79%; "In the 19th century, . . . " 0%	0.22/0.10/0.21 (0.54/0.69/0.68)	"BC" is partly lexical ("The BC Lions" 45%)
9.resid.13945 (Figure 21)	magnetic-confinement fusion (tokamak, stellarator)	tokamak / stellarator 69–103%; MRI, motor, transformer 0–10%; same wording about a bakery or an architect 0–6%	0.07/−0.05/0.05 (0.73/0.93/0.85)	amplifier: lifts ∼12,000 latents; run alone, same-wording controls ∼50%
10.resid.15497	the possessive after a name that carries a law, before "law" appears	"Snell's" 94%, "Ohm's" 80%; "Darwin's", "Picasso's" 0%; invented "Zorbel's" 76% in electromagnetism, 0% in painting	0.00/−0.12/0.03 (0.54/0.92/0.64)	amplifier: lifts ∼7,900 latents, target rank 1 → 4; also promotes "Law" and "law" directly (4.2 nats at peak)
8.resid.31095	"but also" when a sentence attributes causes	"influenced not only by memory but also" 89%; "involves not only memory but also" 0%	0.19/0.41/0.28 (0.73/0.93/0.81)	one latent alone gives most of the drop
8.attn.29991	acid–base dissociation	acid–base sentences 46–87%; other chemistry 0–16%	0.25/0.11/0.16 (0.61/1.19/0.80)	"Her acid wit" 38%
9.resid.7640	the pivot in "not X, but rather Y"	negated clause, then pivot 37–124%; same comma without negation 0–13%	0.00/0.03/0.00 (0.42/0.80/0.39)	fragile: random removals also hurt; lifts 15–21k latents
9.resid.17596	the second "as" of a comparison about importance	"as important for health as" 79%; "as fast as" 0%	importance group no better than random	not validated; lifts ∼30k latents

Table 24: Seven case studies from the hunt. The first two rows are production-run circuits; the other five are circuits of the protocol-v1 run (Appendix H). Knock-outs remove the case's named story groups together and report faithfulness under zero (Z), activating-mean (A), and contrast-mean (C) ablation; in brackets, the mean over 8 draws of the same number of random members of similar contribution rank. In every validated case, topical control groups (historical period, off-topic members, generic possessive) are no worse than random; the generic core of latents that fire on most text is not a control in this sense, since removing it breaks the circuit (antiquity 0.00/0.27/0.61, fusion 0.00/0.73/0.74, against random 0.70/0.91/0.84 and 0.83/1.05/0.92).

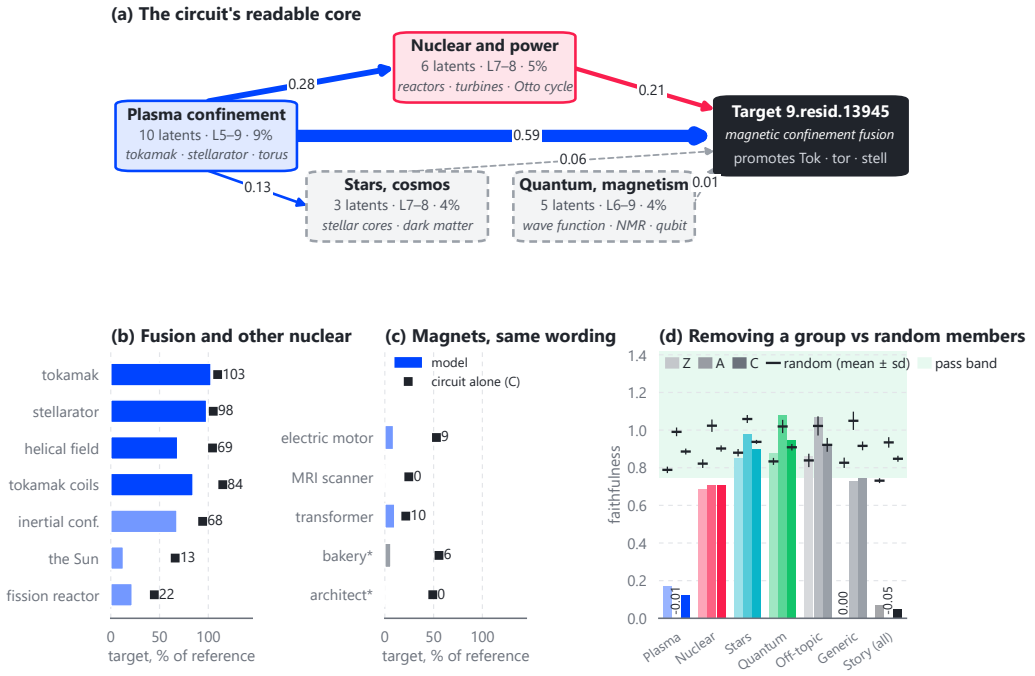


Figure 21: A readable core inside an amplifier: the production-run circuit of a latent for magnetic-confinement fusion (layer 9, residual stream, 1,322 latents). **(a)** The readable core: role groups among the circuit's largest contributors, named from their latents' own top contexts, with each group's size, layers, and share of the circuit's contribution at the target's anchor. Arrows give the total drop in the destination (or the target) when the source group is removed, with all other members recomputed (activating-mean ablation, circuit alone, held-out strongest contexts): plasma confinement lowers nuclear and power by 0.28 and stars by 0.13; the drops in the target are 0.59 (plasma confinement), 0.21 (nuclear and power), 0.06 (stars, dashed), and 0.01 (quantum). Dashed grey groups (stars, quantum) are no more necessary than random members; the off-topic (7 latents) and generic (7) control groups appear only in (d). The target box lists tokens it promotes ("Tok", "tor", "stell"). **(b,c)** The target on hand-written sentences as a percentage of its mean activation on its held-out strongest contexts: (b) fusion and other nuclear sentences, (c) devices with the same mechanics (magnets, coils) and same-wording controls (bakery, architect; marked *). Bars and their number labels are the unmodified model; squares are the circuit run alone under contrast-mean ablation (the empty circuit gives 0). Run alone, the circuit reads ~50–55% on the motor, bakery, and architect sentences, where the model gives 0–9%: the amplifier effect. **(d)** Faithfulness with each of the seven groups removed, including the off-topic and generic controls, against removing the same number of random members of similar contribution rank (mean over 8 draws); shaded, the pass band [0.75, 1.5].

Target	Target label	Size	Z / A / C	Nec.	Ind.	Outcome	Top three members (label)
3.attn.9767	Token following "Capital" in finance terms	257	0.97 / 1.10 / 1.16	1.00	0.58	pass	1.resid.18903: Word-start fragments in Title Case (unclear) 0.resid.4768: Title-case word-start fragment (unclear) 0.mlp.4768: Title-case word-start fragments (unclear)
0.mlp.12557	Control word detector	26	0.88 / 0.95 / 0.99	0.72	0.39	fail	0.attn.16913: Token "control" 0.attn.5256: Token "Control" 0.attn.19725: Title-case word starts (unclear)
0.resid.27712	Cell division terms (chrom, pol, II)	102	0.86 / 0.66 / 0.81	1.00	1.24	fail	0.mlp.205: 'chrom' word-start token 0.mlp.4393: "fect" in "Perfect" tense headings 0.mlp.21122: Arabic mid-word letters (unclear)
5.attn.10168	'men' in "menstrual"	274	0.00 / 0.00 / 0.00	1.00	0.00	fail^{t}	4.resid.19412: Mid-word fragments, unclear 3.resid.12927: Word after "the" (unclear) 3.resid.18667: Mixed tokens, unclear (further/later)
5.mlp.15947	"not limited to" negated limitation	543	0.99 / 1.19 / 1.19	1.00	0.76	pass^{a}	4.resid.21357: Token "limited" 4.resid.25898: "not limited to" phrase 4.resid.26922: Negated verb "do not contain/have"
7.resid.29770	food token detector	721	0.96 / 0.87 / 1.27	1.00	1.24	pass^{a}	6.resid.32006: "red" in ingredient (food prep) 5.resid.3962: Grill/barbecue word fragments (food) 5.resid.17590: Token "food" before noun
9.attn.38891	Connectors in lists of consequences	613	0.88 / – / 0.11	1.00	1.21	fail^{a}	8.resid.33386: Predicts negative-consequence words 7.resid.39521: Subject before past-tense verb 7.attn.15390: Humanism/Renaissance context function words
8.mlp.2453	Structure nouns followed by that/can	870	0.92 / 0.68 / 0.41	1.00	1.46	fail^{a}	8.attn.4808: Engineering-context function words (unclear) 7.attn.27777: Composite materials engineering context (unclear) 7.attn.18401: Plural subject, predicts plural verb
9.resid.10786	"global" as modifier	920	0.92 / 1.01 / 1.08	1.00	1.30	pass^{a}	8.resid.30012: Capitalized "Global" at title start 8.resid.26956: Growing awareness → environment 7.resid.11194: Pre-"international/global" context words

Table 25: Circuits for nine randomly drawn targets of the protocol-v1 run, one per depth band (rows grouped L0–4, L5–7, L8–11) and site kind, seed 20261012. Size is the number of upstream latents; Z, A, C are faithfulness under the three ablation methods, Nec. necessity, and Ind. sufficiency to induce with coefficients, all on held-out contexts; "–" is an undefined score, which counts as a fail. ^{a} flagged as an amplifier; ^{t} near-threshold target. Labels are hypotheses written by Claude Sonnet 5.5 from each latent's own strongest contexts and logit effect; members are ranked by contribution share at the target's anchor, generic hub latents excluded.

O Extended Related Work

Discovery methods and their settings. Each family of discovery methods is built for its own setting. ACDC (Conmy et al., 2023) removes edges of a model's computational graph one at a time while an output metric holds, and Edge Pruning (Bhaskar et al., 2024) optimises an edge mask by gradient descent so that the pruned model stays faithful to the full model's predictions; both work over components such as heads and MLPs, and both are anchored at an output rather than a latent. Attribution patching (Nanda, 2023; Syed et al., 2023) replaces each activation patch with a first-order estimate, AtP* (Kramár et al., 2024) adds two changes that address the two classes of failure behind its false negatives, and EAP-IG (Hanna et al., 2024) scores edges by integrating gradients along a path from corrupted to clean input embeddings. The path-attribution baseline of Appendix C, whose comparison with WCM is pending, adapts this estimator to nodes, integrating jointly over the upstream SAE sites from their mean-ablation values. Transcoders (Dunefsky et al., 2024) replace MLPs with sparse, wider layers whose circuits factorise into input-dependent and input-invariant terms, attribution graphs (Ameisen et al., 2025; Lindsey et al., 2025) build one graph per prompt over cross-layer transcoder features, and PIE (Chen et al., 2026a) prunes cross-layer transcoder features by their gradient-weighted write contributions before interpreting them. The transcoder-based methods need a substrate our SAE bank does not provide, and produce a graph per prompt where we fit one circuit over many contexts, so we do not run them here.

Masks, scalars, and ablations. Learned masks for circuit discovery descend from subnetwork probing (Cao et al., 2021), applied to computational-graph vertices by Conmy et al. (2023) and to edges by Edge Pruning (Bhaskar et al., 2024) and by uniform gradient sampling with optimal ablation (Li and Janson, 2024), on L_0-gate and continuous-sparsification machinery (Louizos et al., 2018; Savarese et al., 2020). Masks over SAE latents have been learned to localise attributes (Chaudhary and Geiger, 2024), a procedure adopted by benchmarks (Mueller et al., 2025; Karvonen et al., 2025), and to find behavioural circuits by continuous sparsification, with a mask per token position and a single per-token mean ablation (Caples et al., 2025); with its coefficients fixed at 1, its objective moved to a single latent, one mask shared across positions, and four training ablations (zero, two means, and a resample ablation over contrast contexts) with the sparsity-preserving fill and rank-keep in place of one mean ablation, the latter is our unweighted circuit masking. Every such mask keeps a node at its natural scale or ablates it. The continuous coefficients of Li and Janson (2024) interpolate between the two and play the role of our gates; what they learn besides is a constant for each *removed* component, the dual of our scale for each kept one. That constant minimises the ablated model's expected loss, so a component's importance is what it carries beyond the best constant; we keep the ablations fixed and defined by data (zero, the means over activating and over contrast contexts, and real contrast contexts), so that what absence means is set before training rather than fitted with the circuit.

Learned per-unit scales appear with other jobs: scales that double as the pruning criterion (Liu et al., 2017; Huang and Wang, 2018; You et al., 2019); scales for the units kept after pruning, fitted after the mask is fixed (Kwon et al., 2022; He et al., 2017); steering multipliers (Stoehr et al., 2024); sparse fitted weights on SAE latents for a task vector (Kharlapenko et al., 2025); regression weights on feature-to-feature edges (Yin et al., 2026); and, after pruning, a fitted logit scale and shift together with manual rescaling of a few nodes (Gao et al., 2025). Gao et al. (2025) warn that rescaling nodes can hide superposition, letting a few rescaled nodes stand in for many; our random circuits receive the same coefficient fitting as the circuits they control for, which tests whether rescaling alone makes a set faithful (Appendix I). CircuitLasso (Yin et al., 2026), run among others on GPT-2 small with OpenAI's sparse autoencoders, the dictionary family of our public run, links features one layer pair at a time by an observational sparse regression, whose coefficients its authors note are not causal effect sizes, and scores its circuits on training data; we do not run it, since our question is which latents suffice under intervention on held-out contexts, though its coefficients could rank members for the comparison of Appendix D. Gate-times-magnitude parametrisations separate membership from strength in sparse training and fine-tuning (Louizos et al., 2018; Guo et al., 2021; Lai et al., 2025), where a closed gate means no change or zero rather than an ablated node. Feature-preserving subnetworks have also been pruned by saliency in vision models (Hamblin et al., 2022).

Circuits are evaluated under one or several ablations (Miller et al., 2024) but discovered under one; MIB compares discovery under counterfactual, mean, and optimal ablations while evaluating under counterfactual patching alone, and circuits discovered under the evaluation's own ablation score best (Mueller et al., 2025). The nearest objectives sum the clean-to-corrupted and corrupted-to-clean directions of one counterfactual (Chen et al., 2025), or certify a circuit against every patch value in a bounded set (Hadad et al., 2026). Training under several ablations is the circuit-level counterpart of attribution averaged over several baselines (Kapishnikov et al., 2019; Erion et al., 2021); since the production protocol, it includes a resample ablation over training contrast contexts, and 69%, 66%, and 41% of targets on the three models also pass the donor-background test, whose donors are never training donors (Appendix L).

Nulls, variance, and non-uniqueness. Random circuits as a null follow Shi et al. (2024), who test whether a candidate is more faithful than 90% of 100 random circuits built from random input-to-output paths, with the random circuits' size varied from the candidate's own up to most of the model, and CircuitKIT (Seth et al., 2026), which compares a circuit with size-matched random subgraphs by an improvement ratio and a z-score. In both, the random circuits are sampled and ablated but not fitted, which matches circuits that are not fitted either. Our circuits fit a coefficient per node, and fitting alone might rescale an arbitrary set into reproducing a target, so our random circuits receive the same coefficient fitting before scoring (Section 3.3); with every coefficient at 1 they reduce to the nothing-fitted null. The broader concern is the one Méloux et al. (2025a) raise: interpretability methods produce plausible explanations even for randomly initialised networks, so findings are better treated as statistical estimates tested against alternative hypotheses. Causal scrubbing (Chan et al., 2022) is a stricter test in this spirit, resampling everything outside the hypothesis from other inputs (Appendix L).

Circuit discovery is also variable. EAP-IG circuits vary with the data batch and the prompt template, and their faithfulness varies widely across individual examples (Wu et al., 2026); on one task, defensible analytic choices change the derived claim for most pairs of specifications (Mahale, 2026); and Certified Circuits stabilise membership by randomised subsampling of the data, keeping only components included consistently (Anani et al., 2026). On small networks where every circuit can be enumerated, many circuits reproduce the same behaviour exactly (Méloux et al., 2025b), and in language models structurally distinct, low-overlap circuits support the same task (Chen et al., 2026b), and circuits for one task found on different inputs differ in structure, yet a shared core recovers their performance (Makou et al., 2026). Our node sets are non-identifiable in the same way, which is why a circuit's function, not its exact member list, is what we report as the finding (Appendix P).

Self-repair and inhibition. Ablation-based tests meet compensation: when one attention layer is ablated, later layers make up part of its effect (McGrath et al., 2023); the self-repair of single heads persists over the full training distribution, partial and variable across prompts, carried in part by the final LayerNorm scale and by sparse anti-erasure neurons (Rushing and Nanda, 2024); and copy-suppression heads, which lower the logit of a token that earlier layers predict when it already appears in the context, account for 39% of the self-repair that follows ablating the name-mover heads of the IOI circuit (McDougall et al., 2024). Conditional co-ablation finds backup components that become important only once a primary set is removed (Gong et al., 2026). Our scores intervene on the circuit as one set (Section 3.3), so they measure what the set does with any compensation inside the model included; necessity in particular is scored with the complement recomputed, so backups outside the circuit are free to respond, while faithfulness training holds the complement at its ablation value and so never rewards including dormant backups; per-member roles, where backups matter most, are not claimed.

Position-aware circuits. Position handling divides prior work into three strategies, each right for its setting: scores summed over positions, with nodes kept per position as an option of the released code (Marks et al., 2025), per-instance graphs built one prompt at a time (Ameisen et al., 2025), and schema-aligned positions across variable-length examples (Haklay et al., 2025). WCM needs none of

these commitments: one mask is shared across positions and nodes are recomputed at every position, which suits the setting none of the three targets, one reusable circuit over arbitrary non-templatic text; our attribution baselines instead select per position and take the union over the causal prefix (Appendix C.2).

Contrast contexts. Retrieved non-activating examples have been used to falsify feature explanations: Ma et al. (2025) select the sentences most similar to a feature's top activations and keep those on which it is zero at every token, and EleutherAI's `delphi` library draws nearest-neighbour hard negatives per activating example for explanation scoring (EleutherAI, 2024b). Explanation scores collapse to chance when negatives come from neighbouring features (EleutherAI, 2024a), convergent with our finding that close contrast contexts generalise worst to other backgrounds (Appendix J). Retrieval-based corruptions have also been used for circuit discovery on a behaviour, as the nearest non-memorised samples in the model's own embedding space (Lasy et al., 2025), and O'Neill and Bui (2024) design corrupted negatives for task-level circuit identification. We use retrieved, silence-checked contexts per latent as the background a circuit is trained and evaluated against (the C mean, the resample donors, and the held-out set for sufficiency to induce), rather than for explanation scoring or as one paired corruption, and treat their construction as a controlled variable with measured consequences.

The SAE basis. Static structure among latents is well documented: co-occurrence and decoder geometry reveal modularity (Li et al., 2025), coactivation-derived components steer behaviour (Deng et al., 2026), meta-decompositions show latents are not atomic (Leask et al., 2025), cross-layer feature-flow graphs track latents through the forward pass (Laptev et al., 2025), though decoder similarity misses most ablation-validated feature transitions through an MLP layer (Droste et al., 2026), and Jacobian SAEs train adjacent dictionaries for sparse computation between them (Farnik et al., 2025). But that structure is correlational, geometric, or weight-based, where the relationships discovered here are context-conditioned and intervention-tested. Weight-based connections need that test: in a one-layer model expanded into virtual weights between transcoder features, many large weights are *interference weights*, irrelevant or harmful to behaviour, and raw weight magnitude separates them poorly from the weights that matter (Turner et al., 2026). Intervention-based latent graphs have been mapped at scale in single-cell foundation models, by ablating one SAE feature at a time and measuring the change in downstream features, finding that most edges are ones whose removal lowers the downstream feature (termed inhibitory there) (Kendiukhov, 2026a) and a heavy-tailed hub structure with redundancy that grows with interaction order (Kendiukhov, 2026b); those graphs measure the effect of one feature at a time, where our circuits are fitted jointly for each target and tested as a set. The basis itself is imperfect: reconstruction errors are pathological (Gurnee, 2024), although much of the error is linearly predictable from the activation (Engels et al., 2025); SAEs trained from different seeds learn largely different features, Top-K ones most of all (Paulo and Belrose, 2025); downstream utility is mixed (Smith et al., 2025), and in causal-variable localisation SAE features are no better than neurons (Mueller et al., 2025); and units are not canonical (Leask et al., 2025). Interventions on the basis are imperfect too: an optimised perturbation can recover a behaviour suppressed by clamping SAE features, through the reconstruction residual and with the features still clamped (Cui et al., 2026); the effect of ablating single-token features on their token's logit differs more across SAE families than across model scales within a family (Cho et al., 2026); and dictionaries trained only on what an affine map from an earlier layer cannot explain recover more of the loss when several layers are replaced at once, at most sparsity levels (Poduval et al., 2026). This conditions our claims: reconstruction residuals stay as explicit anchors, evaluation intervenes on measured activations, and the SAE basis is treated as a useful but imperfect sample of the model's internal variables.

P Node Sets Are Non-Identifiable; Pass Rates Are Stable

The masking engine is bit-deterministic at fixed settings on one GPU type (two identical runs agree exactly, with *zero* node flips), so small perturbations are controlled experiments on the objective, and

Model	Group	Targets	Jaccard (median)	Pass fit 1 (%)	Pass fit 2 (%)	Pass core (%) [95% CI]	Core passes / both pass
TuringLLM	L0-4	47	0.55	87.2	89.4	83.0 [72.3, 93.6]	38 / 40
	L5-7	33	0.42	97.0	100.0	84.8 [72.7, 97.0]	27 / 32
	L8-11	43	0.31	81.4	79.1	30.2 [16.3, 44.2]	12 / 30
	all	123	0.42	87.8	88.6	65.0 [56.9, 73.2]	77 / 102
	residual	43	0.42	97.7	93.0	79.1 [67.4, 90.7]	34 / 40
	MLP	46	0.46	82.6	87.0	63.0 [50.0, 76.1]	28 / 36
	attention	34	0.34	82.4	85.3	50.0 [32.4, 67.6]	15 / 26
Pythia-70m	L0-1	8	0.53	75.0	62.5	75.0 [37.5, 100.0]	5 / 5
	L2-3	12	0.47	66.7	66.7	66.7 [41.7, 91.7]	7 / 7
	L4-5	12	0.37	66.7	83.3	75.0 [50.0, 100.0]	8 / 8
	all	32	0.51	68.8	71.9	71.9 [56.2, 87.5]	20 / 20
	residual	12	0.50	75.0	66.7	75.0 [50.0, 100.0]	7 / 7
	MLP	10	0.52	90.0	90.0	90.0 [70.0, 100.0]	9 / 9
	attention	10	0.31	40.0	60.0	50.0 [20.0, 80.0]	4 / 4
GPT-2 small	L0-4	14	0.45	78.6	71.4	57.1 [28.6, 85.7]	8 / 9
	L5-7	7	0.32	100.0	85.7	57.1 [14.3, 85.7]	4 / 6
	L8-11	11	0.27	72.7	63.6	18.2 [0.0, 45.5]	2 / 6
	all	32	0.39	81.2	71.9	43.8 [28.1, 62.5]	14 / 21
	residual	10	0.43	90.0	70.0	50.0 [20.0, 80.0]	5 / 7
	residual (mid)	6	0.47	100.0	66.7	50.0 [16.7, 83.3]	3 / 4
	MLP	9	0.36	66.7	77.8	33.3 [11.1, 66.7]	3 / 5
	attention	7	0.22	71.4	71.4	42.9 [14.3, 85.7]	3 / 5

Table 26: The stable core by depth band and site kind: targets, median Jaccard between the original fit and the reordered refit, pass rates of both fits and of the core alone (coefficients refitted on the core) with 95% bootstrap intervals, and the number of targets whose core passes among those where both fits pass.

the ones we tried gave the same verdict. In exploratory probes, five on unweighted circuit masking and two on WCM, each perturbation left faithfulness roughly unchanged but exchanged nodes. A float32 and a bfloat16 run differ only in rounding, yet exchange 16.5% of nodes at layer 8 (Jaccard 0.835) at unchanged faithfulness and size. Strengthening the sparsity penalty swaps nodes rather than nesting them (containment 0.49–0.80 between consecutive steps); no gate ever freezes, so the stopping step is another sampling axis; 8–25% of nodes sit within 0.05 of the decision cut and are load-bearing; and short-budget and long-budget runs at matched size share only 0.40–0.53 of their nodes. Recomputing the mean-ablation values without a quarter of their contexts, a change that leaves the pass rate and median faithfulness unchanged, moves node sets as far as a change of hardware (median Jaccard 0.66). The effect holds at scale: under protocol v1, the same settings on a different GPU type give node sets with a median Jaccard of 0.60 (0.32–0.96 over 32 targets), and a refit with reordered training contexts shares a median 64% of the original circuit's latents (below). A discovered node set is one sample from an equivalence class of near-optimal solutions to (P), so we report node sets as equivalence classes and let set-level and functional claims carry the conclusions, since overlap does not track faithfulness (Hanna et al., 2024) and circuits are in general not identifiable (Méloux et al., 2025b,a).

A stable core at scale. In the production run we refitted 126 TuringLLM targets (123 scored) and 32 per public model with the training contexts in a different fixed order and otherwise unchanged settings, and scored the core, the latents in both fits, with its coefficients refitted on the core alone (Table 27). The two fits pass about equally often on TuringLLM and Pythia-70m (87.8% and 88.6%, 68.8% and 71.9%) and less so on GPT-2 small (81.3% and 71.9%, on 32 targets) but share a median Jaccard of only 0.42, 0.51, and 0.39; the refit is smaller (TuringLLM median 564 against 665 latents), and the core holds a median 341, 146, and 178 latents, 54–60% of the original circuit. The core alone passes for 65.0% (80 of 123), 71.9% (23 of 32), and 43.8% (14 of 32) of targets, and, where both fits pass, for 77 of 102, 20 of 20, and 14 of 21. Which latents a circuit contains is thus partly an accident of optimisation, as in the earlier run below; the shared core, about half of the circuit, passes alone for most TuringLLM and Pythia-70m targets but for fewer than half on GPT-2 small. By depth band (Table 26), the core alone mostly fails at depth: on TuringLLM it passes for 83.0%, 84.8%, and 30.2% of targets at layers 0–4, 5–7, and 8–11 (12 of the 30 deep targets where both fits pass), and on GPT-2 small for 57.1%, 57.1%, and 18.2%; Pythia-70m, with 32 targets, shows no trend (75%, 67%, 75%). The two fits also overlap less at depth (TuringLLM median Jaccard 0.55, 0.42, and 0.31).

To measure this on the protocol-v1 run (Appendix H), we refitted 175 of its targets (5 per layer

Model	Targets	Jaccard (IQR)	Size fit 1 / fit 2 / core	Pass fit 1 (%)	Pass fit 2 (%)	Pass core (%)	Core passes / both pass
TuringLLM	123	0.42 (0.31–0.53)	665 / 564 / 341	87.8 [82.1, 93.5]	88.6 [82.9, 93.5]	65.0 [56.9, 73.2]	77 / 102
Pythia-70m	32	0.51 (0.36–0.54)	286 / 214 / 146	68.8 [53.1, 84.4]	71.9 [56.2, 87.5]	71.9 [56.2, 87.5]	20 / 20
GPT-2 small	32	0.39 (0.26–0.47)	401 / 332 / 178	81.2 [65.6, 93.8]	71.9 [56.2, 87.5]	43.8 [28.1, 62.5]	14 / 21

Table 27: Refits pass about as often on two of the three models but share half or fewer of their latents; the shared core alone passes for 44–72% of targets. Stability of circuit membership under a refit with reordered training contexts in the production run: Jaccard (median, interquartile range), median sizes of the two fits and of the core, and pass rates of fit 1, fit 2, and the core alone with refitted coefficients.

and site kind) under that run's protocol with one change: the training contexts visited in a different fixed order. The refit ran on a different GPU from that run's fit, and a refit is bitwise reproducible on the same GPU type but not across types, so the two fits differ by the reordering and by float-level differences in the hardware. The *core* of a target is the set of latents in both circuits; its coefficients are refitted on the core alone, as for the random baseline, and it is scored with the production scorer. Table 28 gives the result. The two fits share a median 64% of the original circuit's latents (Jaccard 0.54), less at depth; component-level EAP-IG circuits are more stable across data resamples (Jaccard 0.80–0.92; Seth et al., 2026), but choose from a candidate pool orders of magnitude smaller. The second fit is smaller (median 402 against 492 latents; smaller for 160 of the 175 targets) but passes about as often as the first (40% against 43%), and where both fits pass, the core alone passes on 48 of 56 targets, and its median necessity is 1.0 in every depth band. Which latents a circuit contains is therefore partly an accident of optimisation, while a core of about two thirds of it passes alone for most targets where both fits pass. A refit does not search for near-disjoint alternatives, which a search that penalises overlap with earlier circuits finds for output tasks (Chen et al., 2026b), so we have not tested them; the contribution-matched null, which excludes the production members, is a substitution test of that kind (Appendix I). The core is a two-fit counterpart, for latent targets, of earlier stable cores: structurally different extractions of an output circuit share a core that recovers its performance (Makou et al., 2026), and certified circuits keep only components selected consistently under subsampling of the data (Anani et al., 2026).

			median size			**pass rate (%)**		
Layers	**targets**	**Jaccard**	**fit 1**	**fit 2**	**core**	**fit 1**	**fit 2**	**core alone**
0–4	70	0.60	241	201	172	67.1	62.9	55.7
5–7	45	0.51	592	499	346	35.6	28.9	26.7
8–11	60	0.45	736	593	405	21.7	21.7	11.7
all	175	0.54	492	402	297	43.4	40.0	33.1

Table 28: Stability of circuit membership under a refit. Protocol-v1 run. Fit 1 is that run's circuit; fit 2 refits the same target with the training contexts in a different order, on a different GPU; the core is the latents of both, with coefficients refitted on the core alone. Jaccard and sizes are medians over targets; pass rates use the pass criterion on held-out strongest contexts.

From circuits to chains. A weighted circuit is one hop; because every circuit is indexed by its target, circuits can in principle be composed by splicing the circuit for an upstream latent into the circuits that use it and re-validating by intervening at the earliest nodes. Non-identifiability sets the boundary condition: each link is one sample from an equivalence class, so composition must manage classes of node sets, not single ones.

Q Pseudocode

Algorithm 1 summarises weighted circuit masking (Section 3.2) as run in the production run.

Algorithm 1 Weighted circuit masking, WCM[Z+C+A] with the resample term and rank-keep, as in the production run (one target f; Table 4)

Require: target f; activating contexts $\mathcal{P}_f$ (train on 32 strongest + 16 mid-band; hold out 16 + 16) and contrast contexts $\mathcal{N}_f$ (48 train / 16 held out); $\lambda = 10^{-3}$; $\gamma_{\mathrm{C}} = \gamma_{\mathrm{A}} = 0.25$; $\gamma_{\mathrm{R}} = 1$; $\gamma_{\mathrm{K}} = 10^{-1}$; steps $S = 400$

target $z \leftarrow$ natural pre-activation of f at the training anchors (no-grad forward); $\nu \leftarrow \mathrm{mean}(z^2)$; $\bar{z} \leftarrow$ its censored read

$r \leftarrow \min(K,$ rank of f among its site's other latents) at each training anchor, unmodified model

ablation means: C $\leftarrow$ site means over the 48 *training* contrast contexts; A $\leftarrow$ over the 48 strongest *training* activating contexts (the evaluation's means); donor codes $\leftarrow$ the 48 training contrast contexts

initialise $\theta_s = 4$, $\psi_s = \mathrm{softplus}^{-1}(1)$ for every upstream site s

for step $= 0, \ldots, S-1$ **do**

  $T \leftarrow 0.05^{\,\mathrm{step}/(S-1)}$; gates $m_s \leftarrow \sigma(\theta_s/T)$; coefficients $\alpha_s \leftarrow \mathrm{softplus}(\psi_s)$ ▷ geometric anneal $1 \to 0.05$

  learning rate: linear warm-up over the first $S/10$ steps to 3.15×0.05, then geometric decay

  for $\pi \in \{\mathrm{Z}, \mathrm{C}, \mathrm{A}\}$ **do** ▷ same θ, ψ under every ablation method

    forward with kept latents at $\alpha\times$ live value, closed latents at their π ablation value; for C and A, the sparsity-preserving fill, straight-through (Appendix S.1)

    $\mathcal{L}_\pi \leftarrow$ MSE of the censored read $\bar{z}_\pi$ against $\bar{z}$; $\mathcal{K}_\pi \leftarrow$ mean $\mathrm{relu}(\tau_\pi^{(r)} - \hat{a}_f)^2$

  end for

  forward with closed latents at a freshly drawn donor's sparse code (members kept, donor fills the Top-K); $\mathcal{L}_{\mathrm{R}} \leftarrow$ MSE of f's pre-activation against z

  AdamW step on $\left[\sum_\pi \gamma_\pi(\mathcal{L}_\pi + \gamma_{\mathrm{K}}\mathcal{K}_\pi) + \gamma_{\mathrm{R}}\mathcal{L}_{\mathrm{R}}\right]/\nu + \lambda \sum_{s,i}\left[\sigma(\theta_{s,i}) + (1-\sigma(\theta_{s,i}))\,|\alpha_{s,i}-1|\right]$ ▷ penalties use $\sigma(\theta)$, not $\sigma(\theta/T)$

end for

circuit $\leftarrow \{\,(i, \alpha_i) : m_i > \frac{1}{2}\,\}$; evaluate on the held-out contexts under Section 3.3

return circuit

R Context Stores and Contrast Contexts

R.1 Context Store Formalism

The collection pass assigns each latent a scored context set $C_f = \{((x,t), a_f(x,t)) : a_f(x,t) > 0\}$. The top-context store retains the K_{top} highest-scoring active contexts,

$$\text{TopCtx}_K(f) = \underset{S \subseteq C_f,\ |S| = K_{\text{top}}}{\arg\operatorname{top}} \sum_{((x,t),s) \in S} s,$$

approximating the upper tail of the conditional activation distribution. The mid-context store is a bounded reservoir sample from the sigma band,

$$\text{MidCtx}_M(f) \sim \text{ReservoirSample}_M\big(\{(x,t) : \mu_f + 0.5\,\sigma_f \le a_f(x,t) \le \mu_f + 1.5\,\sigma_f\}\big),$$

maintained without sorting every active occurrence. Mid contexts distinguish a latent's ordinary operating regime from its most vivid examples and provide probe data where extreme contexts would be too narrow.

Contrast-context retrieval. Each sequence x is represented by $r(x) \in \mathbb{R}^{1024}$, the final layer's residual stream mean-pooled over positions, captured once during collection. The attribution baselines use a retrieval store: for each target, the 64 sequences most similar, by exact cosine search, to the normalised centroid of its activating contexts, excluding those contexts, with no check for silence. Its index samples 2×10^5 sequences, so targets with no activating context in the sample fall back to one shared default set (38% of TuringLLM targets).

Contrast contexts for WCM. Weighted circuit masking and its evaluation instead select contrast contexts per target at fit time. Each of the target's strongest stored contexts $\mathcal{P}_f^{\text{top}}$ is represented, read from the index when present and otherwise computed by a forward pass; candidates from the retrieval store (the union of every latent's stored nearest neighbours, a subset of the 2×10^5-sequence index) are ranked by their highest cosine similarity to any of them, $\max_{x_p \in \mathcal{P}_f^{\text{top}}} \langle r(x_p), r(x) \rangle$, with all of the target's stored activating contexts (strongest and mid-band) excluded; the $128n$ most similar for n requested contexts (8,192 for the 64 of the first protocol; 24,576 for the deduplicated records, which request 192 so that deduplication can refill) are then run through the model in rank order, and each is kept only if the target is outside its site's Top-K at every position, until 64 are kept. A target with fewer than 4 kept is skipped, and one with 4 to 63 uses what it has (19 and 10 TuringLLM targets in the protocol-v1 run). In the production run, 9 TuringLLM targets kept 0–2 and got no record, 6 kept fewer than 64 (at least 6), and every Pythia-70m and GPT-2 small target kept 64. The silence check rejects between 14% and 51% of the closest candidates on the targets we inspected, so unchecked retrieval would place many contrast contexts where the target fires. The same selector with random or maximally dissimilar ranking gives the *random* and *distant* sources of Appendix J.

Logit faithfulness. The output-level score of Appendix E is the faithfulness formula of Section 3.3 with the target's value replaced by the output metric m, the mean log-probability of the actual next token at the anchor: $(m(C) - m(\varnothing))/(m(M) - m(\varnothing))$, with every in-scope latent outside the circuit mean-ablated.

S Evaluation Protocol Studies

S.1 Sparsity-Preserving Mean Ablation

The classic dense mean ablation assigns every latent outside the circuit its mean, producing a stream with tens of thousands of simultaneously active latents per position, outside the state space a Top-K SAE can produce. The sparsity-preserving variant applies the means and then re-imposes

the per-position Top-K (kept latents, then the largest means of the other latents up to k, rest zero), which restores the keep-all identity to exactly 1.0. At scale (32 targets, paired) the sparsity-preserving variant does not systematically shift faithfulness: it raises both a_C and $a_\varnothing$, leaving the ratio a wash (50W/46L). It therefore functions as a cleaner protocol rather than a score inflator; but at small node sets and tight Top-K budgets the two variants can disagree on *rankings*, which is why the body's comparisons name their ablation method.

The same fill in training. Weighted circuit masking trains its A and C terms under this fill, defined exactly as the scorer's. At each position every member (gate above $\frac{1}{2}$) is kept at α times its live value, even when more than k members are active; the remaining budget, k minus the members active there, goes to the non-members with the largest means, at their means; every other latent is zero. The selection is not differentiable, so its gradient is passed straight through: the forward pass uses the filled code and the backward pass treats the selection as the identity, so closed gates still receive a gradient. The means are the evaluation's own: A over the strongest training contexts that serve as the evaluation's reference, C over the training contrast contexts. Trained under the dense mean instead, circuits learn to rely on a background of tens of thousands of weakly active latents that the evaluation's fill removes, so they can pass under a dense mean and fail under the fill.

S.2 Sources of the Mean Ablation

The empty-circuit baseline $a_\varnothing$ is a measured object with its own failure modes. Means over *activating* contexts leak target signal into $a_\varnothing$ increasingly with depth ($a_\varnothing/a_{\text{pos}} = 0/23/30/26\%$ at layers 2/8/9/10); means over the target's *contrast* contexts hold $a_\varnothing = 0.0000$ on every target measured, even under the hardest close contrast contexts. These exploratory measurements, on four targets, are the source of the leakage figure in Section 3.3.

T Dense ReLU Dictionaries

The method is scoped to Top-K dictionaries because of an exploratory test on Pythia-70m with the L1-trained ReLU dictionaries of Marks et al. (2025): six residual-stream targets at layers 2 and 4, an earlier form of weighted circuit masking, and held-out contexts. The method replicated: 11 of 12 fits (six targets at two sparsity penalties) were faithful under zero and mean ablation (0.80–1.00) and necessary (0.81–1.0) on held-out contexts. The random-circuit baseline did not: random sets given the same coefficient fitting failed at up to ~1,100 latents, but at 2,000–6,000 latents they reproduced the target under zero ablation (0.71–1.00) while still failing necessity. The cause is the encoder's sparsity: 6.5–37% of a target's live ReLU latents are nonzero at its anchor, against 0.3% expected under Top-K ($k = 128$ of 40,960), so under ReLU a random latent's fitted coefficient has a nonzero value to scale, and thousands of them can synthesise the target; under Top-K a random latent is almost always zero at the anchor. On dense dictionaries, necessity rather than faithfulness is the discriminating test, and the random baseline needs its own design.

U Building with Weighted Circuit Masking

Weighted circuit masking fixes a problem, (P), and a way of relaxing it, but leaves most of the recipe open. Table 29 lists its slots: what is reproduced, over which nodes, with which parametrisation, under which ablation methods and backgrounds, with which regularisers and optimiser, from which contexts, and how the result is scored. The production choice in each slot was made by an experiment reported in these appendices, most of them on TuringLLM and carried over unchanged to the public models (Table 4, Provenance); none is claimed to be the best setting, and several are known to trade one evaluation against another.

Table 29: The slots of weighted circuit masking, the production choice in each, and the alternatives this paper examined or that the formulation admits.

Slot	Production choice	Alternatives	Evidence
Target	the pre-activation $\hat{a}_f$ of one SAE latent at its anchors; censored read under Z, C, A	an output metric, as in learned-mask circuits for outputs (Bhaskar et al., 2024; Caples et al., 2025); output- and latent-targeted circuits fail each other's objective (exploratory)	Section 3.2, Appendices E and R.1
Nodes	every latent at every SAE site upstream of the target; SAE errors passed through; nodes recomputed	dense ReLU dictionaries (the method replicates; the random baseline needs its own design); members clamped (resample diagnostic)	Section 3.1, Appendices L and T
Parametrisation	gate $\sigma(\theta/T)$ and coefficient $\text{softplus}(\psi)$ per node, $\alpha_0 = 1$	$\alpha \equiv 1$ (unweighted circuit masking)	Section 4.2, Table 13
Training ablations	Z (weight 1), C and A (0.25 each) under evaluation's sparsity-preserving fill; resample D over the training contrast contexts (1)	one method alone; dense means; D in place of C; no D	Appendices L, S.1 and V.3
Contrast background	64 close, silence-checked contexts per target (48 train, 16 held out)	random, distant, or fused sources; the unchecked retrieval store	Appendices J and R
Regularisers	flat penalty $\lambda = 10^{-3}$ per open gate; closed-gate charge $\lambda(1-\sigma(\theta))\lvert\alpha-1\rvert$; rank-keep $\gamma_K = 10^{-1}$; one shared normaliser ν	other penalties λ (sweep); off-target term, Top-K hinge, pull to rank 1, firing margin; per-term normalisers	Section 4.3, Appendices B.1, V and V.1
Optimiser	AdamW, 400 steps, batch 4; gate temperature annealed $1 \to 0.05$; learning-rate warm-up, then geometric decay	straight-through binarisation; 200-step and reach-then-hold schedules; larger batches (exploratory)	Table 4, Appendices U.1 and V
Training contexts	32 strongest and 16 mid-band activating contexts, deduplicated and split by whole groups of copies	a split without deduplication, which leaked copies into held-out sets	Section 4, Appendix B.2
Evaluation	held-out strongest contexts; Z, A, C in $[0.75, 1.5]$ and necessity ≥ 0.9; vacuous targets excluded and counted	reported, not required: mid-band contexts, sufficiency to induce, donor-background test, specificity, $\alpha = 1$ rescoring, random-circuit baselines, stable core	Section 3.3, Appendices B, I and P

Training in the evaluation's domain. Across these experiments, circuits did best on the evaluations their training resembled. When a fit fails an evaluation, the failure is often a property of the recipe, and a slot of Table 29 can put the fit in that evaluation's domain. *Real backgrounds.* Circuits trained only against zero and means failed a resample test almost always; adding a resample term over the training contrast contexts raised the resample pass rate on 72 targets from 0% to 24%, at 1.18× the circuit size and 1.45× the fitting time, and with rank-keep raised it took a 72-target check panel from 19 to 39 targets passing (Appendix L). Using it in place of the contrast-mean term gained as much on the resample test but cost the pass criterion, so terms should be added for a new evaluation rather than swapped for an old one; on the production sweep the term's effect on the pass criterion itself is 5.2 points and not significant (Table 23). *Ablation methods and fills.* A mask trained under one ablation method learns what that method rewards: without the zero-ablation term, coefficients overshoot zero ablation by orders of magnitude, and circuits trained against dense means can fail under the sparsity-preserving fill that evaluation uses (Appendices S.1 and V.3). *Top-K survival.* Reproducing the pre-activation does not keep the target in its site's Top-K; the censored read and rank-keep train for it, though neither removes amplifier circuits (Appendices B.1 and K). *Other backgrounds.* Contrast faithfulness generalises furthest when training uses random or distant contrast contexts (Appendix J). *Size.* Circuits shrink with λ at a nearly constant pass rate up to a plateau's edge (Section 4.3). Two evaluations have no training term yet: sufficiency to induce, which no circuit is trained on (median 0.78–0.88 on held-out contrast contexts, 0.35 for TuringLLM attention outputs; Appendix G), and transfer to weaker, mid-band contexts (Section 4.1). The framework's prescription for them is a term or a training set in their domain, which we have not tested. A fit trained for an evaluation is, for that evaluation, scored on home turf (Appendix D), so the evaluation stays meaningful only if its contexts and donors are never seen in training, as the donor-background test's donors are not (Table 4), and the

evaluations not trained for are reported alongside.

Choosing the sparsity penalty. The penalty sets circuit size and, past a point, the pass rate: on the TuringLLM sweep the pass rate holds at 89–93% from $\lambda = 2.5 \times 10^{-5}$ to 5×10^{-4} while each doubling of λ shrinks circuits 1.4–1.9×, and it declines beyond (Section 4.3). We recommend fixing the selection rule before looking at pass rates, as our size-only pilot rule did on the public models, where it kept $\lambda = 10^{-3}$ on both (Table 15), and reporting a curve over λ rather than one point, since one setting that transfers is not the best setting for each model.

A new model or dictionary. Only the model and dictionary adapters change between our three models (Section 4); four things needed checking. *Precision*: TuringLLM runs in the bfloat16 its pipeline was validated in, but on the public models a bfloat16 data pass did not reproduce the float32 top-context sets, so they run in float32 (Table 4); a precision check belongs before any fit, since node sets also move with rounding (Appendix P). *The dictionary's sparsity*: k sets the fill's budget and the censored read's cut; on a dictionary that keeps every latent ($k = d_{\text{sae}}$) the censored read has no cut to model, and on dense ReLU dictionaries a random-circuit baseline with fitted coefficients stops discriminating, so necessity, not faithfulness, separates real from random circuits there (Appendix T). *Contrast retrieval*: contrast contexts need a pooled representation for every corpus sequence and a silence check, which rejects 14–51% of the closest candidates on the targets we inspected (Appendix R). *Corpus copies*: duplicated windows silently leak into held-out sets unless pools are deduplicated and split by groups of copies (Appendix B.2).

What to report. Because each slot changes the circuits, a result is interpretable only with its recipe: the ablation methods used in training and in evaluation; faithfulness under each method and necessity, on held-out contexts, by depth band and site kind, with intervals over targets; excluded and vacuous targets counted; circuit size with whether coefficients were fitted, and the $\alpha = 1$ rescoring; random-circuit baselines given the same fitting; and the reported-only evaluations (sufficiency to induce, the donor-background test, specificity, mid-band transfer) beside the pass rate (Section 5).

Choices that matter, and what stays open. The ablation method changes not only scores but rankings: the two mean-ablation variants can order small node sets differently (Appendix S.1), and in exploratory comparisons a method tended to win the scores of the ablation method it was trained or ranked under (Appendix D), so comparisons between recipes need the full matrix of recipe against ablation method. Node sets are not identified by any recipe: a refit with reordered training contexts shares a median Jaccard of 0.39–0.51 with the original and its shared core alone passes for 44–72% of targets (Appendix P), so changes to a slot should be judged by sizes and scores, not by which latents moved. Finally, the production choices were calibrated on TuringLLM targets that overlap its evaluation set (Section 4), and on the public models every setting except λ carried over unchanged (Table 4, Provenance), so on a new model they are defaults to be checked, not tuned values.

U.1 How the Production Recipe Was Reached

The production recipe is the end of a sequence of changes, each made because a run exposed a failure, and each tested against the recipe before it on the same targets, mostly under a selection rule stated before the comparison. We list them in order as evidence of which kinds of change have paid off; the panels differ between steps, so numbers compare only within a step.

Contrast contexts built per target. The first contrast contexts came from a retrieval store whose index covered too few sequences, so 38% of TuringLLM targets fell back to one shared list; WCM now selects contrast contexts per target at fit time, ranked by similarity to its strongest contexts and kept only if the target is silent (Appendix R), and a later comparison of contrast sources showed how much that choice shapes the C term (Appendix J). The lesson: inspect the data each training term actually sees.

Rank-keep against off-target lifts. Reproducing the pre-activation left some circuits restoring their target by lifting its whole neighbourhood; of the terms tried, rank-keep alone kept the gains without the alternatives' costs, and entered the first protocol at a small weight (Appendices K and V.1).

The first run at scale. Under that protocol, WCM was fitted to 15,046 TuringLLM targets, and 46% passed, 22% at layers 8–11 and 14% for attention outputs (Appendix H). Analyses of the run exposed three problems: 66.8% of held-out strongest contexts repeated a training context, which led to deduplicated records split by groups of copies (Appendix B.2); with depth, failing circuits increasingly failed only the two mean ablations (Figure 9); and 0.1% of circuits passed a resample test, against 43.1% under the pass criterion (Appendix L).

Training under the fill that evaluation uses. The mean-ablation failures were a mismatch: training replaced non-members with dense means, a code no Top-K dictionary emits, while evaluation re-imposed the Top-K (Appendix S.1). Training the A and C terms under evaluation's own fill and means raised the pass rate on 72 stratified targets from 31 to 50 (paired difference +26.4 points, 95% interval 15.3–38.9), from 2 to 11 of 24 at layers 8–11 and from 3 to 12 of 24 for attention outputs, with zero-ablation faithfulness unchanged (median paired change 0.00), circuits 0.91× the size, and 1.27× the fitting time. The gain came from aligning training with evaluation, not from a new term.

A resample term, and a stronger rank-keep. To survive real backgrounds, training gained the resample term (above and Appendix L). On the previous version's zero-ablation failures, 15 of 18 restored the target's pre-activation but lost it at the Top-K cut, so rank-keep was raised from 3×10^{-3} to 10^{-1} with the resample term at full weight; chosen by a rule stated before the comparison, the pair took a 72-target check panel from 48 to 54 targets passing and from 19 to 39 passing the resample test, at 1.33× the circuit size (Appendix L). In the production sweep, the arm trained without the resample term passes 5.2 points less, not significantly (Table 23).

A learning-rate schedule and the censored read. Under a constant learning rate, the gate step grows twentyfold as the gate temperature anneals, and some fits degraded late in training. A geometric decay alone stabilised them but left gates too little total step to close, so circuits grew 2–4×; raising its peak 3.15× restored the summed step size, and a short warm-up was added for the early spikes the higher peak brought back (Appendix B.1). With the censored read added, so that training also sees whether the target survives its site's Top-K, the check panel went from 54 to 63 of 72 targets passing (10 gained, 1 lost) at about the same circuit size (median ratio 1.04).

Settings left as they were. The penalty was carried to the public models by a size-only rule fixed before a pilot (Section 4.3). In exploratory tests, batches of 16 for the same 400 steps passed 15 of 16 TuringLLM targets against 16, with circuits 0.75× the size where both passed, at 3.4× the fitting time, so the batch of 4 was kept; and policies that adapted λ per target traded passes for size on the check panel, losing passes mostly to necessity, which no training term covers. Other rejected designs are in Appendix V.

Directions not yet taken. The same path suggests where the recipe can go next. Necessity, sufficiency to induce, and transfer to mid-band contexts are scored but not trained for; each could get a term or training contexts in its domain, as the resample test did. Deep and attention-output targets still pass least (Section 4.1), and in the schedule pilot the attention failures kept zero-ablation faithfulness near 0 under every schedule, so the learning rate does not explain them. Amplifier circuits survive rank-keep (Appendix K); the off-target term cut lifts three- to sevenfold but grew circuits (Appendix V.1), so a cheaper specificity term remains open. And circuits explain the target at its anchors, not at every position (Appendix L); training on more positions is untested.

V Weighted Circuit Masking Design Ablations

Rejected or superseded designs for WCM, recorded because each closes a natural reviewer question.

V.1 Loss Terms for Target Rank and Specificity

An *off-target* term charging every other site latent for rising above its unmodified and empty-circuit values cut lifts three- to sevenfold at no faithfulness cost, but addressed a property the method need not guarantee (Appendix K) and grew circuits when it also charged harmless lifts below the cut. A *Top-K hinge* requiring only that the target beat the k-th other latent restored faithfulness but, used alone, let circuits keep the target in the Top-K by lifting it together with its neighbourhood. A convex pull towards rank 1 inflated circuits by an order of magnitude at every weight tried, since it never switches off and dominates once the reproduction error is small. The *firing-margin* objective, fitting the target's lead over the Top-K cut, needs its loss normalised by the target's pre-activation scale rather than the margin's, which is near zero for targets at the cut. Rank-keep, which switches off once the target holds its natural rank, is the single term that kept the gains without these costs, and its effect is flat across a thirtyfold range of weights.

V.2 Normalisation

The shared normaliser is target-scaled and bounded by construction because its predecessor, in an earlier two-method objective, was not: normalising each term by its own fully-closed-mask loss, which is unbounded far from the data, silently annihilated the zero-ablation term at depth (the two closed-mask losses differ there by up to eight orders of magnitude), reducing WCM[Z+C] training to the C term alone; the design rule is that a normaliser must never be able to annihilate its own term.

V.3 Training under One Ablation Method

In exploratory calibration the A term roughly halves the smallest passing node set on layer-2 targets and raises the share of layer-9 targets with any passing circuit. A mask trained only under mean ablation over contrast contexts has zero-ablation faithfulness 0.0 at depth: it learns the difference from the contrast-context mean, never a set that suffices alone. Removing the zero-ablation term from training explodes zero-ablation faithfulness to 36–1,681 at depth while the trained ablation methods stay in band: coefficients trained without it calibrate against the mean ablations' background and overshoot by one to three orders of magnitude once that background is removed.

V.4 Binarisation Sweep

The gates are optimised directly rather than sampled (Louizos et al., 2018; Li and Janson, 2024); the local minima that sampling avoids appear here as the equivalence class of near-optimal node sets (Appendix P). The straight-through estimator makes interior gate values forward-irrelevant, so the sparsity penalty drifts nodes toward just-above-threshold: the decision boundary becomes an *attractor*, and per-step node flips grow through training (218 per step at layer 10), a stopping-step lottery. Annealing instead yields the tightest zero-ablation faithfulness in the programme (0.002–0.10 deviation from unity) at slightly worse mean-ablation scores, another home-turf demonstration. Schedule compression does not help: a 200-step run matches the 400-step metrics at ~2× the per-step churn (churn is set by per-step pressure, not schedule shape); reach-then-hold leaves churn unchanged and inflates size 21–48%; and truncating a 400-step run at 200 yields node sets 57–72% oversized with the gate still half-soft.